

Bereichsschaltplan single-zone circuit diagram

index:

D0129411 6

CS48a-01-R-V1-06

Bereich Rotor
rotor zone

ENERCON E-44, E-48, E-53

800kW/900kW

3/N/PE 50Hz 400V

CS 48a

2014



Alle nicht gekennzeichneten Neutralleiter 1mm² hellblau.
All not marked neutral conductor 1mm² light blue.

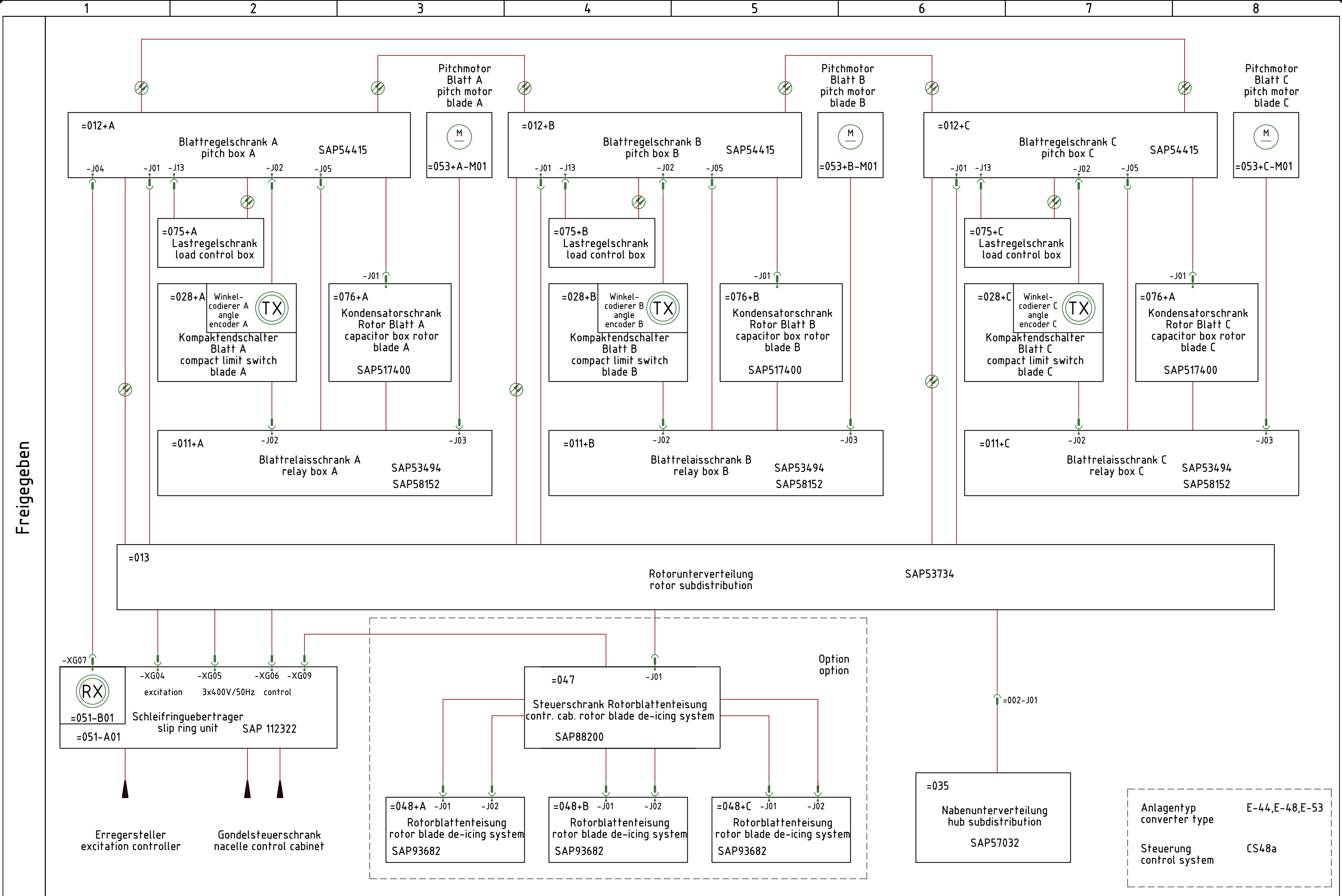
Technische Änderungen vorbehalten!
Technical subject to alteration!

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6		29.03.2011	14056	date	28.03.2014 04.09.2008	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Deckblatt cover	Bereich Rotor rotor zone	order no.: CS48a-01-R-V1-06		= COVER	
				prep.by	Rülander						ass. ID:		+	
				arr.by										
no.	revision	date	name	appr.	2467	origin: germany	repl. for	repl. by			© Copyright ENERCON GmbH. All rights reserved.	document ID: D0129411	index: 6	page 1 of 1

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	Baugruppeneübersicht/ assembly overview										Verwendung/ application												
	Bgr. Nr. assembly no.	Baugruppen- bezeichnung assembly designation			Version version	SAP-Artikel Nr. SAP-article no.	Zeichnungsnr. Schaltplan drawing no. circuit diagram		Schaltplan siehe Bgr.Nr. circuit diagram look assembly no.		Bemerkung remark		E-48 CS48a	E-44 CS48a	E-53 CS48a								
	001	Nabe hub				diverse			035				x	x	x								
	002	Rotor rotor				diverse			035				x	x	x								
	011	Blattrelaisschrank relay box			V1	53494	011-53494-V1-xx				Anhang attachment		x	x	x								
	011	Blattrelaisschrank relay box			V2	58152	011-58152-V2-xx						x	x	x								
	012	Blattregelschrank pitch box			V1	54415	012-54415-V1-xx						x	x	x								
	013	Rotorunterverteilung rotor subdistribution			V1	53734	013-53734-V1-xx						x	x	x								
	028	Kompaktendschalter compact limit switch				diverse			011				x	x	x								
	035	Nabenunterverteilung hub subdistribution			V1	57032	035-57032-V1-xx						x	x	x								
	047	Steuerschrank Rotorblattenteisung contr. cab. rotor blade de-icing system			V1		047-88200-V1-xx				optional optional		x		x								
	048	Rotorblattenteisung rotor blade de-icing system			V1		048-93682-V1-xx				optional optional		x		x								
	051	Schleifringuebertrager slip ring unit			V1	16600	051-16600-V1-xx				Anhang attachment		x										
	051	Schleifringuebertrager slip ring unit			V1	58635	051-58635-V1-xx				Anhang attachment		x										
	051	Schleifringuebertrager slip ring unit			V2	65830	051-65830-V2-xx				Anhang attachment		x	x	x								
	051	Schleifringuebertrager slip ring unit			V3	83268	051-83268-V3-xx				Anhang attachment		x	x	x								
	051	Schleifringuebertrager slip ring unit			V5	112322	051-112322-V5-xx						x	x	x								
	053	Pitchmotor pitch motor				diverse			011				x	x	x								
	075	Lastregelschrank load control box			V1	27530	075-27530-V1-xx								x								
076	Kondensatorschrank Rotor capacitor box rotor			V2	517400	076-517400-V2-xx						x	x	x									
6	diverse	29.03.2011	14056	date	17.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Baugruppeneübersicht assembly overview Bereich Rotor rotor zone			order no.: ass. ID: CS48a-01-R-V1-06		Y98		= 000Y98 +							
5	C000593	26.08.2009	ehmen	prep.by	A.Rülander																		
04	diverse	18.08.08	ETD	arr.by																			
no.	revision	date	name	appr.	K.Goeman	origin: germany	repl. for	repl. by				© Copyright ENERCON GmbH. All rights reserved.		document ID: D0129411		index: 06		page 1 of 3					

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Freigegeben		Baugruppeneübersicht/ assembly overview										Verwendung/ application											
		Bgr. Nr. assembly no.	Baugruppen- bezeichnung assembly designation			Version version	SAP-Artikel Nr. SAP-article no.	Zeichnungsnr. Schaltplan drawing no. circuit diagram		Schaltplan siehe Bgr.Nr. circuit diagram look assembly no.		Bemerkung remark		E-48 CS48a	E-44 CS48a	E-53 CS48a							
		096	Zentralschmiereinheit central lubrication system				diverse			013, 035				x	x	x							
		6	diverse	29.03.2011	14056	date	17.01.2014	Schaltplan circuit diagram		CS 48a Windenergieanlage wind energy converter				Baugruppeneübersicht assembly overview Bereich Rotor rotor zone		order no.: CS48a-01-R-V1-06		Y98		= 000Y98			
		5	C000593	26.08.2009	ehmen	prep.by	A.Rülander									ass. ID:		+					
04	diverse	18.08.08	ETD	arr.by		© Copyright ENERCON GmbH. All rights reserved.										document ID: D0129411		index: 06		page 2 of 3			
no.	revision	date	name	appr.	K.Goeman	origin: germany		repl. for		repl. by													



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	000Y98						Baugruppeneübersicht assembly overview					3			
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	011			A			Sicherheitskreis Blatt A security circuit blade A					6			
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	011Y68						Kabelliste wire list					1			
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	012			A			Blattsicherheitskreis, Ausgaenge blade security circuit, outputs					3			
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				prep.by	A.Rülander					ass. ID:					
				arr.by						© Copyright ENERCON GmbH.		document ID: D0129411		index: 6	
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	013Y52				Klemmenleiste terminal strip										3				
	013Y52				Aufbauplan construction plan										4				
	013Y58				Anschlussplan connecting plan										1				
	013Y64				Stueckliste parts list										1				
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				prep.by	A.Rülander					ass. ID:		+							
				arr.by															
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Bereich Rotor rotor zone	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0129411	index: 6	page 2 of 5						

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	047Y66					Mengen-Stückliste quantity parts list								2			
	047Y68					Kabelliste wire list								1			
	048COVER					Deckblatt cover								1			
	048		A			Blatt A blade A								1			
	048		B			Blatt B blade B								2			
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	051COVER					Deckblatt cover								1			
	051					Versorgung supply								1			
	051					Rotorblattenteisung rotor blade de-icing system								2			
	051					Steuerung control								3			
	051Y52					Aufbauplan construction plan								1			
	051Y58					Anschlussplan connecting plan								1			
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	075Y52					Aufbauplan construction plan								1			
	075Y58					Anschlussplan connecting plan CS44A								1			
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	076COVER					Deckblatt cover								1			
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	076		C			Kondensatorschrank Rotor Blatt A capacitor box rotor blade A								3			
	076Y52					Aufbauplan construction plan								1			
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	000Y44		A			=011+A-X01								1			
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			prep.by	A.Rülander	ass. ID:							+					
			arr.by		© Copyright ENERCON GmbH. All rights reserved.					document ID: D0129411		index: 6		page 3 of 5			
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000Y44				=013-X01B													4				
000Y44				=013-X01B													4.1				
000Y44				=013-X01C													5				
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000Y48		A		=011+A-J03													4				
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000Y48		B		=012+B-J01													10				
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				prep.by	A.Rülander					ass. ID:				+							
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	000Y48				=015-XG03										21					
	000Y48				=018-XG01										22					
	000Y48				=047-J01										23					
	000Y48				=047-J01										23.1					
	000Y48		A		=048+A-J02										24					
	000Y48		B		=048+B-J02										25					
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	000Y48				=051-XG04										27					
	000Y48				=051-XG05										28					
	000Y48				=051-XG06										29					
	000Y48				=051-XG07										30					
	000Y48				=051-XG09										31					
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	000Y48		B		=076+B-J01										33					
	000Y48		C		=076+C-J01										34					
	000Y48				=096-J01										35					
	000Y48				=096-J02										36					
	000Y48				=096-J04										37					
	000Y48		A		=12+A-XG04										38					
	000Y48		A		=R01+A-J01										39					
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000Y48		B		=R01+B-J03										42						
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				prep.by	A.Rülander															
				arr.by																
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Blattrelaisschrank relay box

SAP article no.:
58152

index:
5

Reifegrad / level of maturity:
Serie
series

Zusatzinfo / additional info:

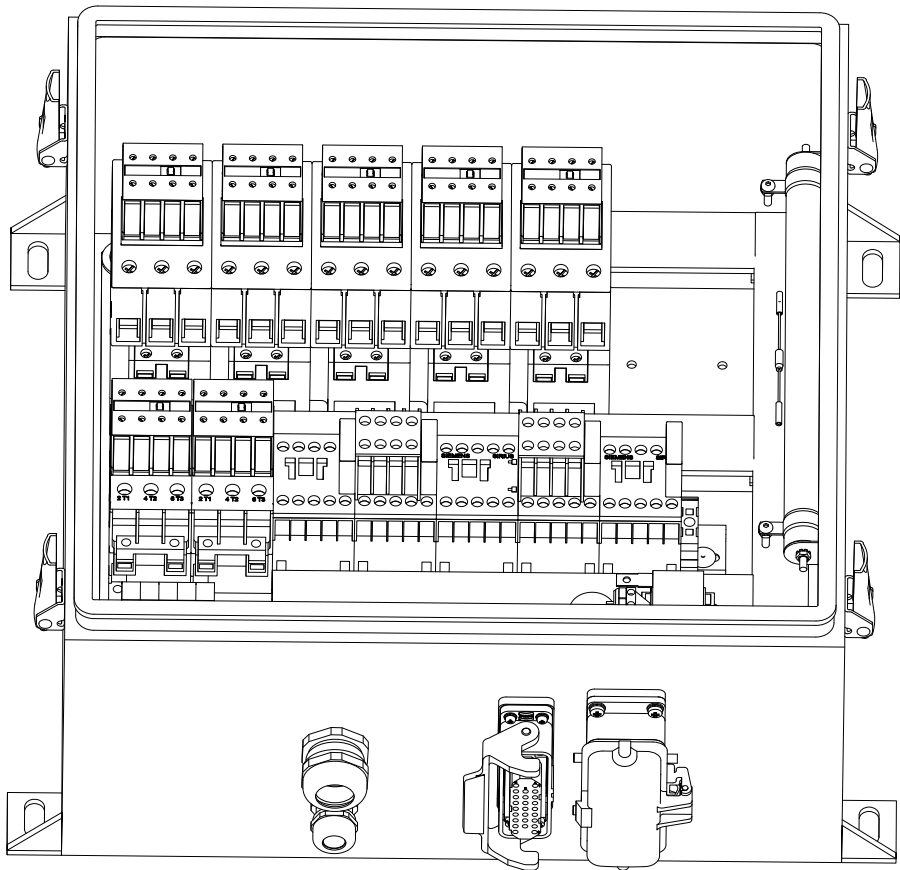
Anlagentyp / converter type:
E-48, E-53, E-44

Steuerung / control system:
CS48a

Baugruppennr. / assembly no.:
011

Version Baugr. / version of assembly:
2

Baugr. Index / component index:
07



Baugruppenschaltplan assembly circuit diagram

document ID:
D0132394

index:
3b

title or assembly ID:
011-58152-V203b

Alle nicht gekennzeichneten Leitungen 1mm² schwarz.
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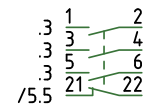
Alle nicht gekennzeichneten Neutralleiter 1mm² hellblau.
All not marked neutral conductor 1mm² light blue.

Technische Änderungen vorbehalten!
Technical subject to alteration!

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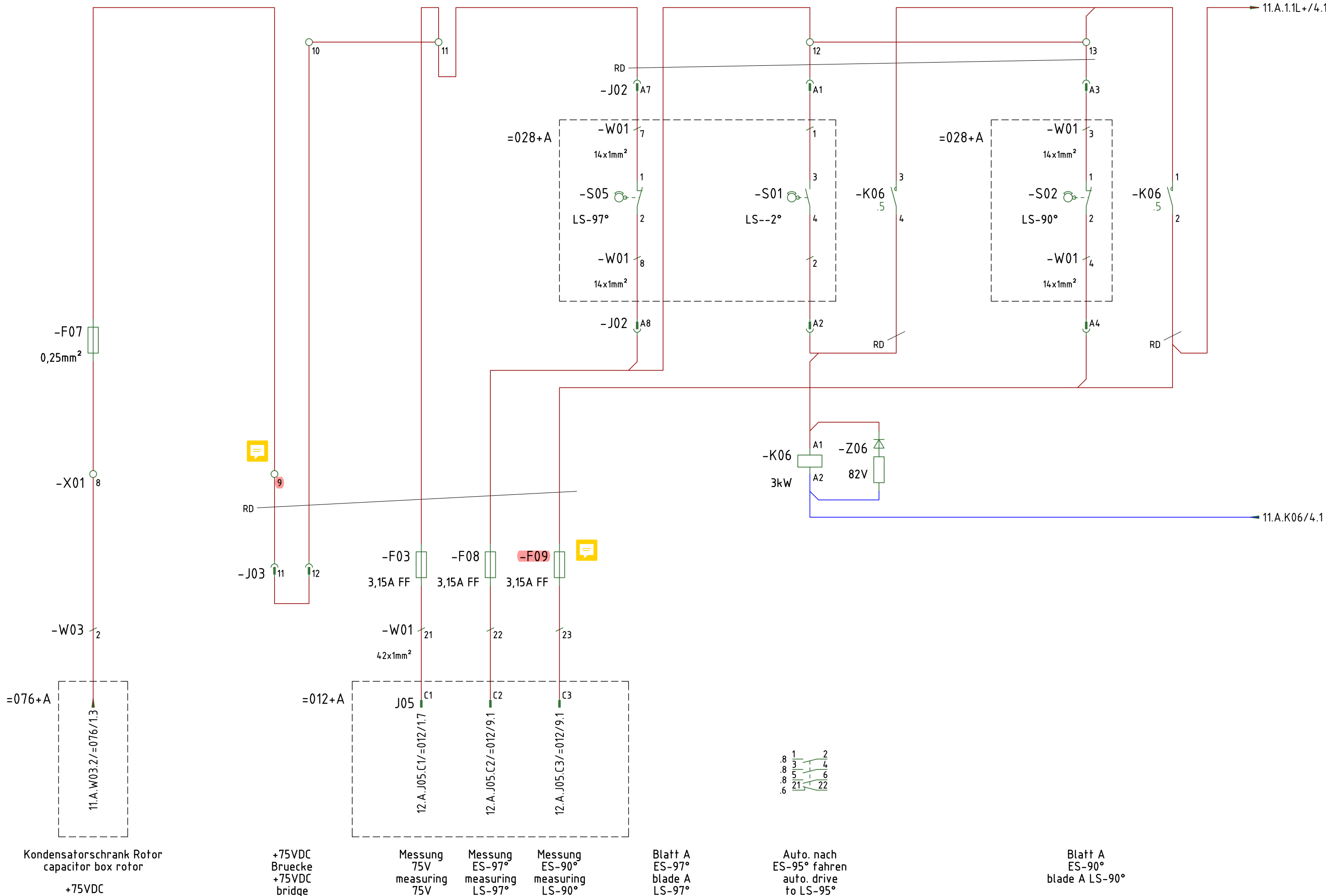
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3b		26.01.2010	dnewiger	date	19.01.2005	27.01.2009	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	Deckblatt cover Blattrelaisschrank relay box	order no.: 011-58152-V2-03b		= 011COVER					
3a	C000581	04.08.2009	ehmen	prep.by	K.Langer						ass. ID:		+					
3	C000507	19.03.2009	ehmen	arr.by	D.Newiger						© Copyright ENERCON GmbH. All rights reserved.		document ID: D0132394		index: 3b		page 1 of 1	
no.	revision	date	name	appr.	14056	KG	origin: germany	repl. for	repl. by									

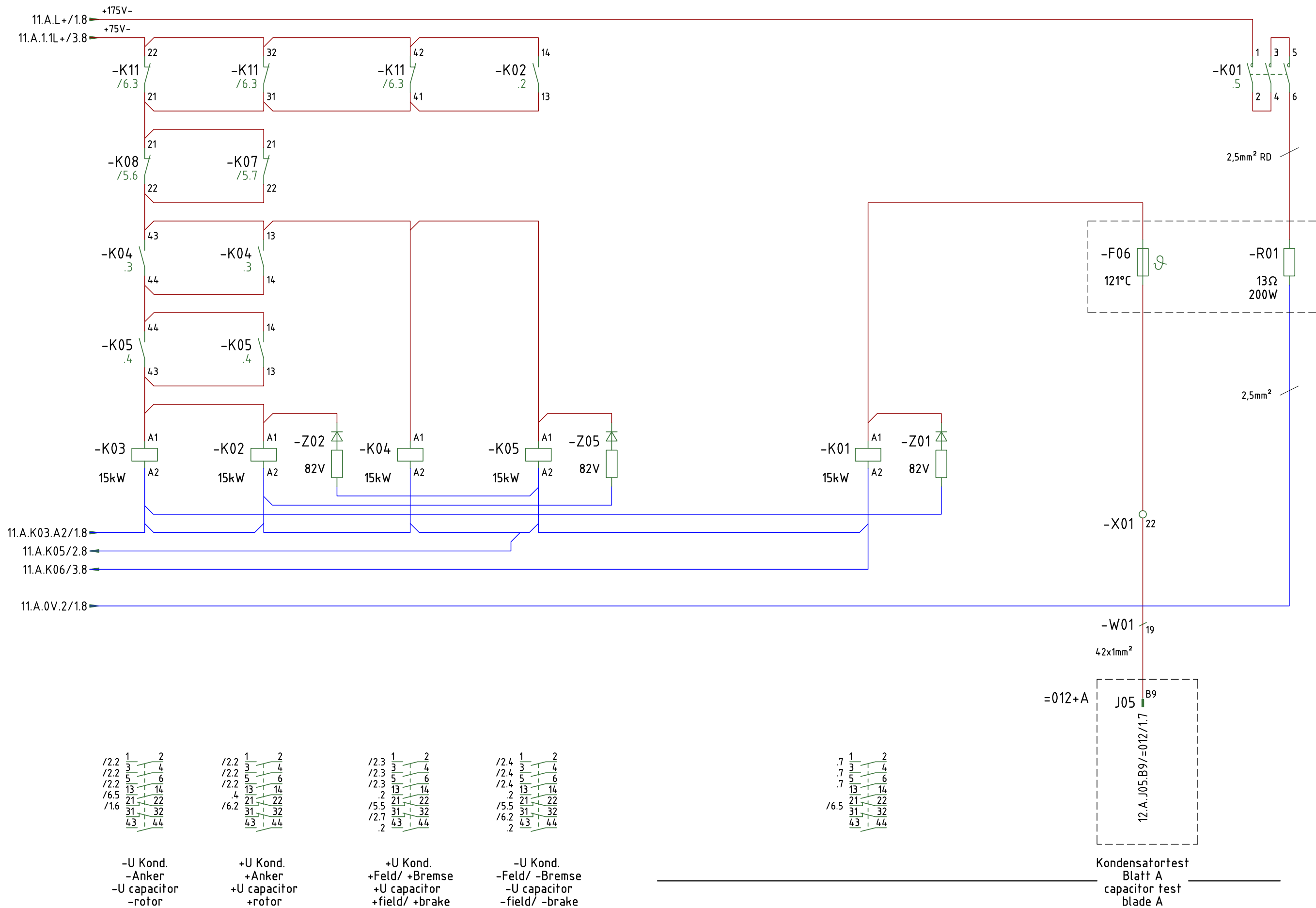


3	C000507	19.03.2009	ehmen GEH	date	19.01.2005	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Ladeschaltung charging circuit	order no.: = 011			
1	wiring	17.01.06	KL	prep.by	K.Langer					ass. ID: 011-58152-V2-03b Y32		+ A	
				arr.by	D.Newiger								
no.	revision	date	name	appr.	W.de Boer	origin: germany	repl. for	repl. by	Blattrelaisschrank relay box	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0132394	index: 3b	page 1 of 8

Freigegeben



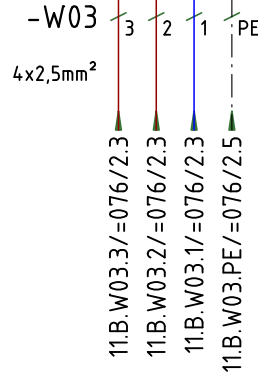
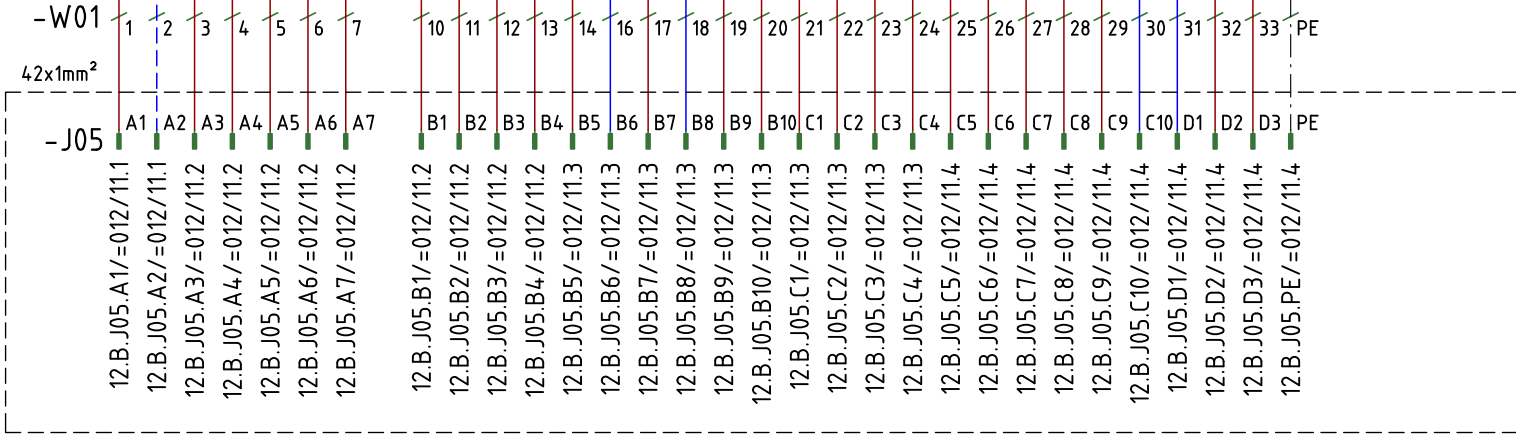
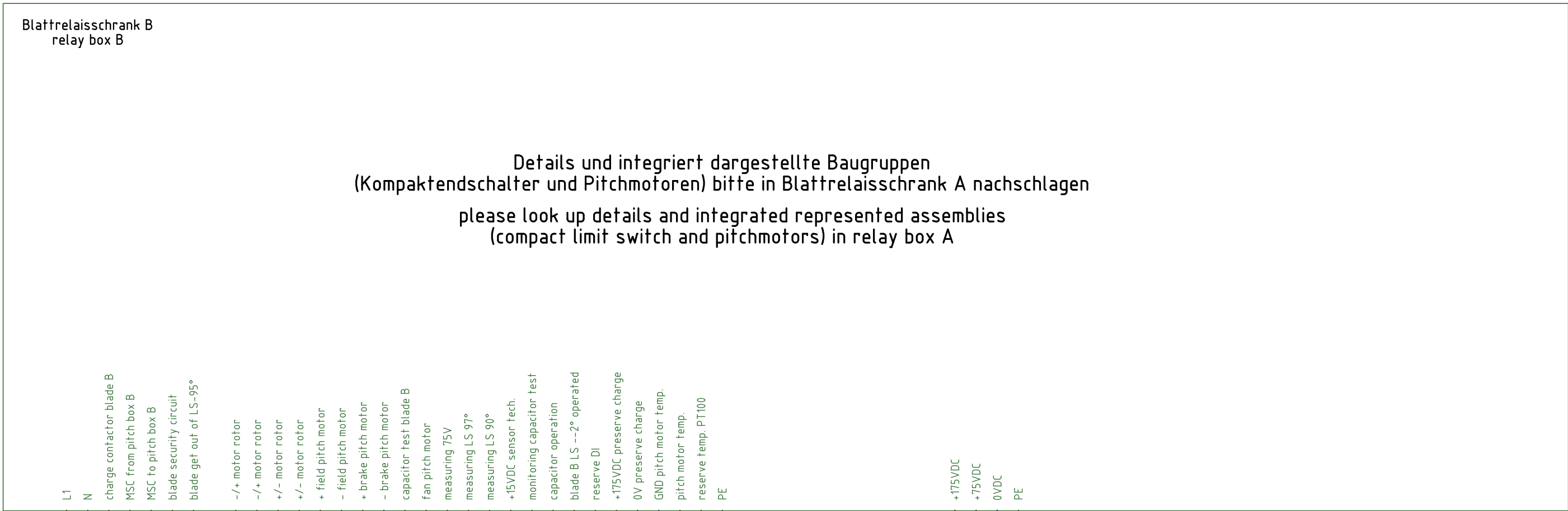
1	wiring	17.01.06	KL	GE	date	19.01.2005	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Kompaktendschalter Blatt A compact limit switch blade A Blattrelaisschrank relay box	order no.: 011-58152-V2-03b Y32 = 011			
				prep.by	K.Langer	ass. ID: + A								
				arr.by	D.Newiger									
no.	revision	date	name	appr.	W.de Boer	origin: germany	repl. for	repl. by			© Copyright ENERCON GmbH. All rights reserved.	document ID: D0132394	index: 3b	page 3 of 8



3	C000507	19.03.2009	ehmen GEH	date	19.01.2005	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Schuetzsteuerung fuer Kondensatorbetrieb contactor control for capacitor in operation	order no.: Y32		= 011	
1	wiring	17.01.06	KL	prep.by	K.Langer					ass. ID: 011-58152-V2-03b		+ A	
				arr.by	D.Newiger								
no.	revision	date	name	appr.	W.de Boer	origin: germany	repl. for	repl. by	Blattrelaisschrank relay box	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0132394	index: 3b	page 4 of 8



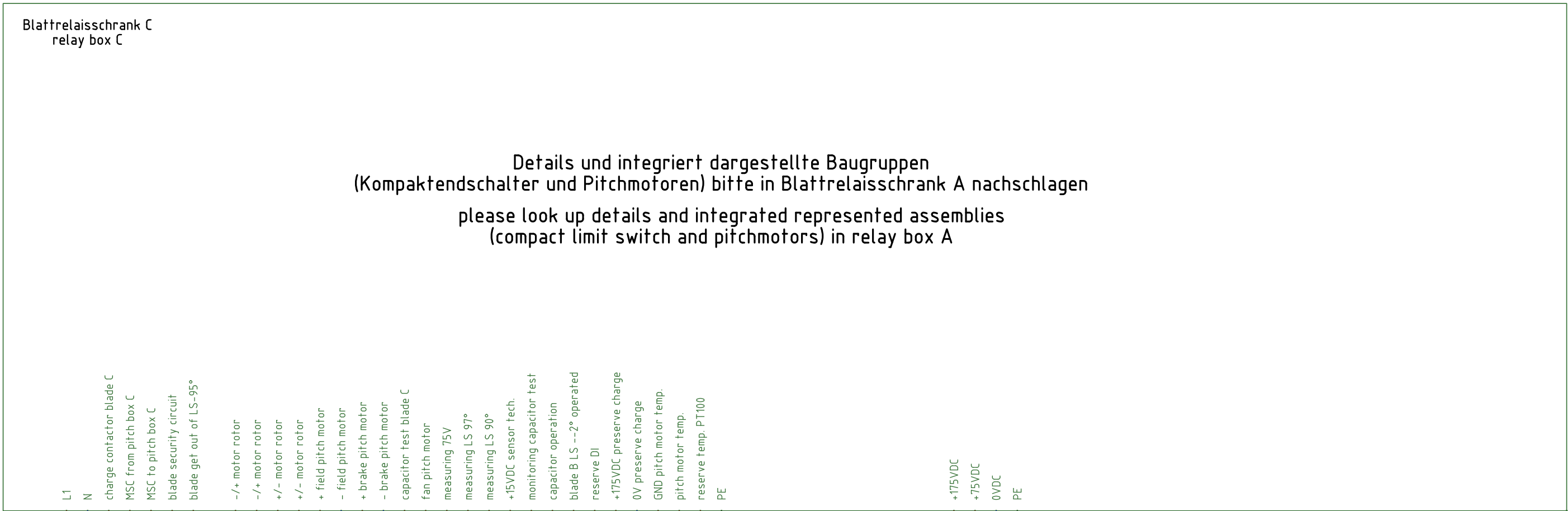


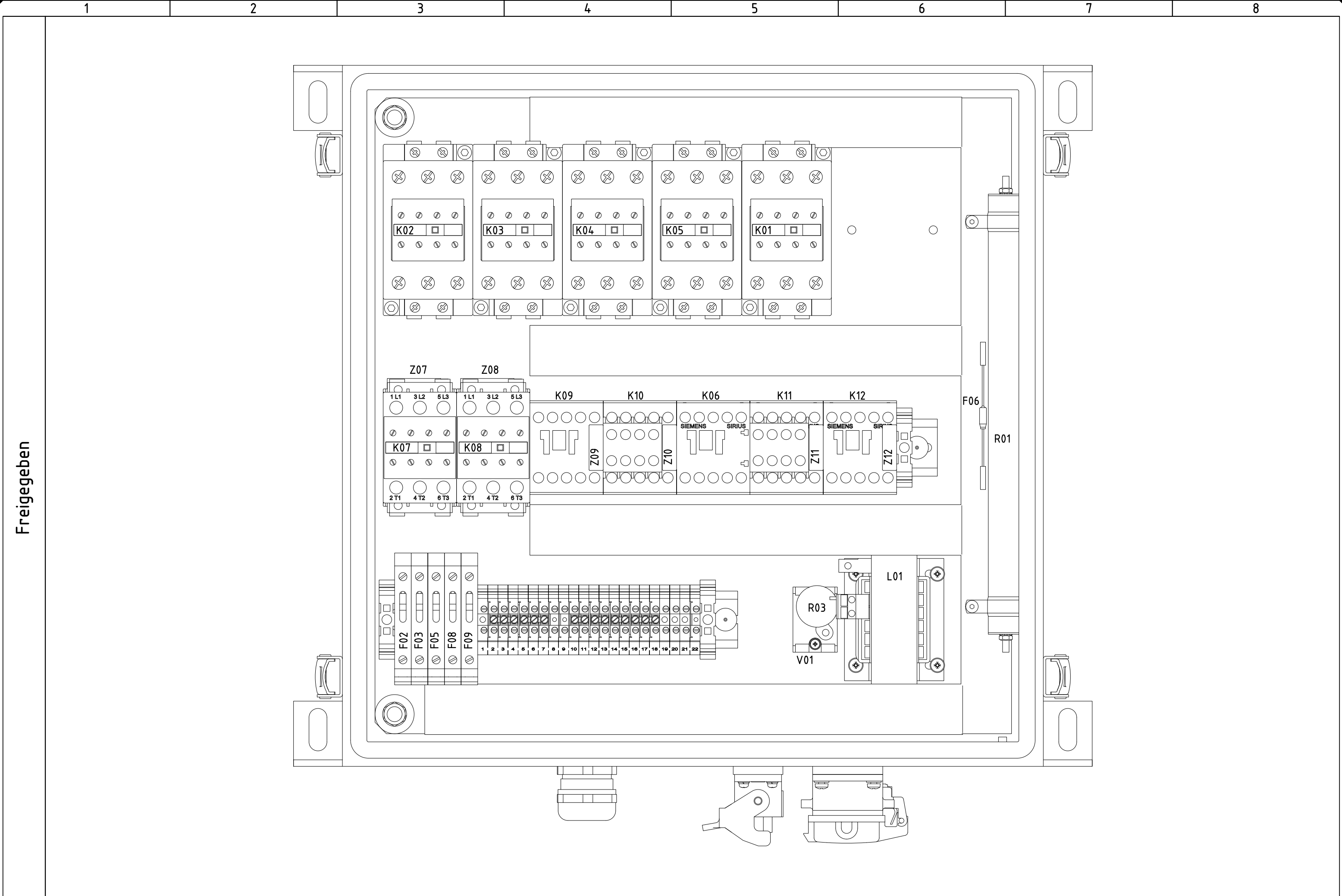


Blattregel-
schrank B
pitch
box B

Kondensator-
schrank Blatt B
capacitor
box blade B

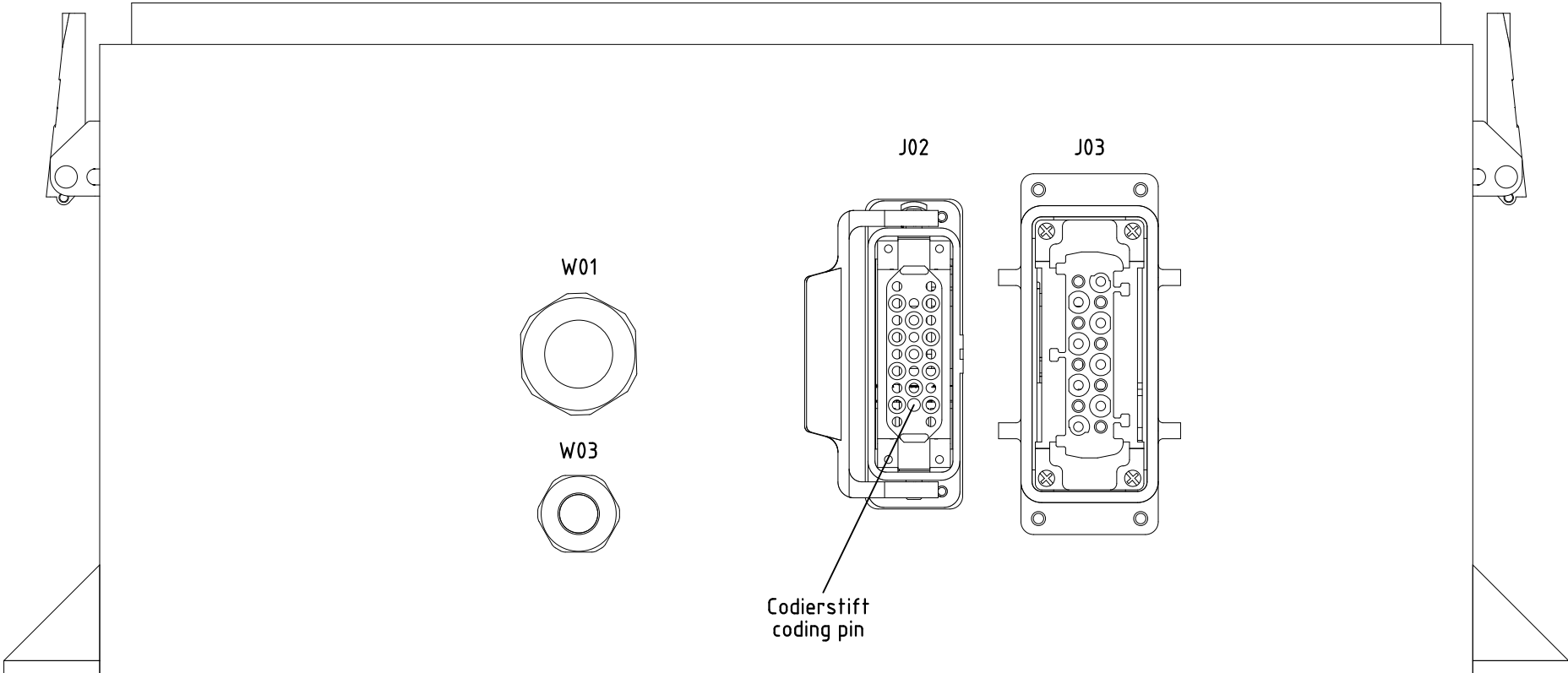
				date	19.01.2005	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Blattrelaisschrank B relay box B	order no.: 011-58152-V2-03b		= 011	
				prep.by	K.Langer					ass. ID:		+ B	
				arr.by	D.Newiger					© Copyright ENERCON GmbH. All rights reserved.		document ID:	index:
no.	revision	date	name	appr.	W.de Boer	KG	origin: germany	repl. for	repl. by	D0132394		3b	page 7
												of	8





3	C000507	19.03.2009	ehmen	date	19.01.2005	Schaltplan	CS 48a		Aufbauplan	order no.:	= 011Y52	
				prep.by	K.Langer	circuit diagram	Windenergieanlage		construction plan	ass. ID:	011-58152-V2-03b	
				arr.by	D.Newiger		wind energy converter		Blattrelaisschrank	document ID:	D0132394	index: 3b
no.	revision	date	name	appr.	KG	origin: germany	repl. for	repl. by	relay box	© Copyright ENERCON GmbH. All rights reserved.		page 1 of 2

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3a	C000581	04.08.2009	ehmen	date	19.01.2005	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Aufbauplan construction plan Blattrelaisschrank relay box	order no.: 011-58152-V2-03b		= 011Y52	
				prep.by	K.Langer					ass. ID: 011-58152-V2-03b		+	
				arr.by	D.Newiger					© Copyright ENERCON GmbH. All rights reserved.		document ID: D0132394	index: 3b
no.	revision	date	name	appr.	KG	origin: germany	repl. for	repl. by					



				date	19.01.2005	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	Anschlussplan connecting plan	order no.: 011-58152-V2-03b			= 011Y58	
			prep.by	K.Langer	Blattrelaisschrank relay box					ass. ID: Y58		+		
			arr.by	D.Newiger						© Copyright ENERCON GmbH. All rights reserved.		document ID: D0132394	index: 3b	page 1
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by					of 1	

Freigegeben

Mengen-Stückliste/ quantity parts list				
Artikel Nr.- Index/ article no.-index	Menge/ quantity	Benennung deutsch/ title german Benennung englisch/ title english	SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english	Hersteller/manufacturer Hersteller Teile-Nr. / manufacturer part no.
11243-0	1	Kontakteinsatz Buchse socket insert	Einsatz Buchse 16p 380VAC 16A Crimpan. socket insert 16p 380VAC 16A crimp term.	Harting Deutschl. GmbH&Co.KG 09330162702
1584-0	1	Varistor varistor	Varistor S20K275 varistor S20K275	
1701-0	1	Anbaugehäuse mounting case	Gehäuse Anbau- 40p 113x 43 SWF 1 Kappe housing,ext.mount.40p 113x43 SWR 1 cover	Harting Deutschl. GmbH&Co.KG 09370160302
1712-0	1	Sicherung fuse	Sicherung Temperatur 121°C SF-119-E1-10A fuse temperature- 121°C SF-119-E1-10A	
18035-0	1	Sicherung fuse	Kabel Sicherungs-Relaisschrank E44/48/82 fuse conductor relay box E44/48/82	
4525-0	1	Anbaugehäuse mounting case	Gehäuse Anbau- 16p 96x 29 SWF 1 Bügel housing,ext.mount.16p 96x29 SWR 1 lever	Harting Deutschl. GmbH&Co.KG 09370250303
53569-0	2	Leistungsschütz power contactor	Schütz kl. 07A 1Ö 230V 3RT1015-1AP02 contactor sm. 07A 1NC 230V 3RT1015-1AP02	Siemens AG A&D 3RT1015-1AP02
53571-0	2	Leistungsschütz power contactor	Schütz Leistungs- 230VAC 3RT1025-1AP00 contactor power- 230VAC 3RT1025-1AP00	Siemens AG A&D 3RT1025-1AP00
53589-0	5	Leistungsschütz power contactor	Schütz Leistungs- 48VDC 3RT1034-1BW40 contactor power- 48VDC 3RT1034-1BW40	Siemens AG A&D 3RT1034-1BW40
53590-0	1	Leistungsschütz power contactor	Schütz Leistungs- 48VDC 3RT1017-1BW42 contactor power- 48VDC 3RT1017-1BW42	Siemens AG A&D 3RT1017-1BW42
53595-0	7	Hilfsschalter auxiliary switch	Schalter Hilfs- Aufb.2S/2Ö 3RH1921-1HA22 switch auxiliary 2NO/2NC 3RH1921-1HA22	Siemens AG A&D 3RH1921-1HA22
53596-0	1	Hilfsschalter auxiliary switch	Schalter Hilfs- Aufb.2S/2Ö 3RH1911-1HA22 switch auxiliary 2NO/2NC 3RH1911-1HA22	Siemens AG A&D 3RH1911-1HA22
53599-0	4	Schutzbeschaltung contact suppression	Schutzbeschalt.127-240VAC 3RT1916-1CD00 protective dev.127-240VAC 3RT1916-1CD00	
53602-0	2	Schutzbeschaltung contact suppression	Schutzbeschalt.240-400VAC 3RT1926-1CE00 protective dev.240-400VAC 3RT1926-1CE00	
55222-0	2	Kleinschütz mini contactor	Schütz kl. 09A 1S 230V 3RT1016-1AP01 contactor sm. 09A 1NO 230V 3RT1016-1AP01	Siemens AG A&D 3RT1016-1AP01

3b		26.01.2010	dnewiger	date	19.01.2005	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Stueckliste parts list Blattrelaisschrank relay box	order no.: 011-58152-V2-03b Y66 = 011Y66			
				prep.by	K.Langer					ass. ID: +			
				arr.by	D.Newiger					© Copyright ENERCON GmbH. document ID: D0132394 index: 3b page 1			
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by		All rights reserved. of 2			

Freigegeben	Mengen-Stückliste/ quantity parts list															
	Artikel Nr.- Index/ article no.-index	Menge/ quantity	Benennung deutsch/ title german Benennung englisch/ title english		SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english				Hersteller/manufacurer Hersteller Teile-Nr. / manufacturer part no.							
	55225-0	1	Hilfsschalter auxiliary switch		Schalter Hilfs- Aufb.1S/3Ö 3RH1911-1HA13 switch auxiliary 1N0/3NC 3RH1911-1HA13				Siemens AG A&D 3RH1911-1HA13							
	55236-0	1	Lastwiderstand load resistor		Widerstand Leistung 13R 200W 10% glas. resistor power 13R 200W 10% glaz.											
	55280-0	1	Sicherung fuse		Kabel Sicherungs- Relaissschrank E48/82 fuse conductor relay box E48/82											
	55788-0	1	Diode diode		Diode T40HFL100S05 40A diode T40HFL100S05 40A											
	56134-0	1	Drossel choke		Drossel Lade- 28mH EDR6800 Bv5629a choke charging- 28mH EDR6800 Bv5629a				Schneefuss GmbH & Co. KG Bv.5629a							
	57071-0	5	Sicherungsklemme fuse terminal		Klemme Sicherungs- UK6,3-HESI terminal fuse UK6,3-HESI				Phoenix Contact GmbH & Co. KG 3004171							
	6706-0	4	Schutzbeschaltung contact suppression		Schutzbeschaltung Freilaufdiode protective circuit invers diode											
	68943-0	5	Sicherungseinsatz fuse link		Sicherungseinsatz.500VAC 3,15A FF 7012540 fuse micro- 500VAC 3,15A FF type 7012540											
	798-0	1	Kontakteinsatz Buchse socket insert		Einsatz Buchse 25p 250VAC 10A Crimpan. socket insert 25p 250VAC 10A crimp term.				Harting Deutschl. GmbH&Co.KG 09210253101							
	885-0	15	Reihenklemme terminal		Klemme Reihen- 4mm ² 800V 41A gr UK5N terminal series 4mm ² 800V 41A gr UK5N				Phoenix Contact GmbH & Co. KG 3004362							
	886-0	3	Reihenklemme terminal		Klemme Reihen- 4mm ² 800V 41A bl UK5NBU terminal series 4mm ² 800V 41A bu UK5NBU				Phoenix Contact GmbH & Co. KG 3004388							
	959-0	4	Reihenklemme terminal		Klemme Reihen- 4mm ² gn/ge USLKG5 terminal series 4mm ² gr/ye USLKG5				Phoenix Contact GmbH & Co. KG 0441504							
3b		26.01.2010	dnewiger	date	19.01.2005	Schaltplan circuit diagram		CS 48a Windenergieanlage wind energy converter		Stueckliste parts list Blattrelaissschrank relay box	order no.: 011-58152-V2-03b		Y66		= 011Y66	
				prep.by	K.Langer						ass. ID:				+	
				arr.by	D.Newiger						© Copyright ENERCON GmbH. All rights reserved.		document ID:		D0132394	
no.	revision	date	name</													

Freigegeben	Kabelliste / cable list																
	Anlagenkennung/ installation ID Ortskennung/ location ID BMK/ device ID			Artikel Nr. - Index/ article no. - index	Länge gesamt / total length belegte Adern / wires used	SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english			Funktionstext deutsch - englisch/ function remarks german - english								
	011	A	W01	600-0	1.85 31	Leitung Steuer- Y-JZ 42x 1,0mm ² grau cable control- Y-JZ 42x1,0mm ² grey			Steuerleitungen control cables								
	011	A	W03	13612-0	1,55m 4	Leitung Steuer- Y-JZ 4x 2,5mm ² grau cable control- Y-JZ 4x2,5mm ² grey			Kondensatorschrank Rotor Blatt A capacitor box rotor blade A								
	011	B	W01	600-0	1,85m 31	Leitung Steuer- Y-JZ 42x 1,0mm ² grau cable control- Y-JZ 42x1,0mm ² grey			Steuerleitung control line								
	011	B	W03	13612-0	1,55m 4	Leitung Steuer- Y-JZ 4x 2,5mm ² grau cable control- Y-JZ 4x2,5mm ² grey			Kondensatorschrank Rotor Blatt B capacitor box rotor blade B								
	011	C	W01	600-0	1,85m 31	Leitung Steuer- Y-JZ 42x 1,0mm ² grau cable control- Y-JZ 42x1,0mm ² grey			Steuerleitungen control cables								
	011	C	W03	13612-0	1,55m 4	Leitung Steuer- Y-JZ 4x 2,5mm ² grau cable control- Y-JZ 4x2,5mm ² grey			Kondensatorschrank Rotor Blatt C capacitor box rotor blade C								
	028	A	W01	608-0	11	Leitung Steuer- Y-JZ 14x 1,0mm ² grau cable control- Y-JZ 14x1,0mm ² grey											
	53	A	W01	4532-0	14	Leitung Steuer- Y-JZ 14x 1,5mm ² grau cable control- Y-JZ 14x1,5mm ² grey											

Blattregelschrank pitch box

SAP article no.:
54415

index:
7

Reifegrad / level of maturity:
Serie
series

Zusatzinfo / additional info:

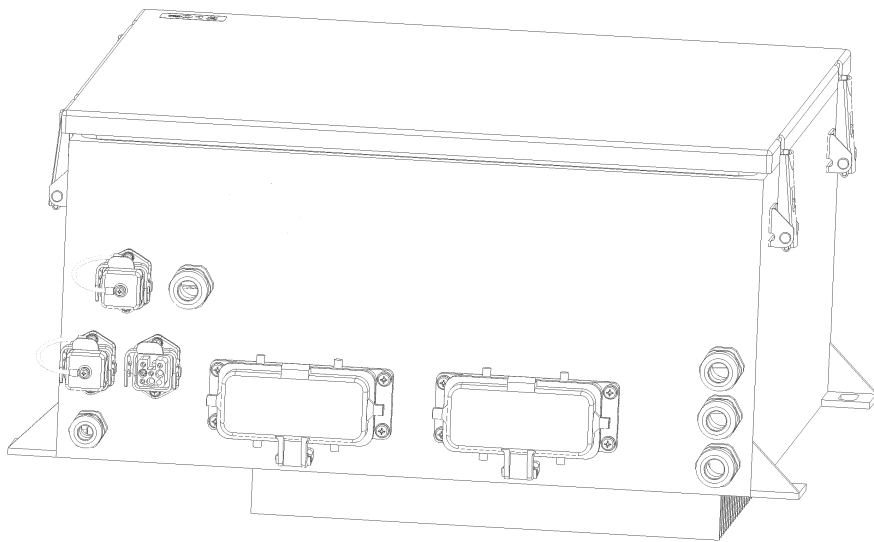
Anlagentyp / converter type:
E-44; E-48; E-53

Steuerung / control system:
CS48a

Baugruppennr. / assembly no.:
012

Version Baugr. / version of assembly:
1

Baugr. Index / component index:
11

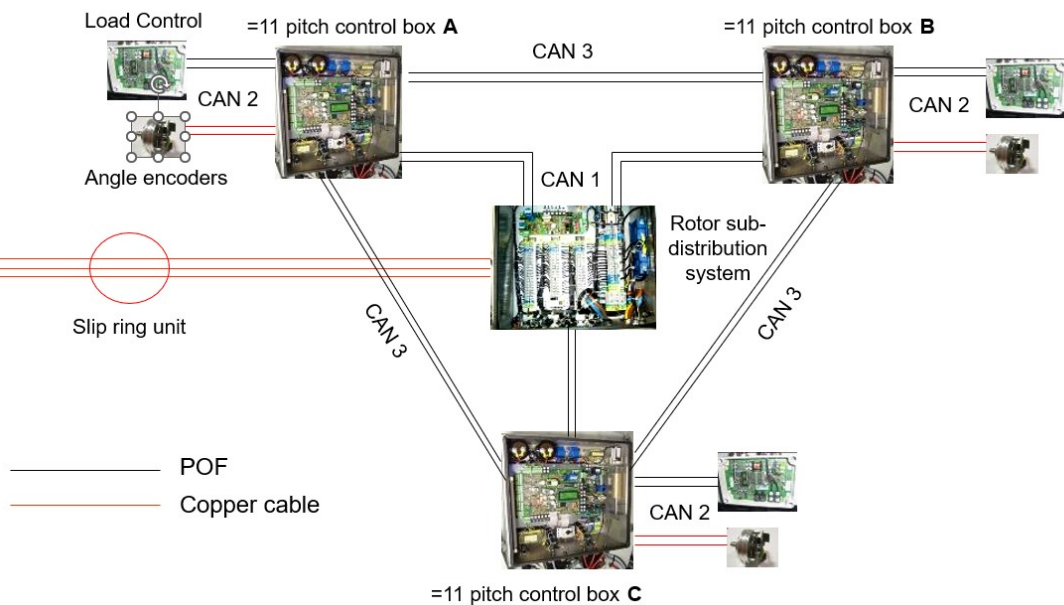


Baugruppens Schaltplan assembly circuit diagram

document ID:
D0125961

index:
7

title or assembly ID:
012 54415 V1-07



Alle nicht gekennzeichneten Leitungen 1mm² schwarz.
All not marked lines 1mm² black.

Alle nicht gekennzeichneten Neutralleiter 1mm² hellblau.
All not marked neutral conductor 1mm² light blue.

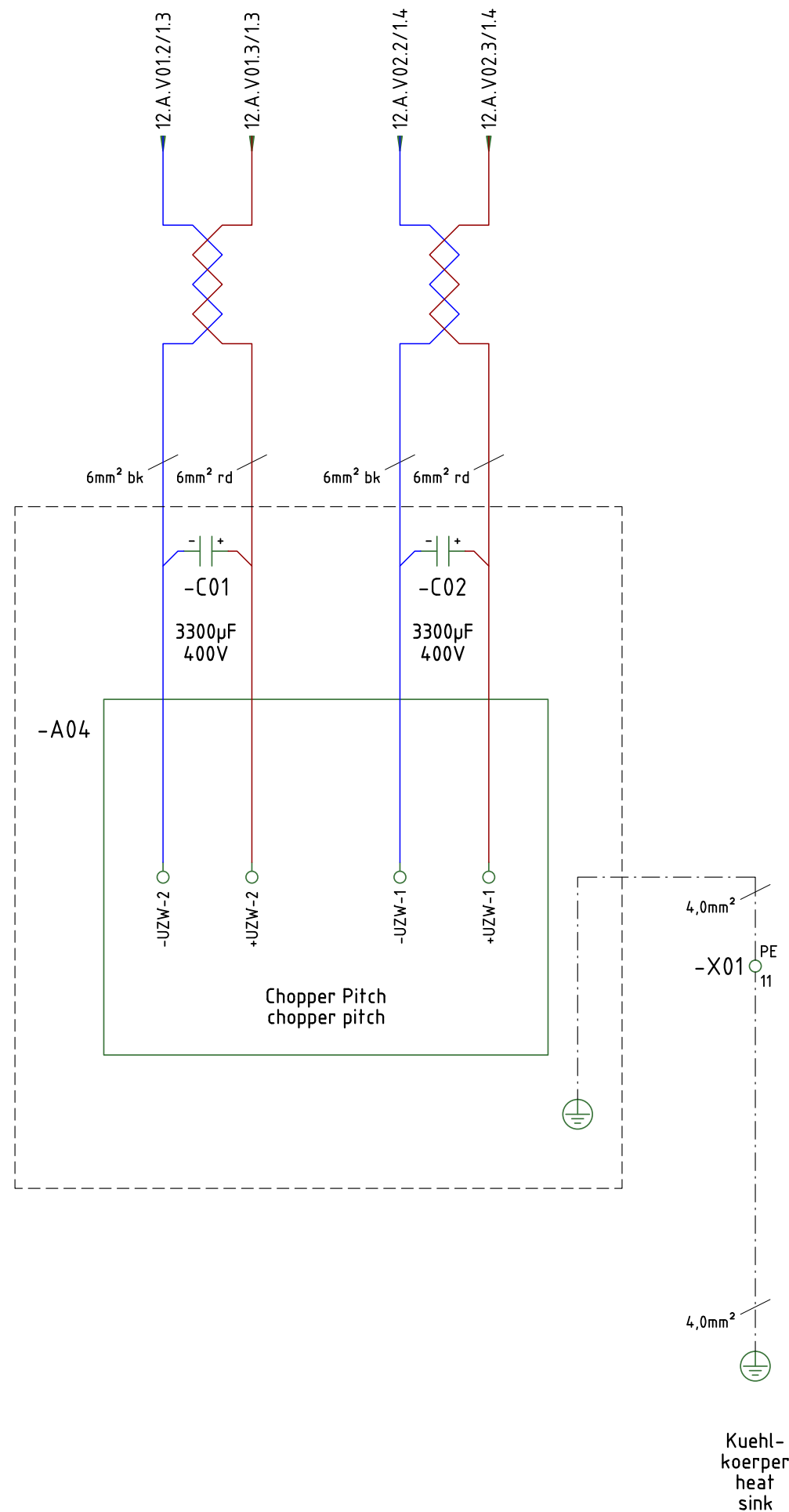
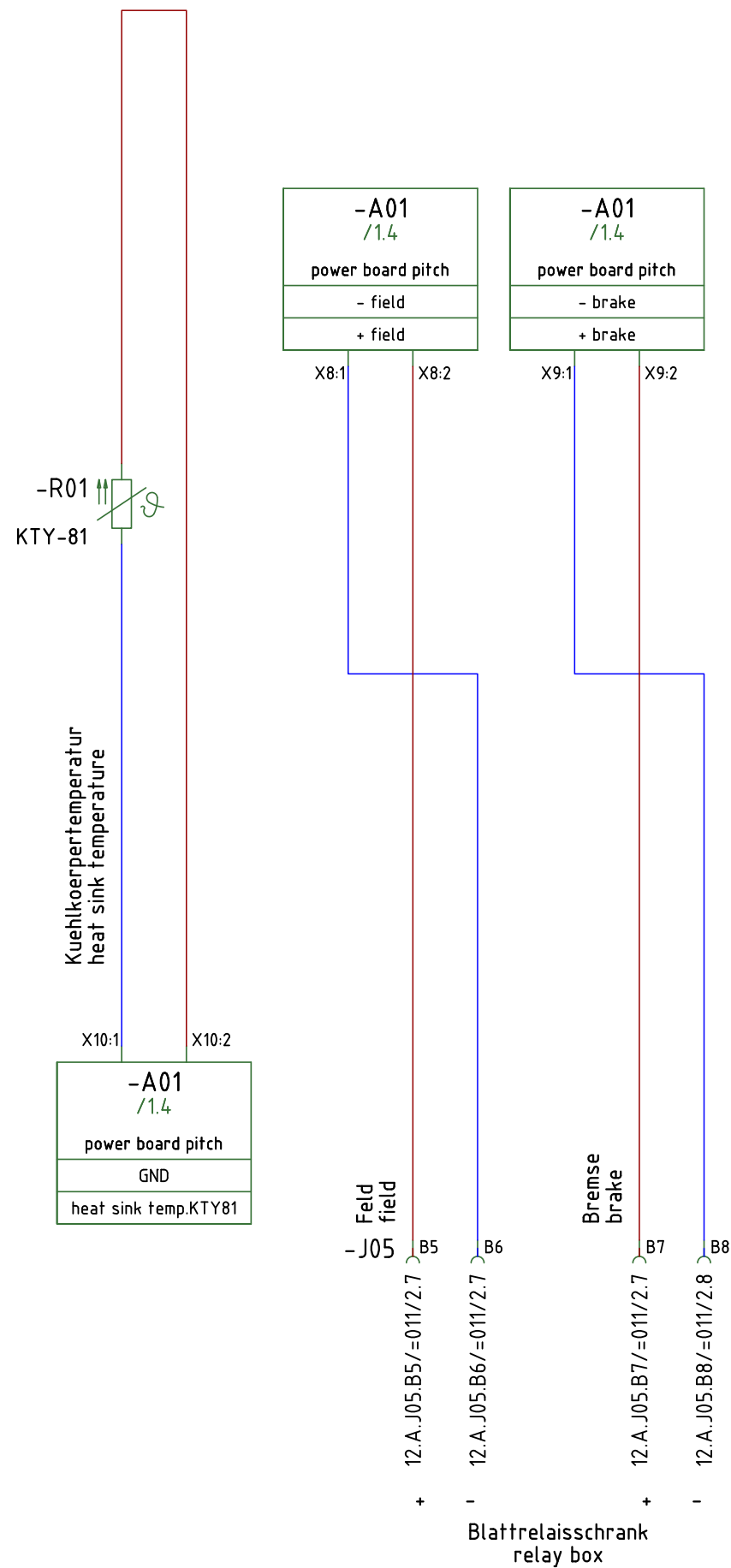
Technische Änderungen vorbehalten!
Technical subject to alteration!

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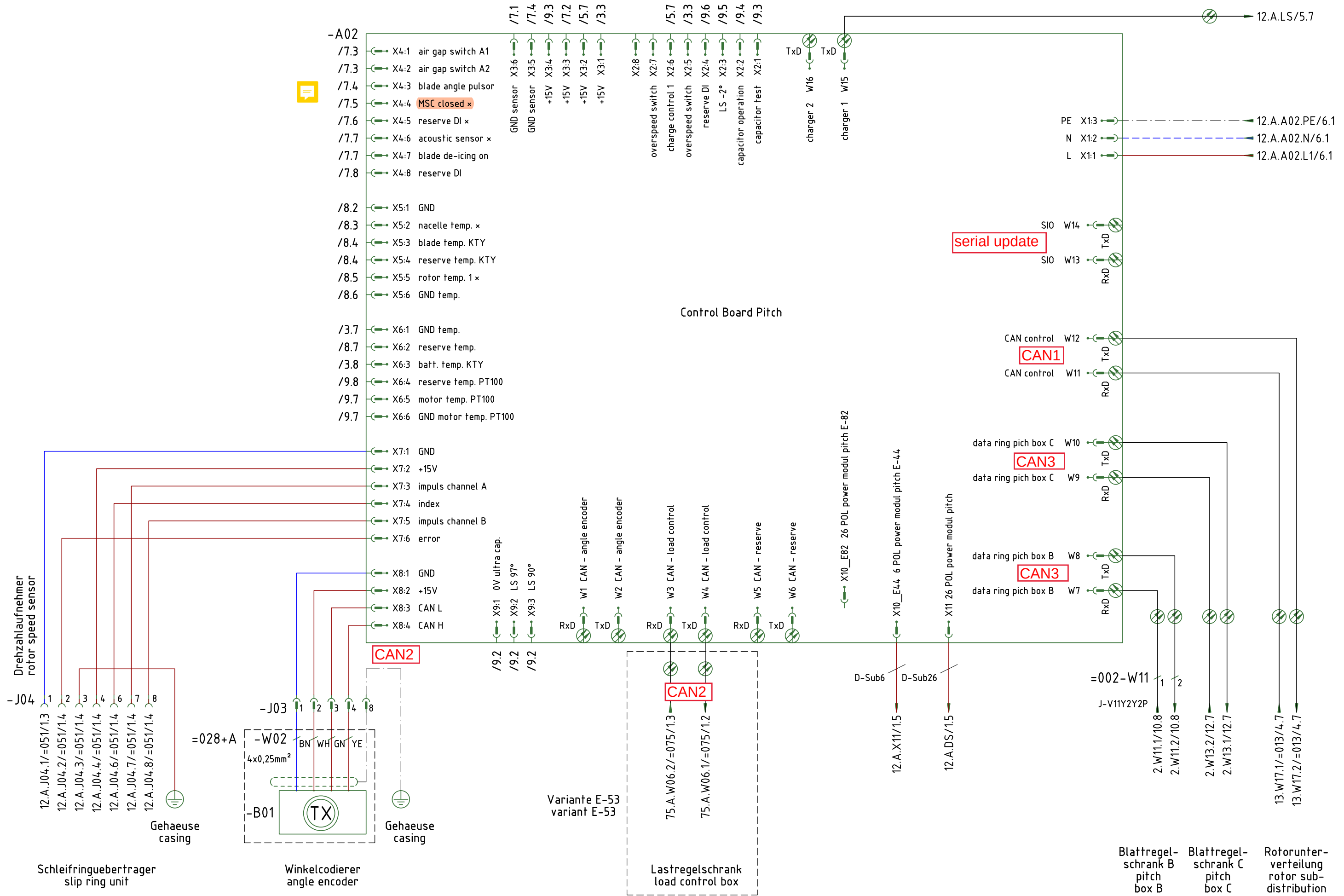
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7	EC00002102	03.05.2013	2467	date	15.01.14	03.05.2013	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Deckblatt cover	order no.: 012-54415-V1-07		Y04	= 012COVER	
6a	BGR Index	15.05.2009	ehmen	prep.by	K.Langer						ass. ID:				
6	C000493	30.03.2009	ehmen	arr.by	G.Ehmen						Blattregelschrank pitch box		© Copyright ENERCON GmbH. All rights reserved.		document ID: D0125961
no.	revision	date	name	appr.	P00028557 KG		origin: germany	repl. for	repl. by						

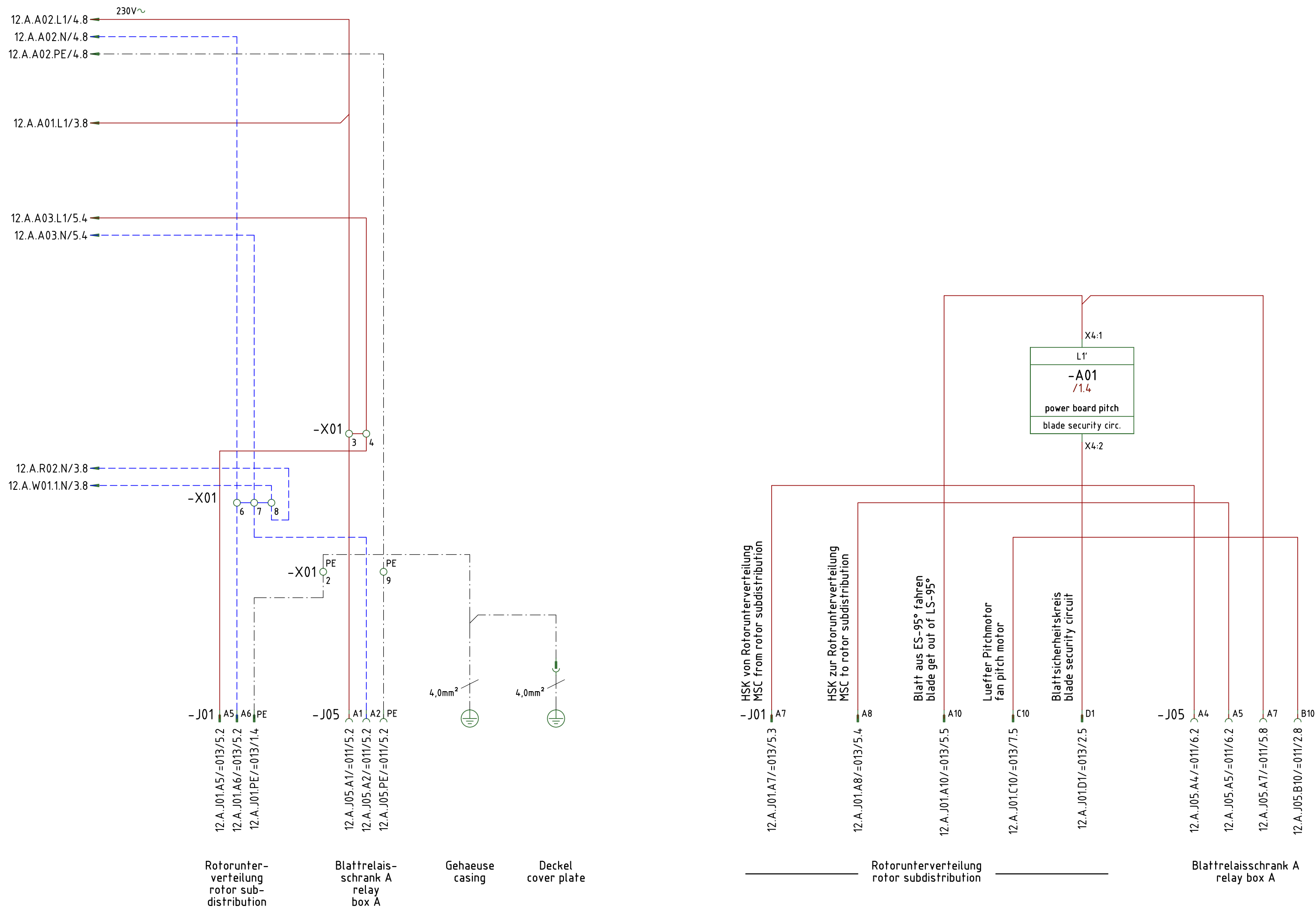


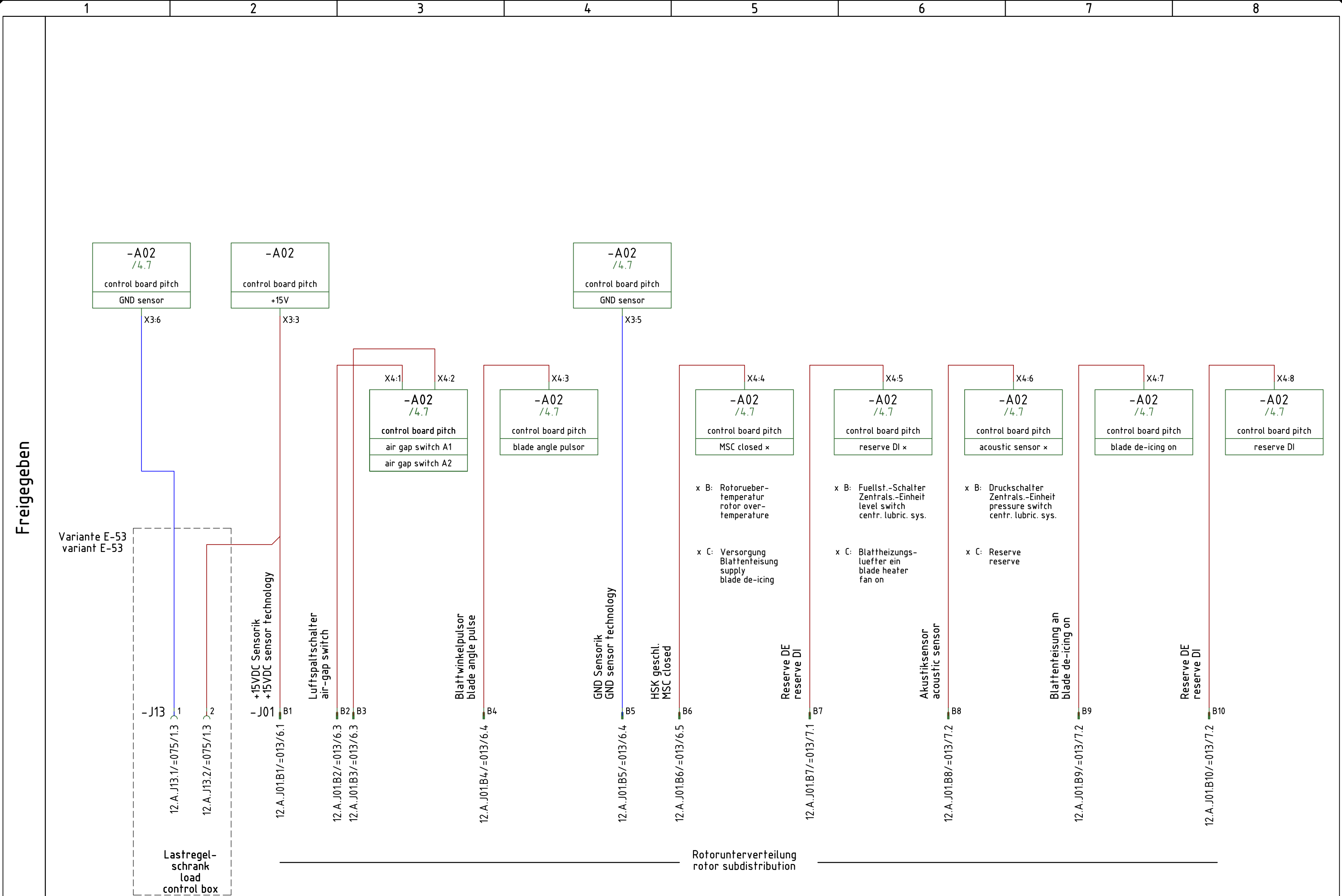


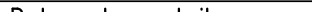


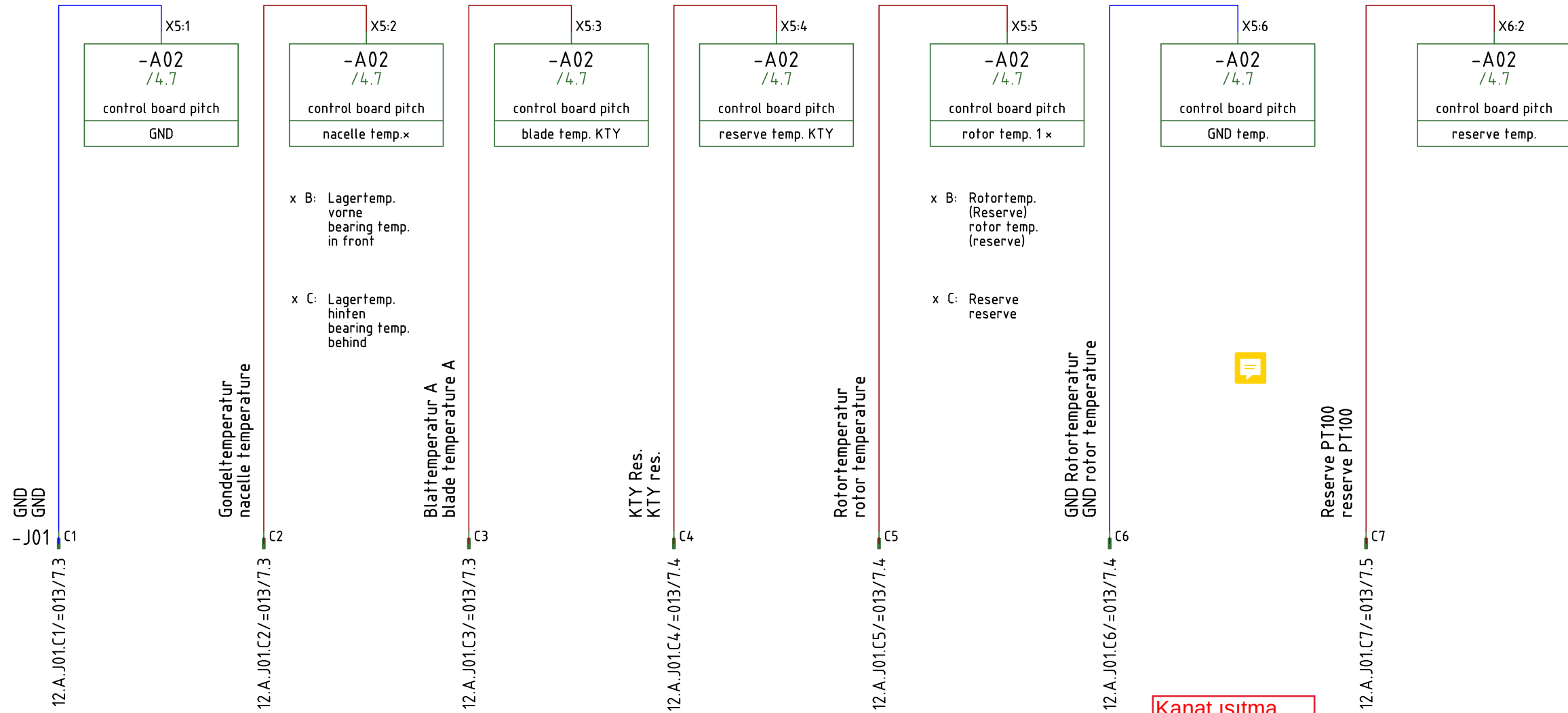




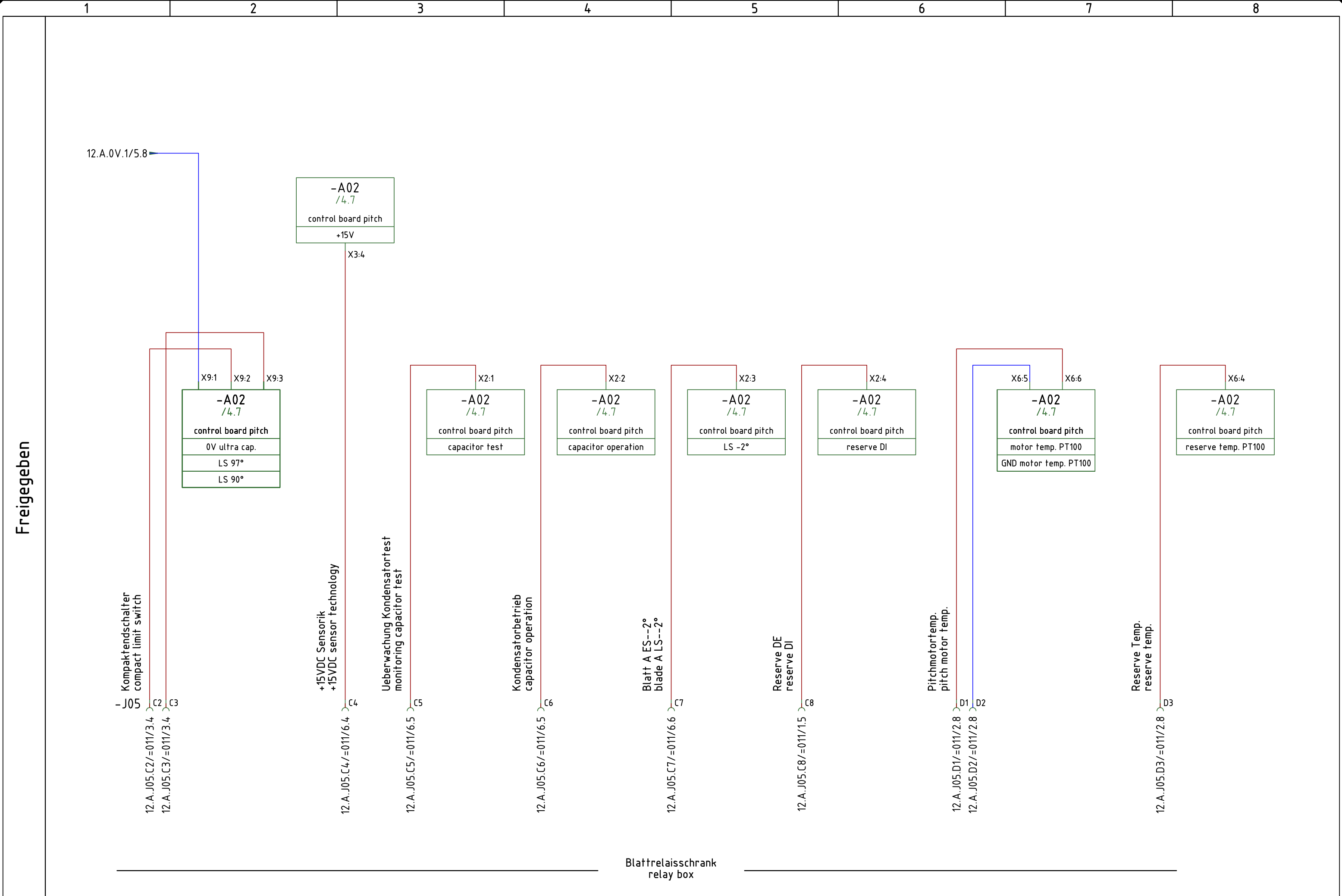




6	J01.B8	30.03.2009	ehmen	date	15.01.14	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Verbindung Rotorunterverteilung connection rotor subdistribution	order no.: ass. ID: 012-54415-V1-07 Y32 = 012 + A			
4	X3:3;6	30.03.06	KL	prep.by	K.Langer								
				arr.by	G.Ehmen								
no.	revision	date	name	appr.	J.Rohlfesen KG	origin: germany	repl. for	repl. by	Blattregelschrank pitch box	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0125961	index: 7	page 7 of 13

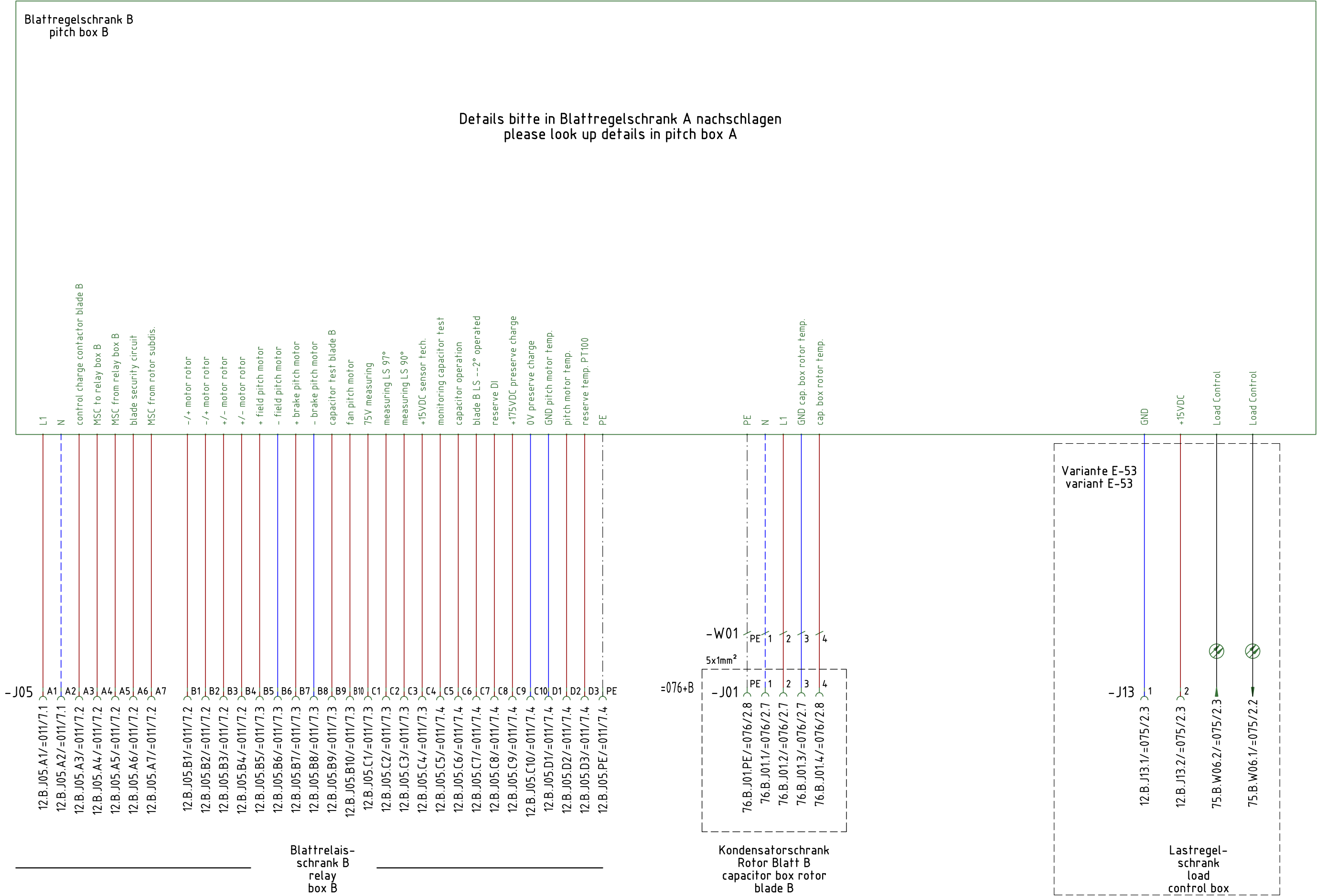


Rotorunterverteilung
rotor subdistribution



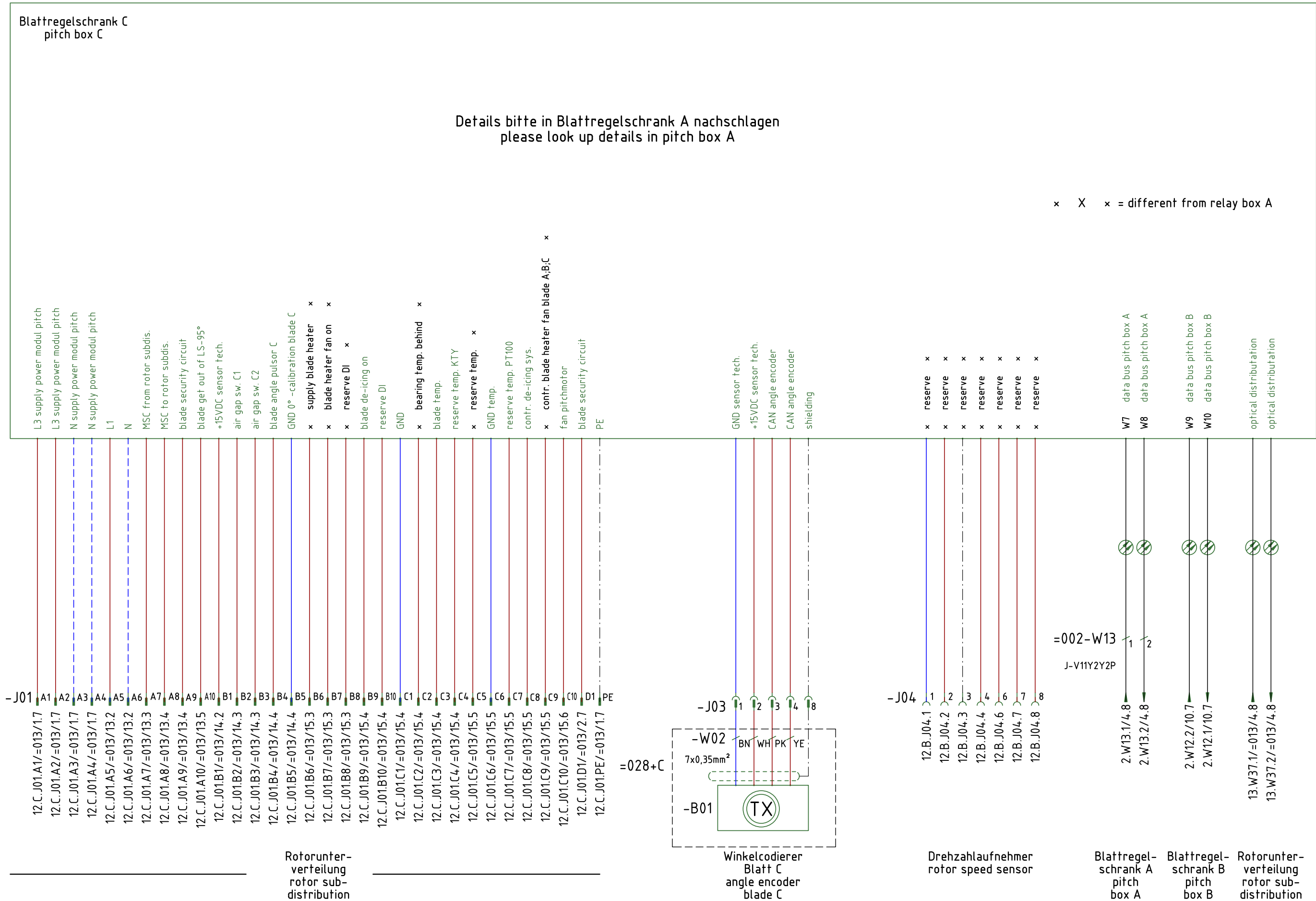


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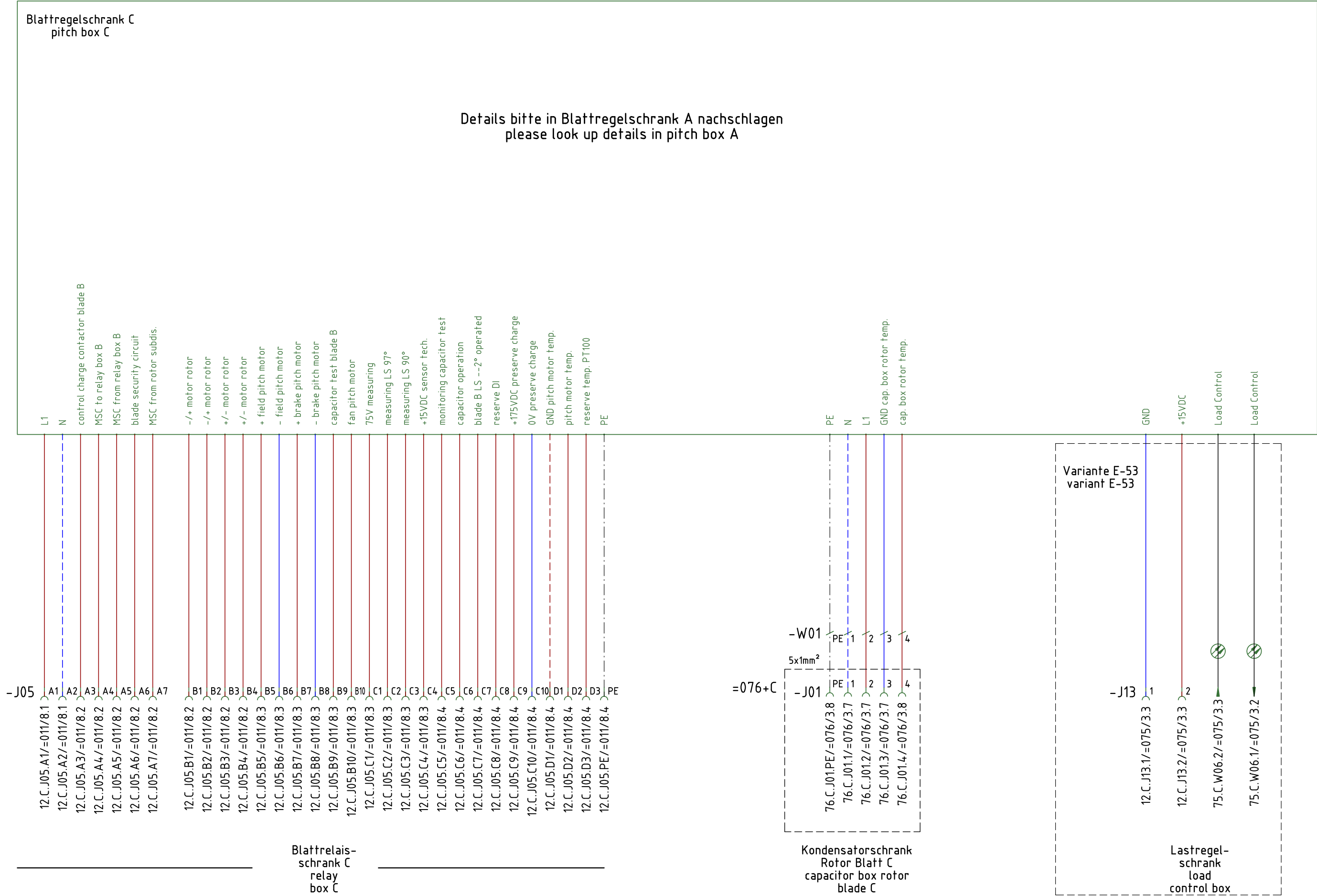
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				prep.by	K.Langer								© Copyright ENERCON GmbH. All rights reserved.	document ID: D0125961	index: 7	page 11
				arr.by	G.Ehmen											of 13
no.	revision	date	name	appr.	J.Rohlfesen KG	origin: germany	repl. for	repl. by	Blattregelschrank pitch box							

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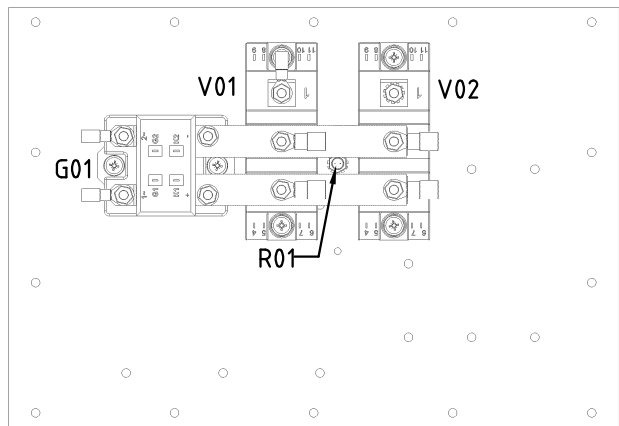
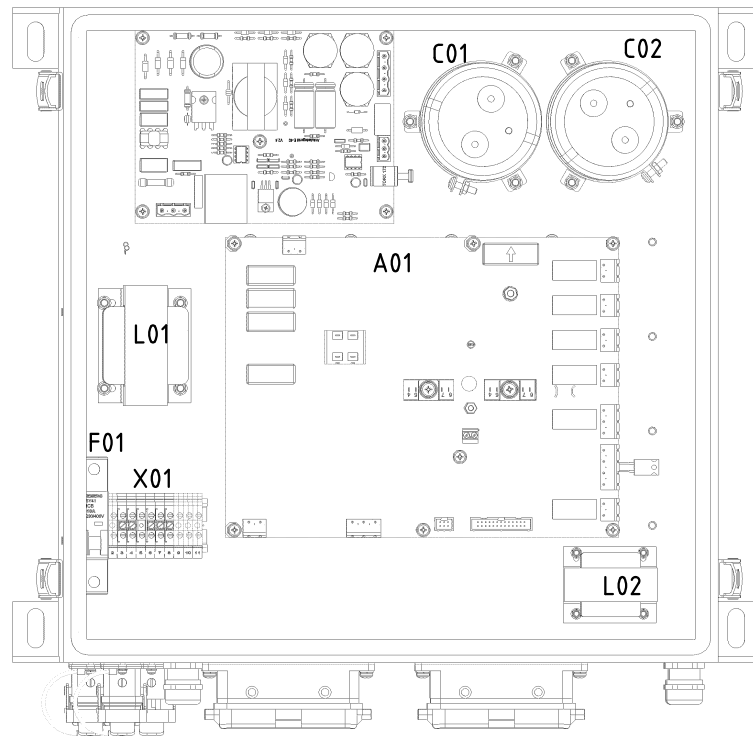
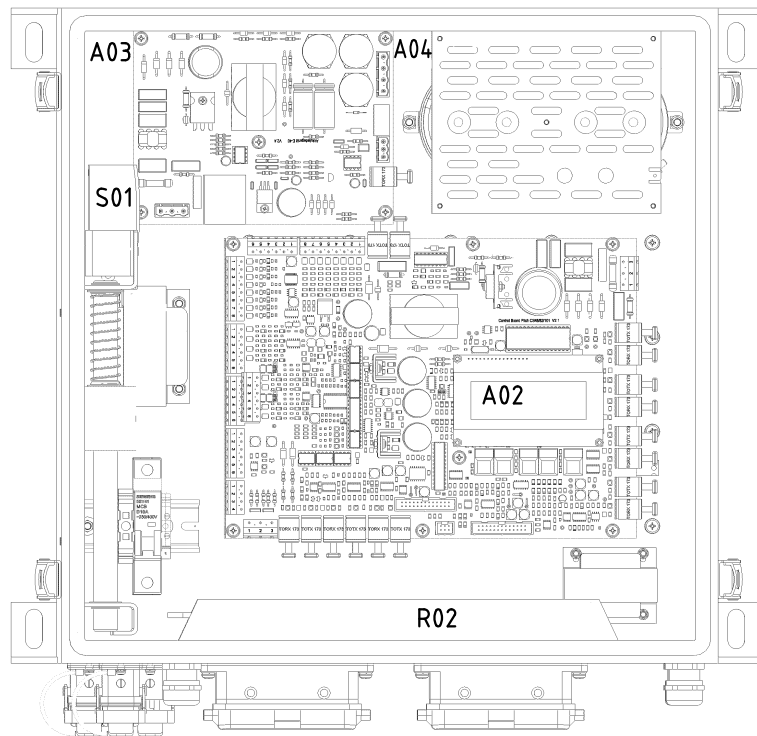
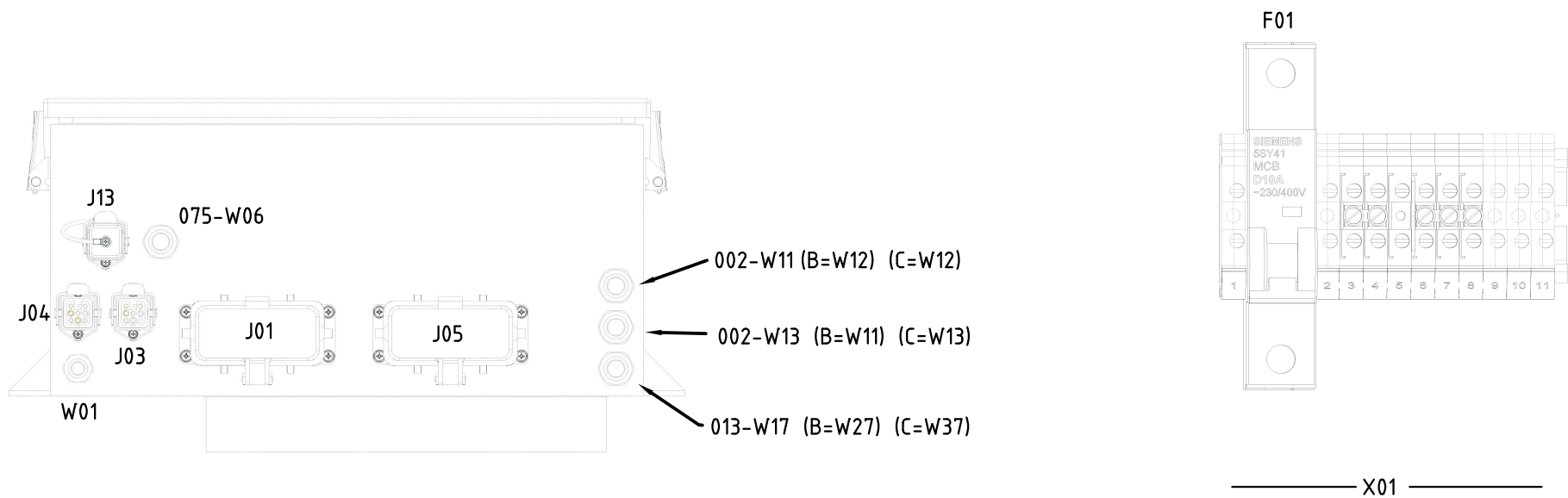


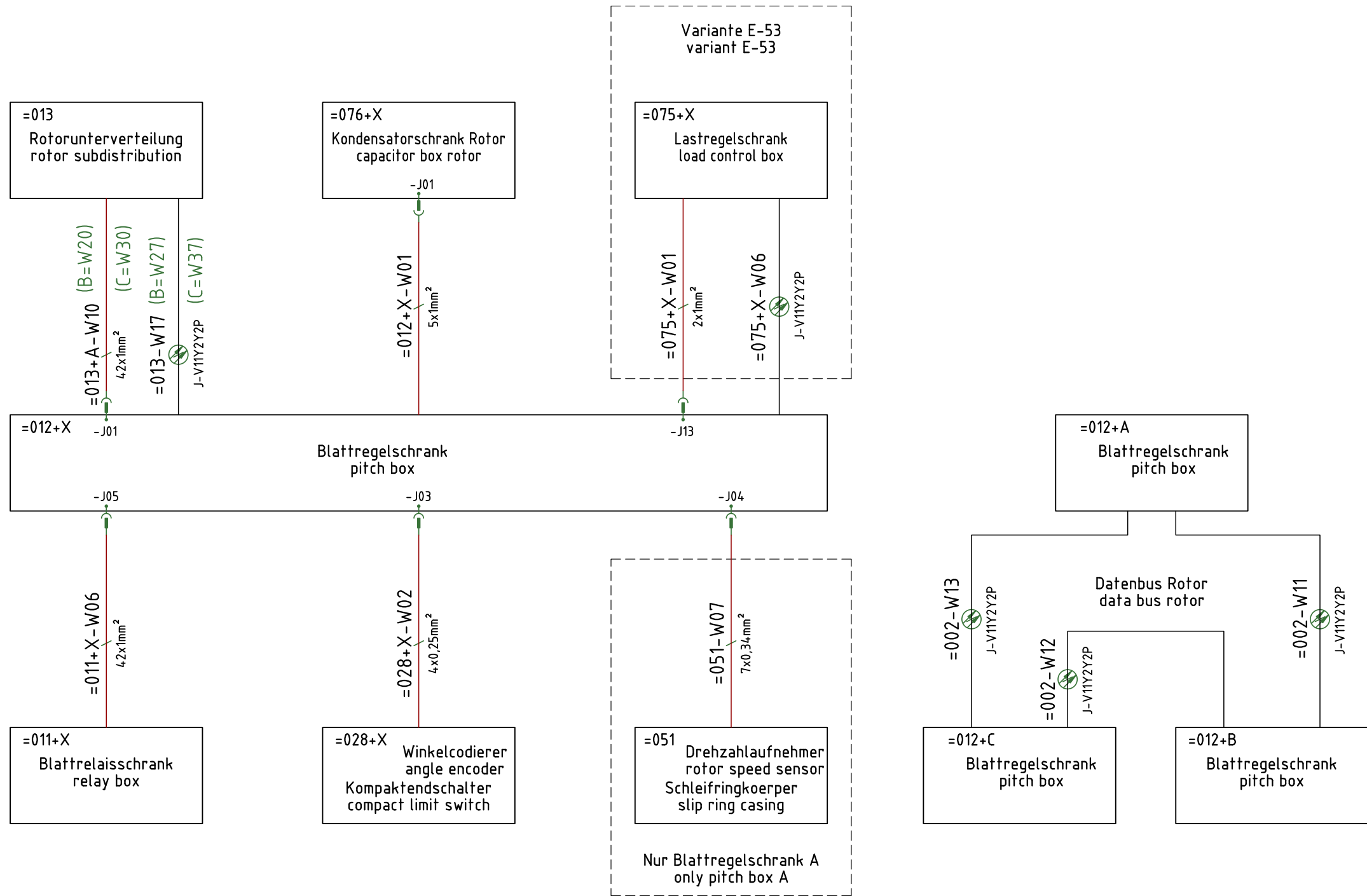
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				prep.by	K.Langer												
				arr.by	G.Ehmen												
no.	revision	date	name	appr.	J.Rohlfesen	KG	origin: germany	repl. for	repl. by	Blattregelschrank pitch box							

Freigegeben



				date	15.01.14	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Blattregelschrank C pitch box C Blattregelschrank pitch box	order no.: ass. ID: 012-54415-V1-07 © Copyright ENERCON GmbH. All rights reserved.	Y32 document ID: D0125961 index: 7	= 012 + C	page 13 of 13
				prep.by	K.Langer								
				arr.by	G.Ehmen								
no.	revision	date	name	appr.	J.Rohlfesen	origin: germany	repl. for	repl. by					





X = A, B, C

Freigegeben

Stückliste/ bill of material														
Anlagenkennung/ installation ID Ortskennung/ location ID BMK/ device ID			Artikel Nr. – Index/ article no. – index	SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english				Funktionstext deutsch/ function remarks german Funktionstext englisch/ function remarks english				Seite. Pfad page. column		
012	A	A01	55812–0	Platine Leistungst.Pitch V2.2 E40/44 PCB power board pitch V2.2 E40/44				Platine Power Board Pitch PCB power board pitch				1.4		
012	A	A02	71576–3	Platine Controlboard Pitch V1.6 CS48/82 PCB controlboard pitch V1.6 CS48/82				Steuerkarte Pitch control board pitch				4.7		
012	A	A02	507710–1	Platine Contr. Bo.Pitch CS48/82/101 V2.0 PCB Control Board Pitch CS48/82/101 V2.0				Steuerkarte Pitch control board pitch				4.7		
012	A	A02	511910–1	Platine Controlboard Pitch V2.1 PCB controlboard pitch V2.1				Steuerkarte Pitch control board pitch				4.7		
012	A	A03	78442–1	Platine Akkuladegerät V2.4 E40 PCB battery charger V2.4 E40				Akkuladegeraet battery charger				5.3		
012	A	A04	11835–0	Platine Chopper–Pitch V1.2a E66 PCB chopper–pitch V1.2a E66				Chopper Pitch chopper pitch				2.6		
012	A	C01	75955–0	Kondensator 3300µF/450V B43564–S5338–M6 capacitor 3300µF/450V B43564–S5338–M6				Chopper Pitch chopper pitch				2.5		
012	A	C02	75955–0	Kondensator 3300µF/450V B43564–S5338–M6 capacitor 3300µF/450V B43564–S5338–M6				Chopper Pitch chopper pitch				2.6		
012	A	F01	28854–0	Sicherungsautomat 1p.D10A 400VAC SIE® autom.circuit break.1p. D10A 400VAC SIE®				Versorgung Leistungsteil Pitch supply power module pitch				1.1		
012	A	G01	1717–0	Gleichrichter Brücken– 40A SKCH40/12 rectifier, bridge 40A 1200V SKCH40/12				Versorgung Leistungsteil Pitch supply power module pitch				1.3		
012	A	L01	55229–0	Drossel Einphasen– 2x0,6mH EDR92 Bv.5546 choke one–phase 2x0,6mH EDR92 Bv5546				Eingangsdrossel input choke				1.1		
012	A	L02	55230–0	Drossel Einphasen– 2x0,12mH EDR17 Bv5547 choke one–phase 2x0,12mH EDR17 Bv5547				Ausgangsdrossel output choke				1.8		
012	A	R01	4424–1	Sensor BLRG4012 KTY81–110 EXX sensor BLRG4012 KTY81–110 EXX				Kuehlkoerpertemperatur heat sink temperature				2.2		
012	A	R02	507847–0	Widerstand Leist.820R 155W GMAD 300x20 resistor power 820R 155W GMAD 300x20				Heizung heater				3.4		
012	A	R04	551603–0	Widerstand MP925 68,0K 30W 1% T0220 resistor MP925 68,0K 30W 1% T0220				Symmetrierwiderstand balancing resistor				1.7		

				date	15.01.14	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Stueckliste parts list Blattregelschrank pitch box	order no.: 012–54415–V1–07		Y64		= 012Y64	
				prep.by	K.Langer					ass. ID:				+	
				arr.by	G.Ehmen					© Copyright ENERCON GmbH. All rights reserved.		document ID: D0125961		index: 7	
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by						page of	1 2

Freigegeben	Stückliste/ bill of material														
	Anlagenkennung/ installation ID Ortskennung/ location ID BMK/ device ID			Artikel Nr. - Index/ article no. - index	SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english				Funktionstext deutsch/ function remarks german Funktionstext englisch/ function remarks english				Seite. Pfad page. column		
	012	A	S01	52130-0	Grenztaster-Norm LS-11S Kuppenstößel limit switch LS-11S plunger				Ueberdrehzahlschalter A overspeed switch A				3.2		
	012	A	S01	12057-1	Überdrehzahlschalter m.Clips-L. BGr E40 overspeed switch w.clips-bear. ass. E40				Ueberdrehzahlschalter A overspeed switch A				3.2		
	012	A	V01	2981-0	IGBT-Modul 1200V 35A BSM35GB120DN2 IGBT-module 1200V 35A BSM35GB120DN2				+/- Motor-Anker +/- motor rotor				1.4		
	012	A	V02	2981-0	IGBT-Modul 1200V 35A BSM35GB120DN2 IGBT-module 1200V 35A BSM35GB120DN2				-/+ Motor-Anker -/+ motor rotor				1.5		
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Mengen-Stückliste/ quantity bill of material														
Artikel Nr.- Index/ article no.-index		Menge/ quantity	Benennung deutsch/ title german Benennung englisch/ title english		SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english				Hersteller/manufacturer Hersteller Teile-Nr. / manufacturer part no.					
11835-0		1	Platine PCB		Platine Chopper-Pitch V1.2a E66 PCB chopper-pitch V1.2a E66				ENERCON GmbH 11835					
12057-1		1	Überdrehzahlschalter overspeed switch		Überdrehzahlschalter m.Clips-L. BGr E40 overspeed switch w.clips-bear. ass. E40									
1701-0		4	Anbaugehäuse mounting case		Gehäuse Anbau- 40p 113x 43 SWF 1 Kappe housing,ext.mount.40p 113x43 SWR 1 cover				Harting GmbH 09370160302					
1717-0		1	Brückengleichrichter bridge rectifier		Gleichrichter Brücken- 40A SKCH40/12 rectifier, bridge 40A 1200V SKCH40/12									
28854-0		1	Leitungsschutzschalter line safety switch		Sicherungsautomat 1p.D10A 400VAC SIEⓈ autom.circuit break.1p. D10A 400VAC SIEⓈ				Siemens AG A&D 5SY4110-8					
2981-0		2	IGBT IGBT		IGBT-Modul 1200V 35A BSM35GB120DN2 IGBT-module 1200V 35A BSM35GB120DN2				Eupec GmbH 15277					
3442-0		5	Anbaugehäuse mounting case		Gehäuse Anbau- 4p 28x 40 SWF Gerade housing ext.mount. 4p 28x40 SWR str.				Harting GmbH 09370030301					
4424-1		1	Temperatursensor temperature sensor		Sensor BLRG4012 KTY81-110 EXX sensor BLRG4012 KTY81-110 EXX				ENERCON GmbH 4424					
507710-1		1	Platine PCB		Platine Contr. Bo.Pitch CS48/82/101 V2.0 PCB Control Board Pitch CS48/82/101 V2.0				ENERCON GmbH 507710					
507847-0		1	Lastwiderstand load resistor		Widerstand Leist.820R 155W GMAD 300x20 resistor power 820R 155W GMAD 300x20				Frizlen GmbH u. Co.KG GMAD 300x20-820					
511910-1		1	Platine PCB		Platine Controlboard Pitch V2.1 PCB controlboard pitch V2.1				ENERCON GmbH 511910					
52130-0		1	Positionsschalter position switch		Grenztaster-Norm LS-11S Kuppenstößel limit switch LS-11S plunger				Moeller Electric GmbH 266105					
551603-0		1	Widerstand resistor		Widerstand MP925 68,0K 30W 1% T0220 resistor MP925 68,0K 30W 1% T0220				Diverse Markenhersteller MP925-68.0k-1%					
55229-0		1	Drossel choke		Drossel Einphasen- 2x0,6mH EDR92 Bv.5546 choke one-phase 2x0,6mH EDR92 Bv5546				Schneefuss GmbH & Co. KG Bv.5546					
55230-0		1	Drossel choke		Drossel Einphasen- 2x0,12mH EDR17 Bv5547 choke one-phase 2x0,12mH EDR17 Bv5547				Schneefuss GmbH & Co. KG Bv.5547					

				date	15.01.14	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Mengen-Stückliste quantity parts list Blattregelschrank pitch box	order no.: 012-54415-V1-07		Y66	= 012Y66		
				prep.by	K.Langer					ass. ID:			+		
				arr.by	G.Ehmen							© Copyright ENERCON GmbH. All rights reserved.		document ID:	D0125961
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by							

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Mengen-Stückliste/ quantity bill of material				
Artikel Nr.- Index/ article no.-index	Menge/ quantity	Benennung deutsch/ title german Benennung englisch/ title english	SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english	Hersteller/manufacturer Hersteller Teile-Nr. / manufacturer part no.
55812-0	1	Platine PCB	Platine Leistungst.Pitch V2.2 E40/44 PCB power board pitch V2.2 E40/44	ENERCON GmbH 55812
71576-3	1	Platine PCB	Platine Controlboard Pitch V1.6 CS48/82 PCB controlboard pitch V1.6 CS48/82	ENERCON GmbH 71576
75955-0	2	Kondensator capacitor	Kondensator 3300µF/450V B43564-S5338-M6 capacitor 3300µF/450V B43564-S5338-M6	EPCOS-TDK B43564-C5338-M6
78442-1	1	Platine PCB	Platine Akkuladegerät V2.4 E40 PCB battery charger V2.4 E40	ENERCON GmbH 78442
793-0	3	Kontakteinsatz Stift pin insert	Einsatz Stift 40p 250VAC 10A Crimpan. pin insert 40p 250VAC 10A crimp term.	Harting GmbH 09210403001
799-0	1	Kontakteinsatz Buchse socket insert	Einsatz Buchse 40p 250VAC 10A Crimpan. socket insert 40p 250VAC 10A crimp term.	Harting GmbH 09210403101
802-0	5	Kontakteinsatz Buchse socket insert	Einsatz Buchse 8p 42VAC 10A Crimpan. socket insert 8p 42VAC 10A crimp term.	Harting GmbH 09360083101
885-0	3	Reihenklemme terminal	Klemme Reihen- 4mm ² 800V 41A gr UK5N terminal block 4mm ² 800V 41A gr UK5N	Phoenix Contact GmbH & Co. KG 3004362
886-0	3	Reihenklemme terminal	Klemme Reihen- 4mm ² 800V 41A bl UK5NBU terminal block 4mm ² 800V 41A bu UK5NBU	Phoenix Contact GmbH & Co. KG 3004388
959-0	5	Reihenklemme terminal	Klemme Reihen- 4mm ² gn/ge USLKG5 terminal block 4mm ² gr/ye USLKG5	Phoenix Contact GmbH & Co. KG 0441504

				date	15.01.14	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Mengen-Stückliste quantity parts list Blattregelschrank pitch box	order no.:	012-54415-V1-07	Y66	= 012Y66	
				prep.by	K.Langer					ass. ID:			+	
				arr.by	G.Ehmen									
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by		© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0125961	index: 7	page 2 of 2

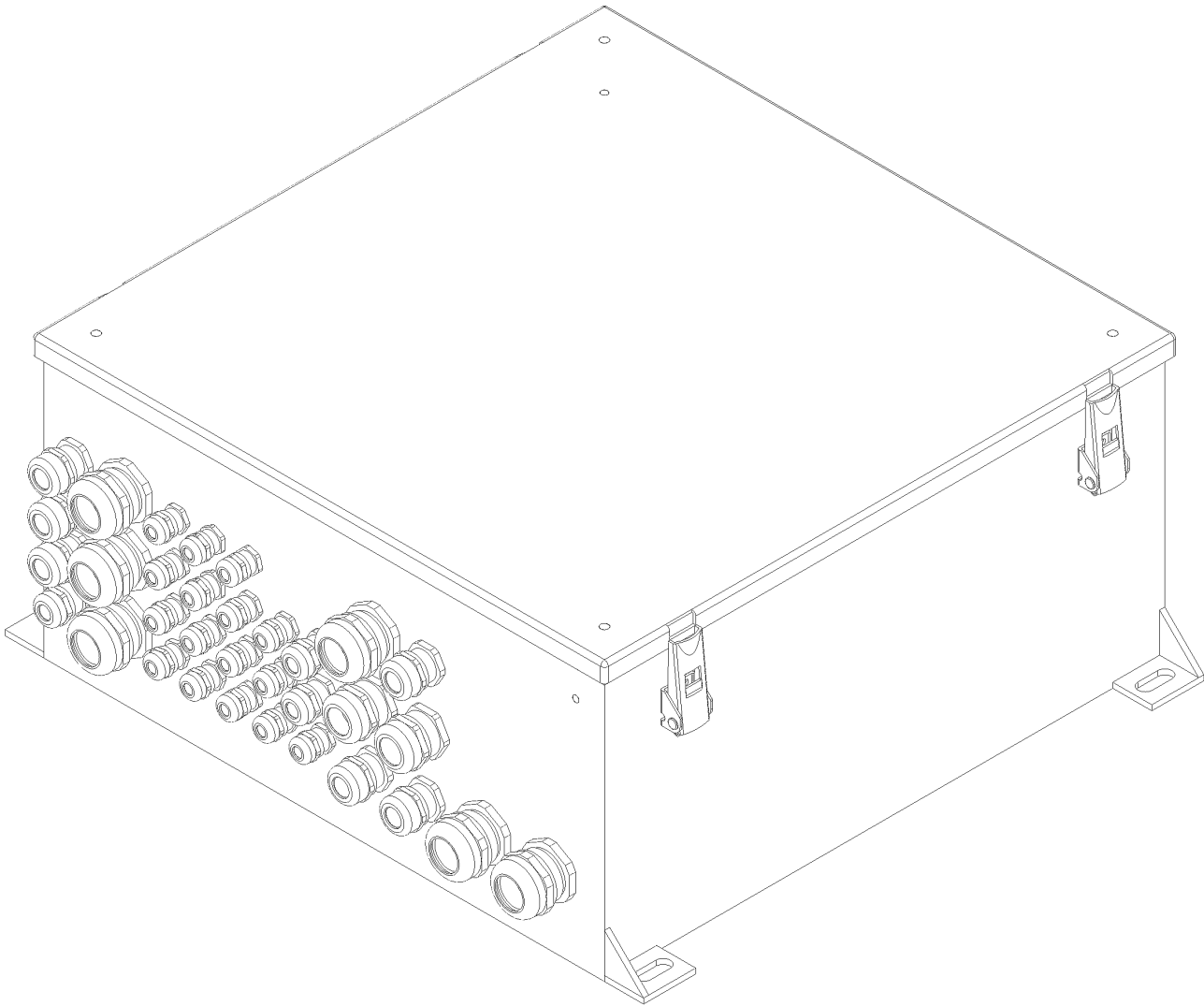
Rotorunterverteilung
rotor subdistribution

SAP article no.: index:
53734 13

Reifegrad / level of maturity:
Serie
series

Zusatzinfo / additional info:

Anlagentyp / converter type:
E-44; E-48
Steuerung / control system:
CS 48a
Baugruppennr. / assembly no.:
013
Version Baugr. / version of assembly:
1
Baugr. Index / component index:
14



Baugruppens Schaltplan
assembly circuit diagram

document ID: index:
D0116345 13

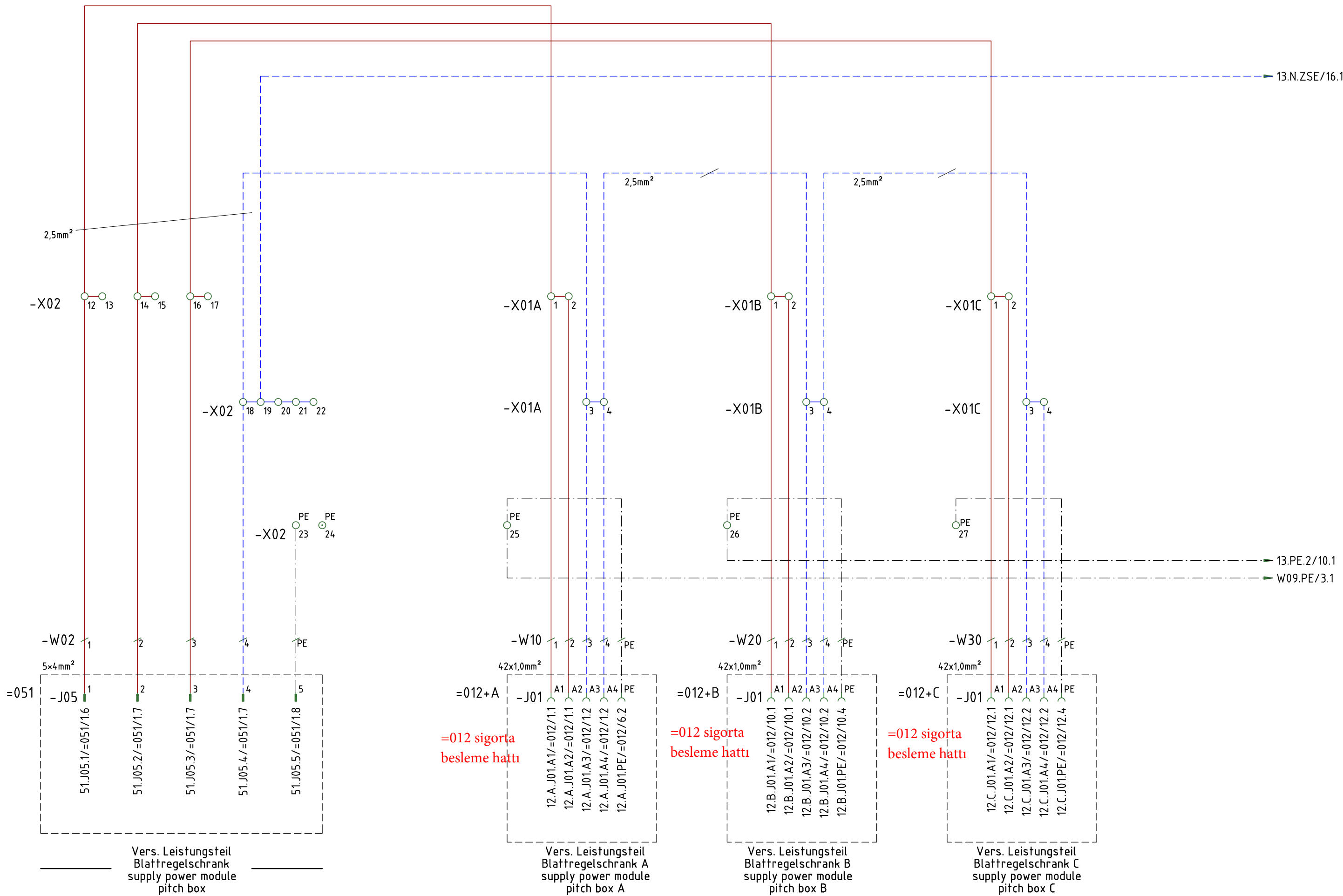
title or assembly ID:
013-53734-V1-13

Alle nicht gekennzeichneten Leitungen 1mm² schwarz.
All not marked lines 1mm² black.
Alle nicht gekennzeichneten Neutralleiter 1mm² hellblau.
All not marked neutral conductor 1mm² light blue.

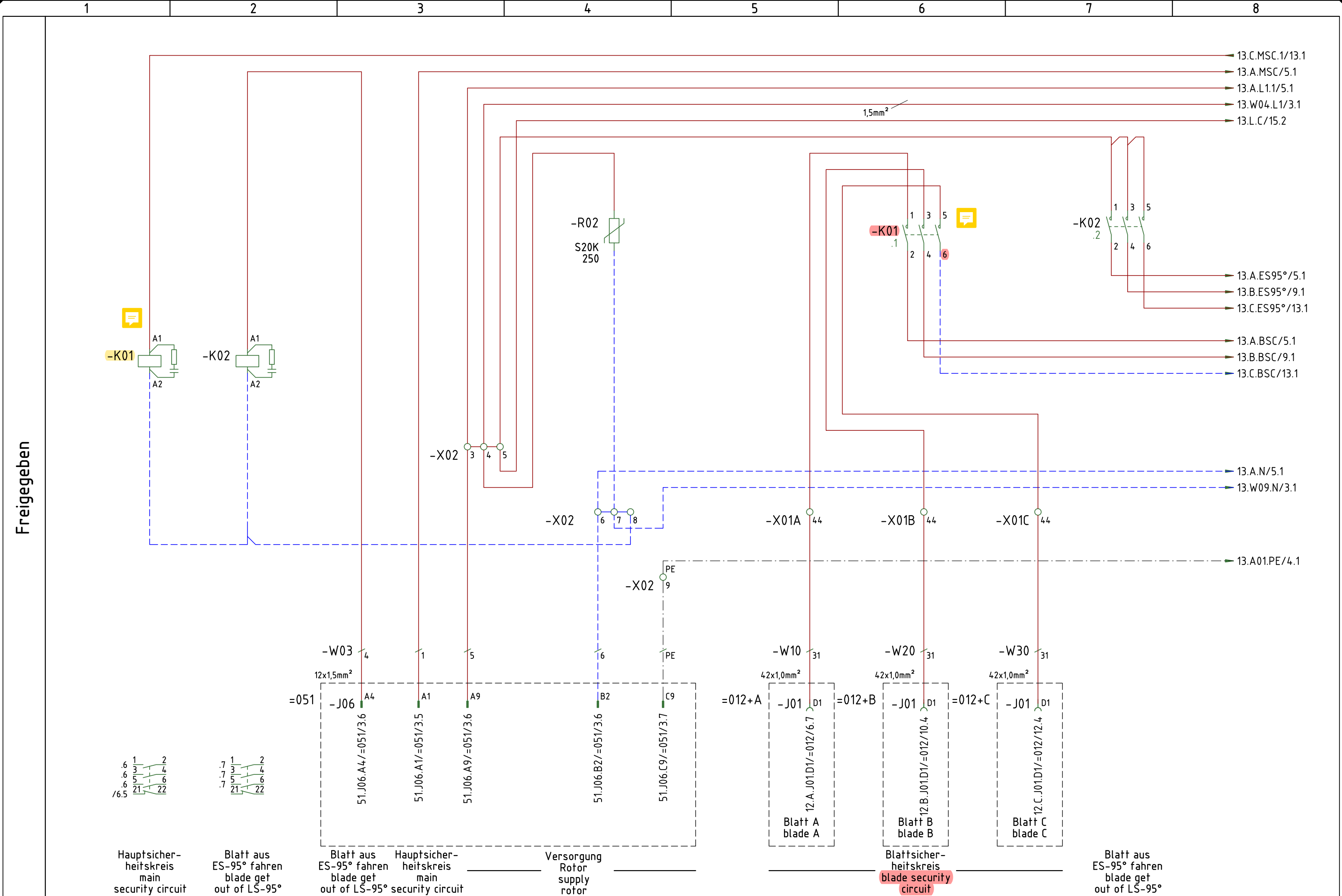
Technische Änderungen vorbehalten!
Technical subject to alteration!

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13	EC00002298	30.07.2013	40681	date	15.01.14	30.07.2013	Schaltplan	CS 48a	 ENERCON ENERGY FOR THE WORLD	Deckblatt	order no.:	013-53734-V1-13		= 013COVER			
12	EC00000527	20.06.2011	12257	prep.by	K.Langer		circuit diagram	Windenergieanlage		cover	ass. ID:			+			
11a		24.11.2010	14056	arr.by	A.Rülander			wind energy converter		Rotorunterverteilung	© Copyright ENERCON GmbH.	document ID:	D0116345	index:	13	page	1
no.	revision	date	name	appr.	P00030628		origin: germany	repl. for		rotor subdistribution	All rights reserved.					of	1



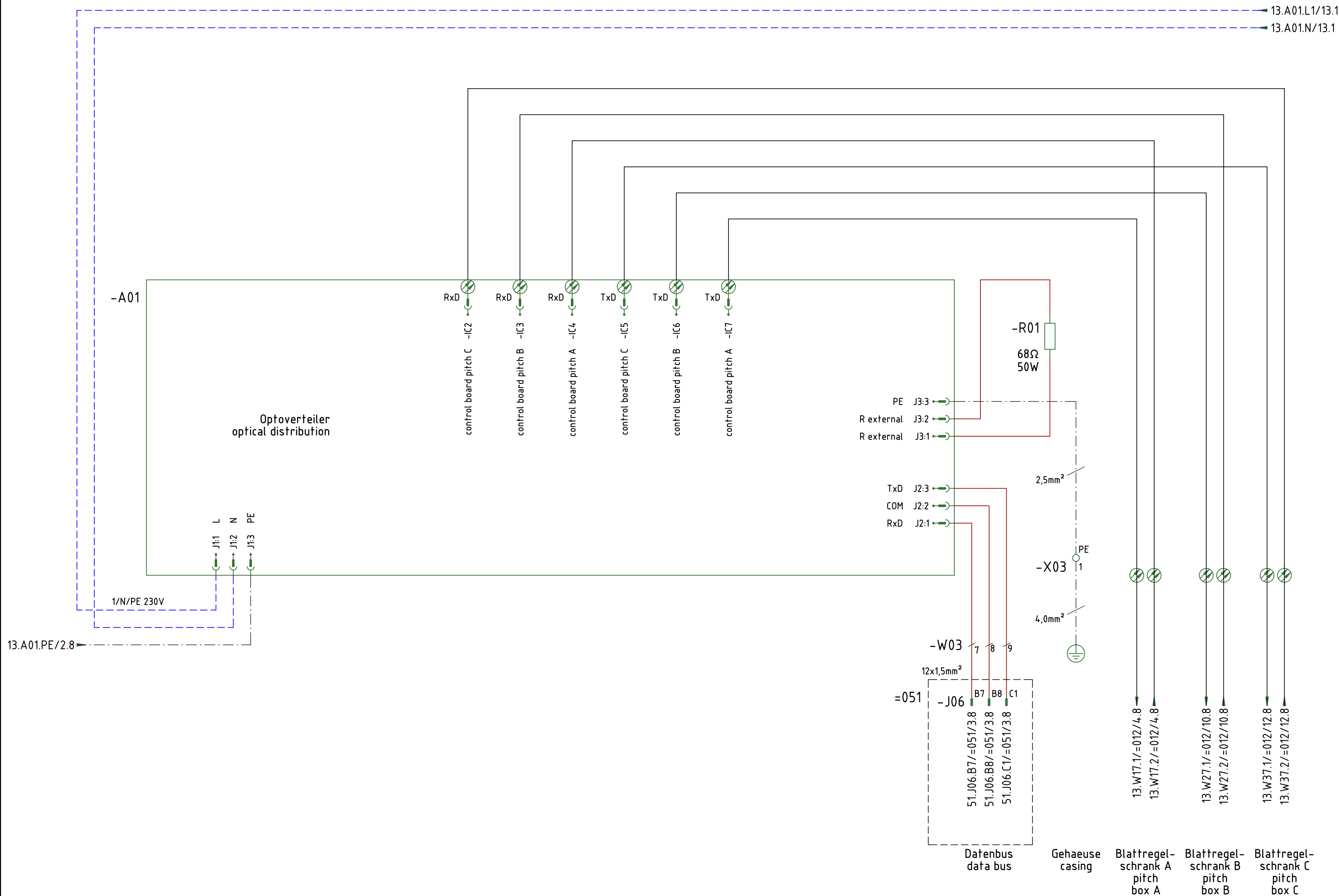
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				prep.by	K.Langer	circuit diagram	Windenergieanlage		supply	ass. ID:			
				arr.by	A.Rülander		wind energy converter		Rotorunterverteilung	© Copyright ENERCON GmbH.			
no.	revision	date	name	appr.	J.de Boer	origin: germany	repl. for	repl. by	rotor subdistribution	document ID:	D0116345	index: 13	page 1 of 16

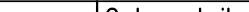


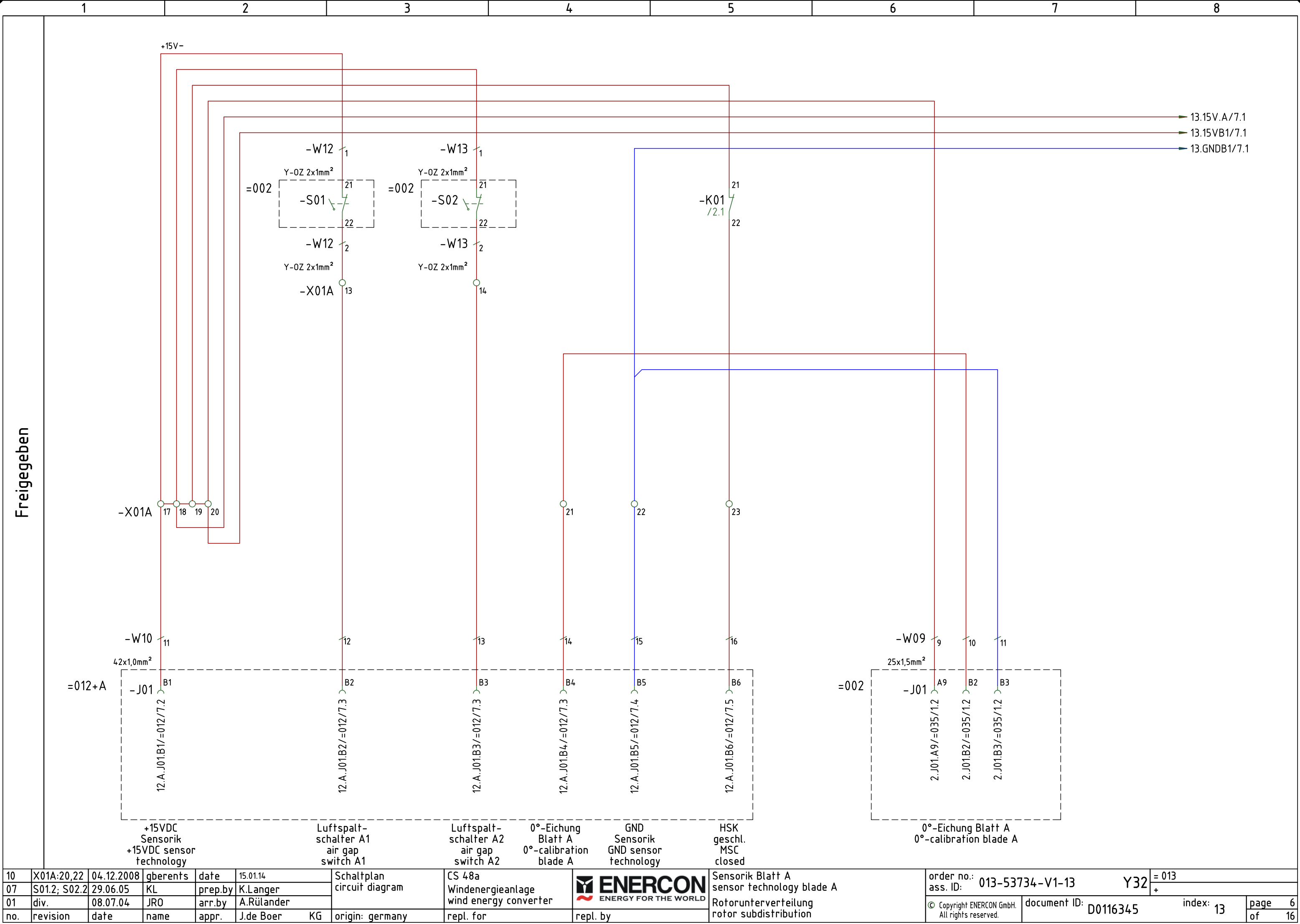
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06	-X01ABC:44	9.05.05	KL	prep.by	K.Langer	circuit diagram	Windenergieanlage		main security circuit	ass. ID:			
02	K01, W03	14.09.04	TL	arr.by	A.Rülander		wind energy converter		Rotorunterverteilung	© Copyright ENERCON GmbH.	document ID: D0116345	index: 13	page 2
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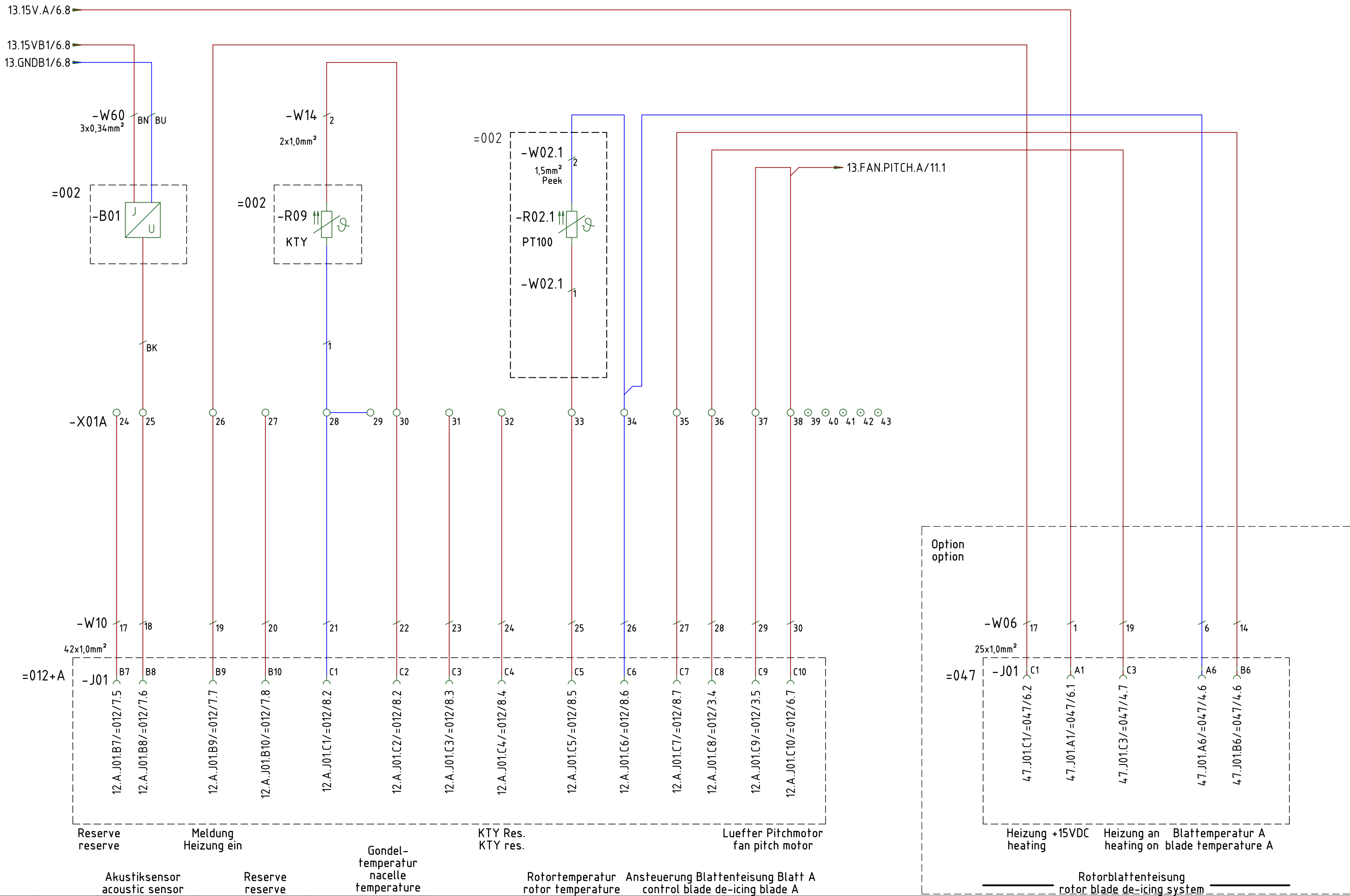
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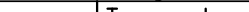


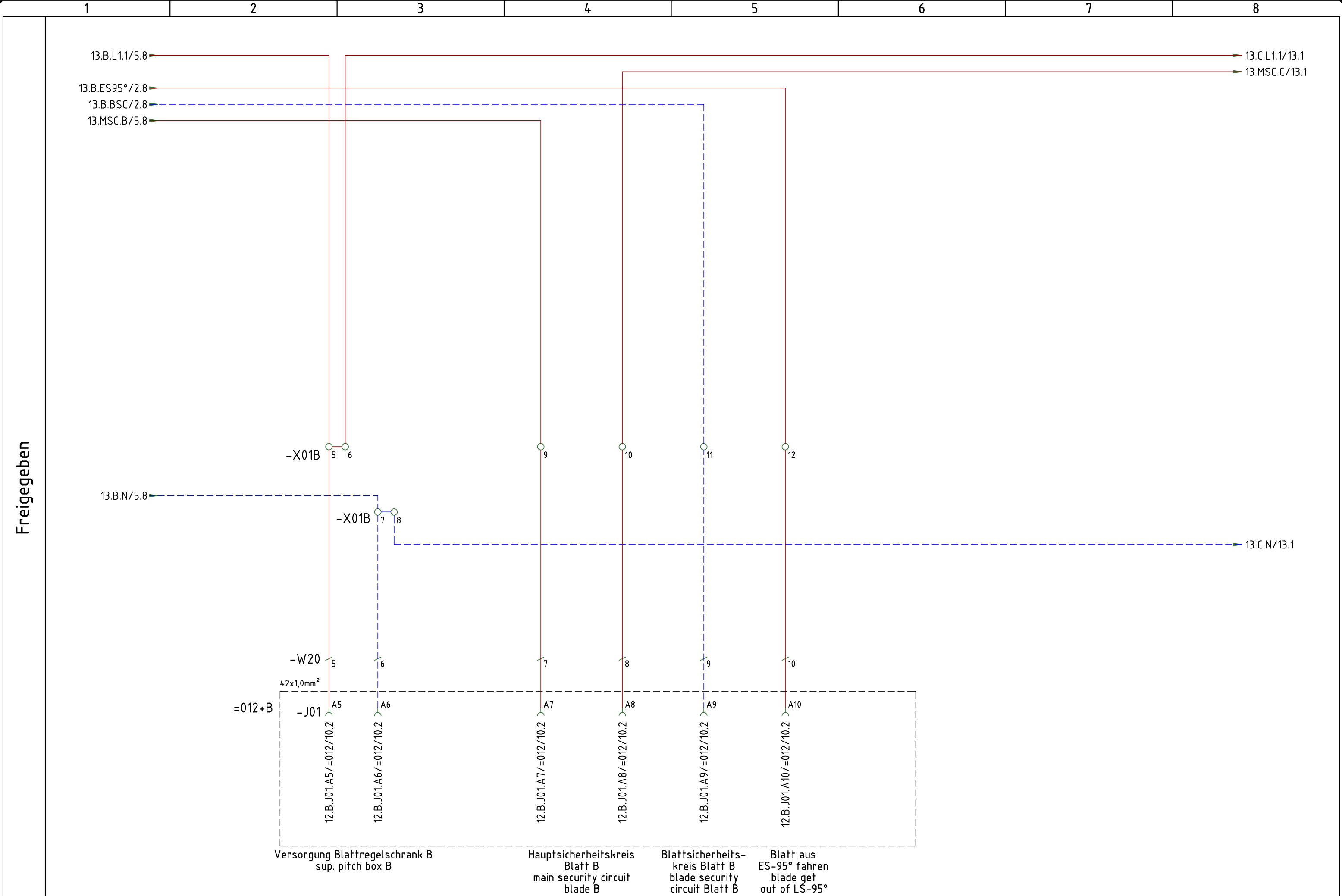
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				prep.by	K.Langer								
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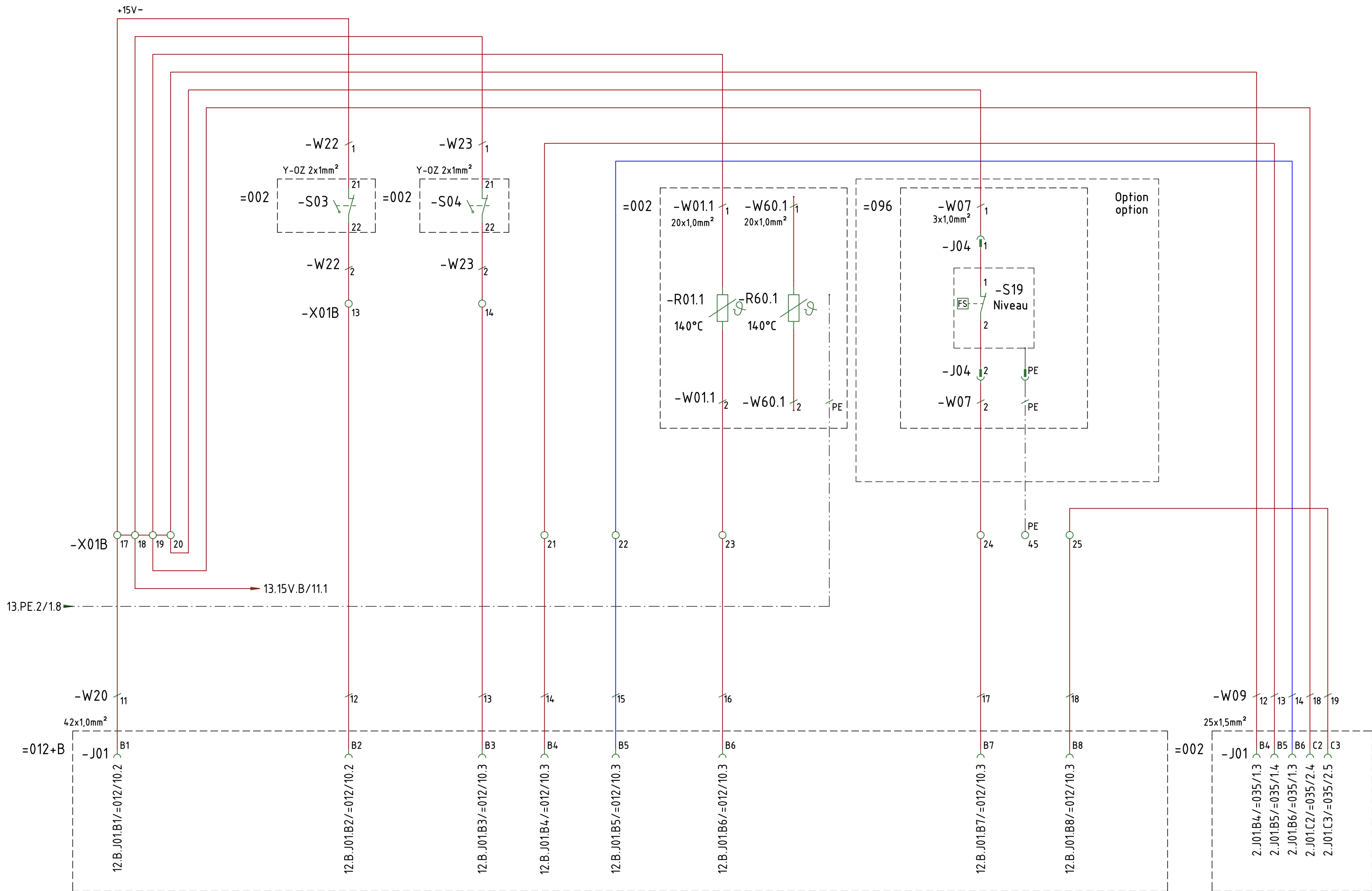


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12	=047 option	20.06.2011	12257	prep.by	K.Langer	circuit diagram	Windenergieanlage						
11	=047	17.06.2009	gberents	arr.by	A.Rülander		wind energy converter						
no.	revision	date	name	appr.	J.de Boer KG	origin: germany	repl. for	repl. by	Rotorunterverteilung rotor subdistribution	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0116345	index: 13	page 7 of 16



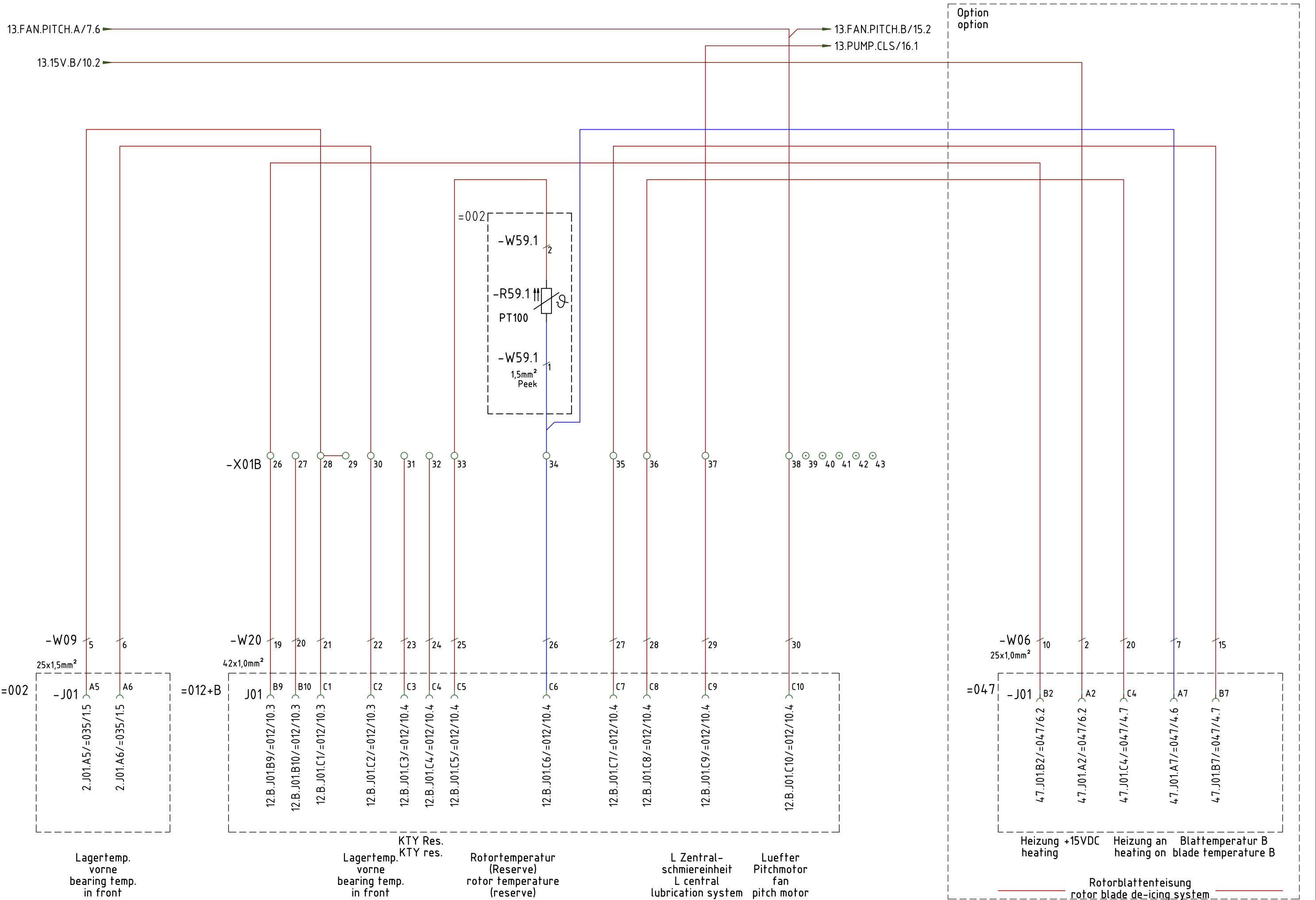
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				prep.by	K.Langer	ass. ID: 013-53734-V1-13 Y32 +								
				arr.by	A.Rülander	© Copyright ENERCON GmbH. All rights reserved.								
no.	revision	date	name	appr.	J.de Boer	KG	origin: germany	repl. for	repl. by		document ID: D0116345	index: 13	page 9 of 16	

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13	EC00002298	30.07.2013	40681	date	15.01.14	Schaltplan	CS 48a		Sensorik Blatt B	order no.:	013-53734-V1-13	Y32	= 013
12	=096 option	20.06.2011	12257 FN	prep.by	K.Langer	circuit diagram	Windenergieanlage		sensor technology blade B	ass. ID:			
10	=096-J04	04.12.2008	gberent	arr.by	A.Rülander		wind energy converter		Rotorunterverteilung	© Copyright ENERCON GmbH.	document ID: D0116345	index: 13	page 10
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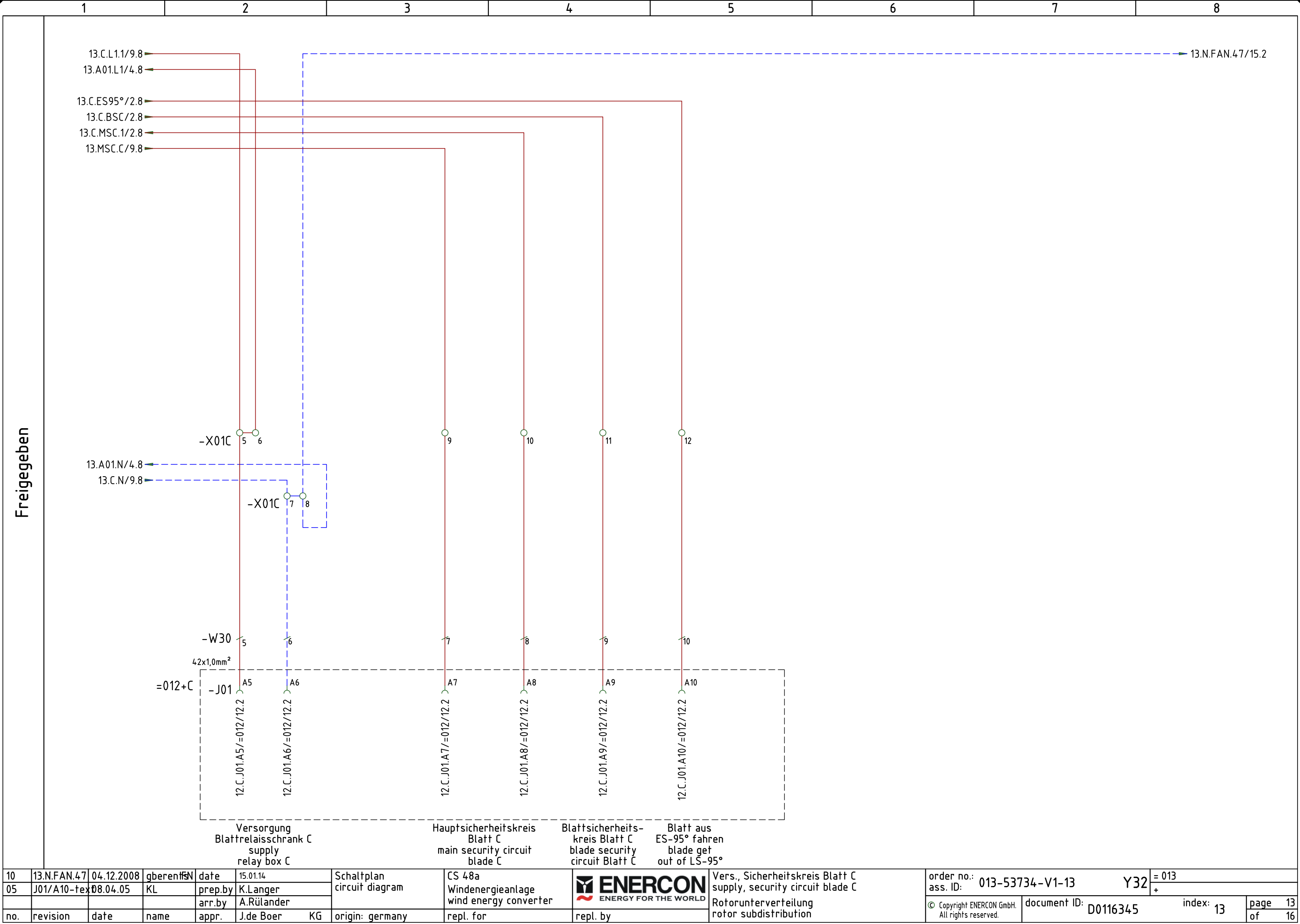
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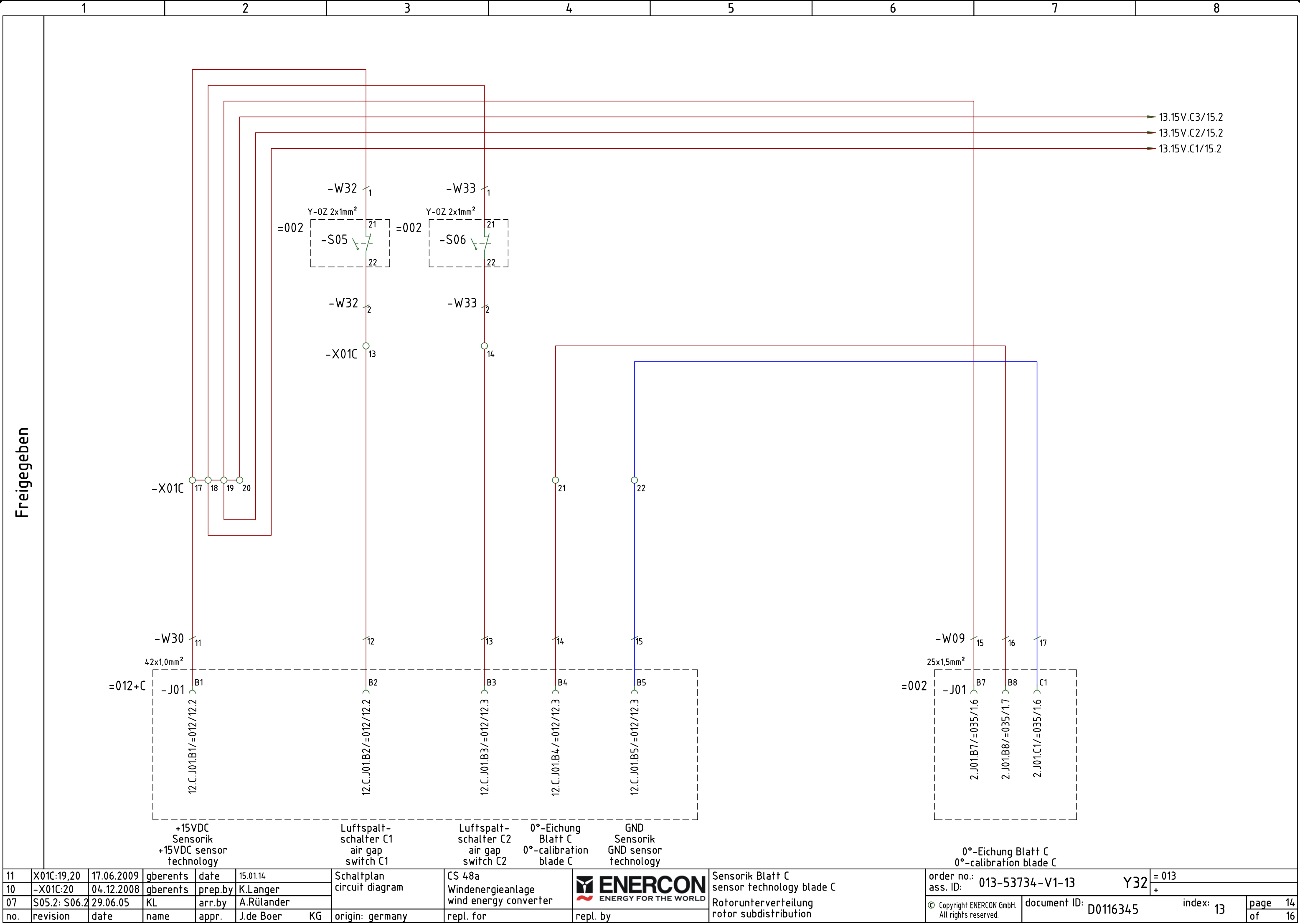


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12	=047 option	20.06.2011	12257	prep.by	K.Langer	circuit diagram	Windenergieanlage		temperature monitoring	ass. ID:				+
11	=047	17.06.2009	gberents	arr.by	A.Rülander		wind energy converter		Rotorunterverteilung	© Copyright ENERCON GmbH.	document ID:	D0116345	index:	13
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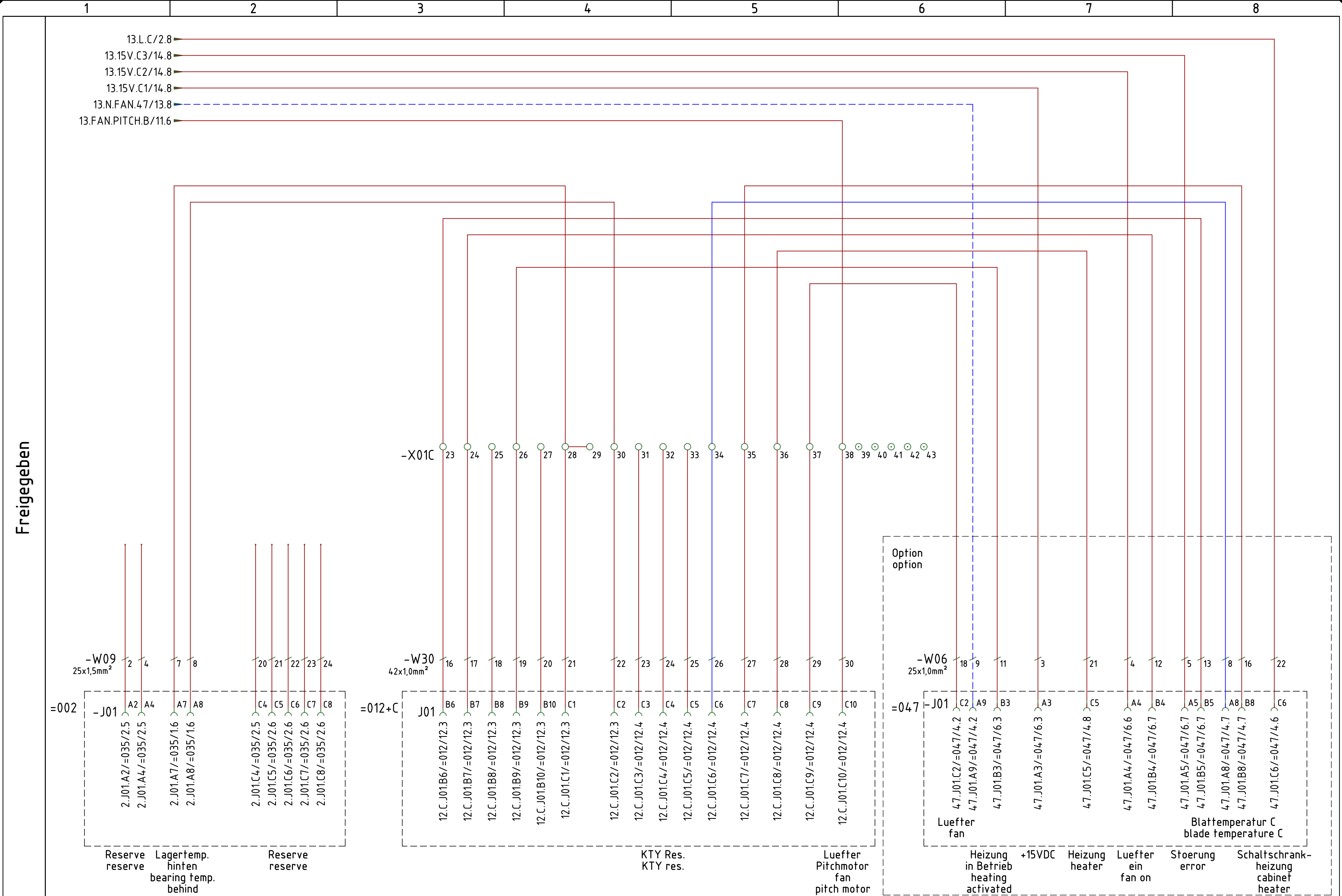


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				arr.by	A.Rülander								
no.	revision	date	name	appr.	J.de Boer KG	origin: germany	repl. for	repl. by	Rotorunterverteilung rotor subdistribution	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0116345	index: 13	page 12 of 16

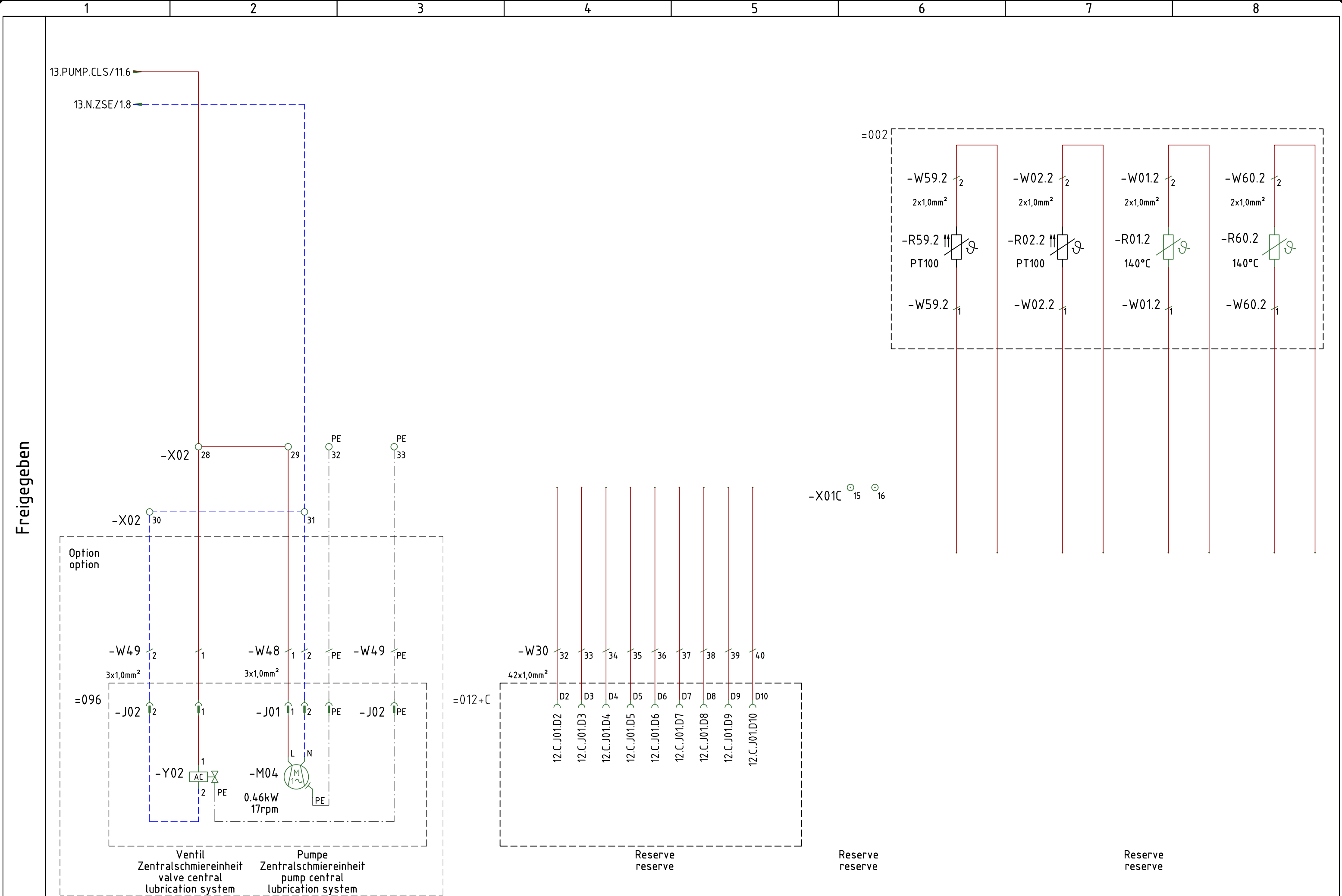


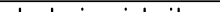


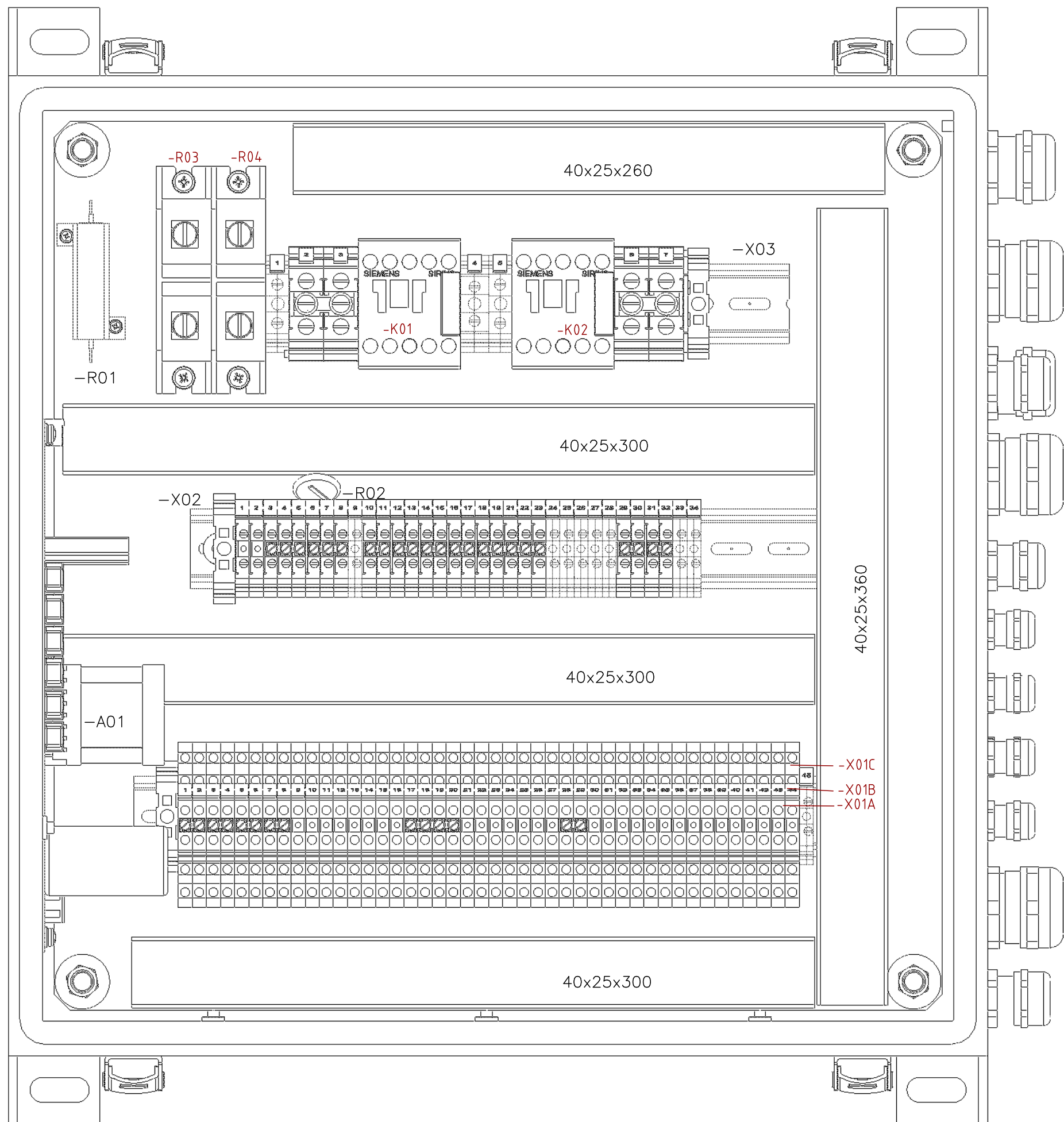
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07	S05.2: S06.2	29.06.05	KL	arr.by	A.Rülander									
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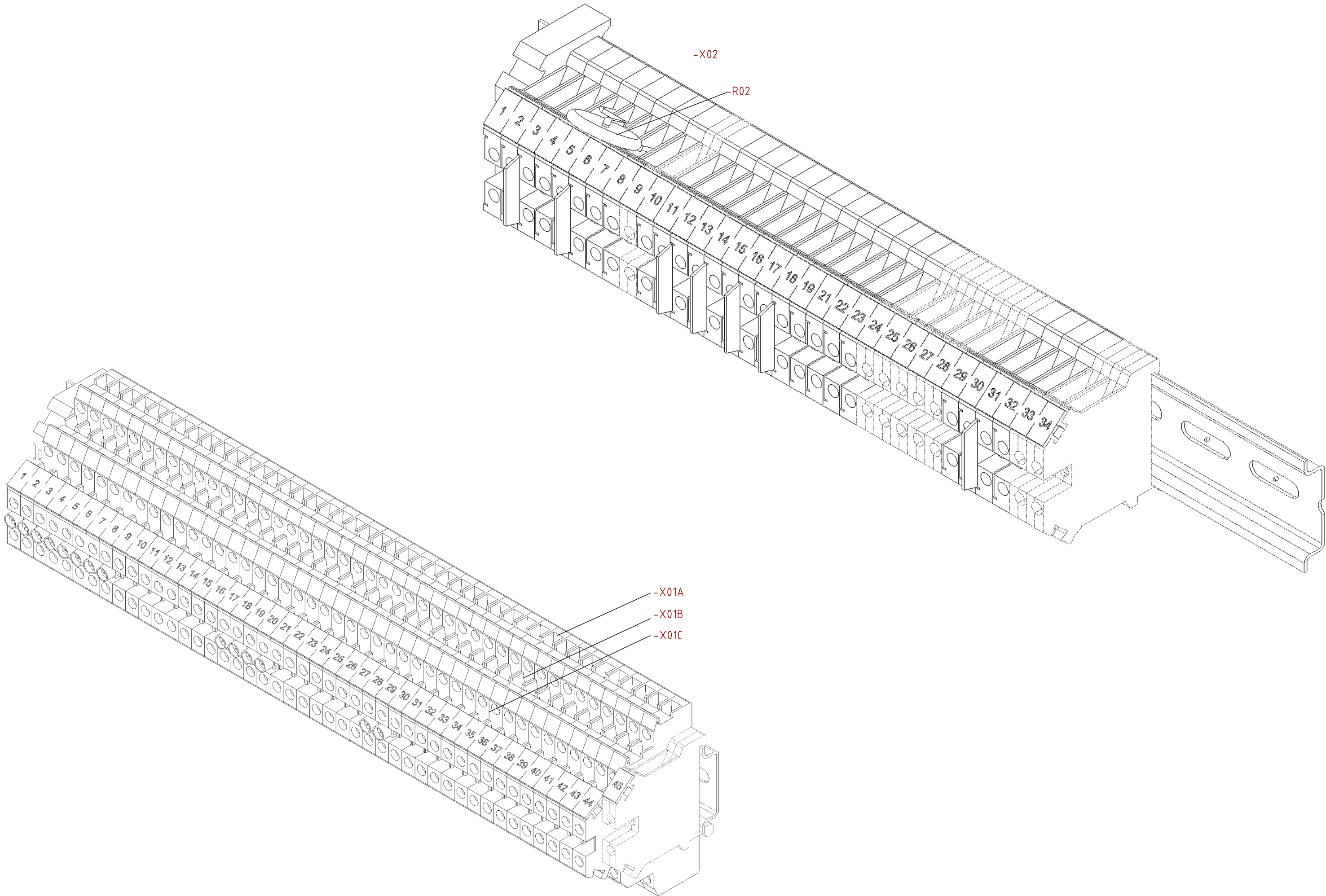
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11	=047	17.06.2009	gberents	prep.by	K.Langer								
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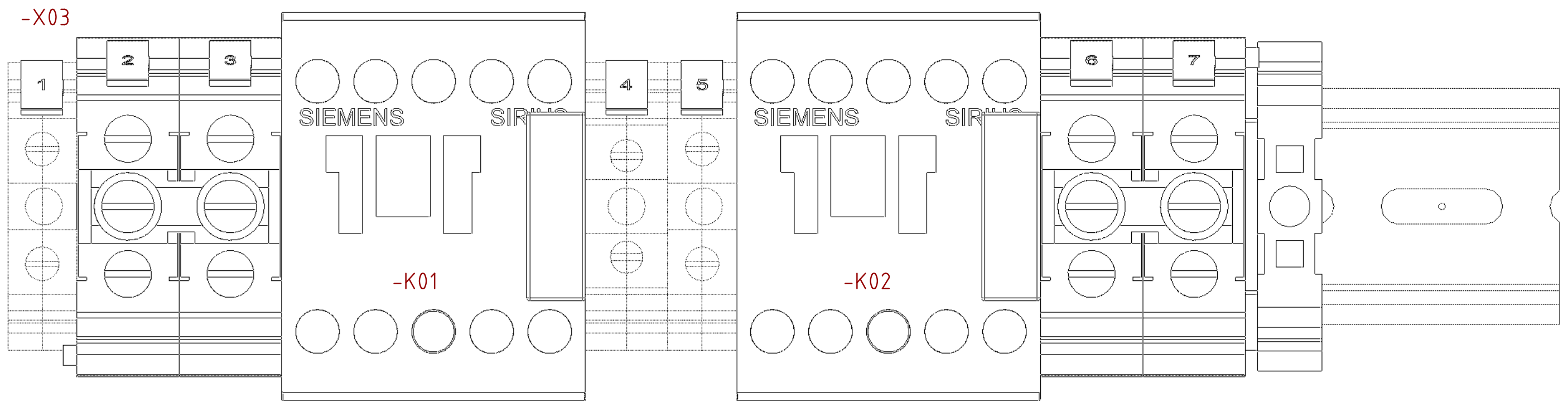
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10	=096-J	04.12.2008	gberentf	arr.by	A.Rülander		wind energy converter				© Copyright ENERCON GmbH.		document ID:	D0116345	index:	13	page	16
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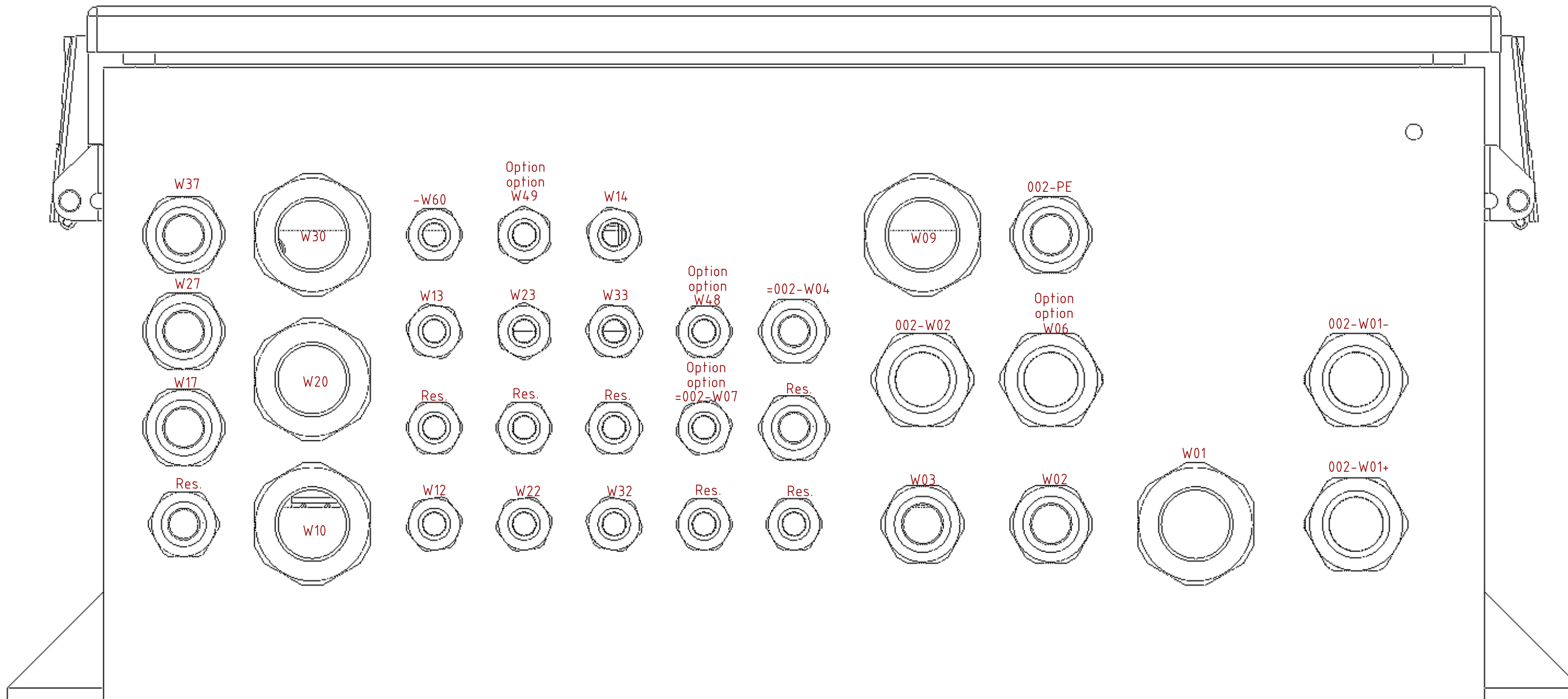


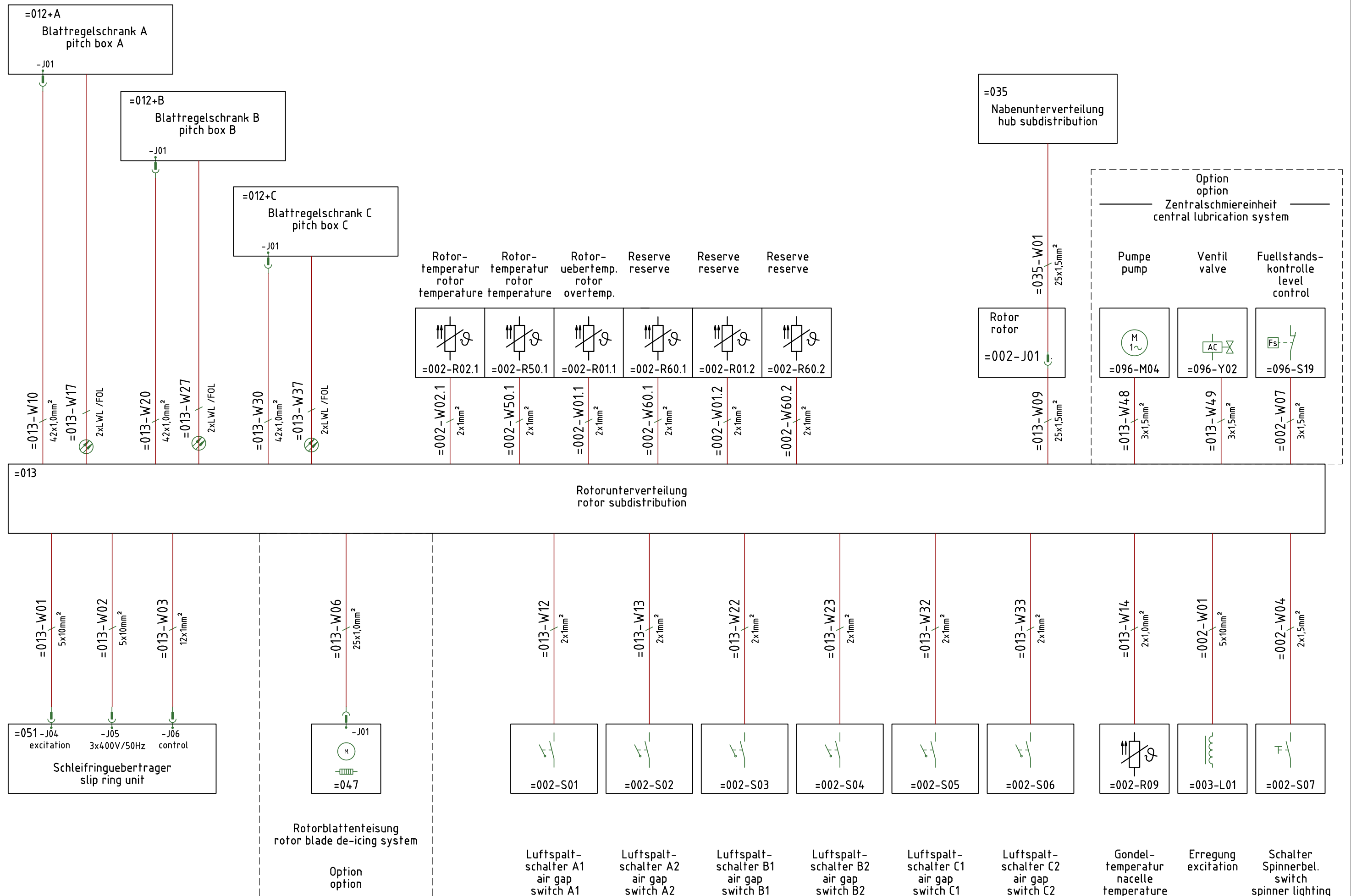
13	EC00002298	30.07.2013	40681	MB	date	15.01.14	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Aufbauplan construction plan Rotorunterverteilung rotor subdistribution	order no.: 013-53734-V1-13		= 013Y52	
06	-X01:44	09.05.05	KL	FN	prep.by	K.Langer					ass. ID: 013-53734-V1-13		+	
05	Kanal	08.04.05	KL	FN	arr.by	A.Rülander								
no.	revision	date	name	appr.	K.Goeman	KG	origin: germany	repl. for	repl. by		© Copyright ENERCON GmbH. All rights reserved.	document ID: D0116345	index: 13	page 1 of 4



13	EC00002298	30.07.2013	40681 MB	date	15.01.14	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Aufbauplan construction plan Rotorunterverteilung rotor subdistribution	order no.: ass. ID: 013-53734-V1-13	= 013Y52		
09	-X01C; -X02	27.02.06	GBE	prep.by	K.Langer						+		
06	-X01:44	09.05.05	KL	arr.by	A.Rülander						© Copyright ENERCON GmbH. All rights reserved.		
no.	revision	date	name	appr.	K.Goeman KG	origin: germany	repl. for	repl. by		document ID: D0116345	index: 13	page 2	of 4







Freigegeben	Stückliste/ bill of material																
	Anlagenkennung/ installation ID Ortskennung/ location ID BMK/ device ID			Artikel Nr. - Index/ article no. - index	SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english				Funktionstext deutsch/ function remarks german Funktionstext englisch/ function remarks english					Seite. Pfad page. column			
	002		B01	82787-0	Sensor Geräusch- KA1-Q25KB-DPÖ-X0409 sensor noise KA1-Q25KB-DPÖ-X0409				Akustiksensorm acoustic sensor					7.2			
	013		A01	53787-0	Platine Optovert.Rotor Typ CAN V1.2 E112 PCB opt.distrib.rotor type CAN V1.2 E112				Optoverteiler optical distribution					4.2			
	013		K01	53569-0	Schütz kl. 07A 1Ö 230V 3RT1015-1AP02 contactor sm. 07A 1NC 230V 3RT1015-1AP02				Hauptsicherheitskreis main security circuit					2.1			
	013		K02	53569-0	Schütz kl. 07A 1Ö 230V 3RT1015-1AP02 contactor sm. 07A 1NC 230V 3RT1015-1AP02				Blatt aus ES-95° fahren blade get out of LS-95°					2.2			
	013		R01	7364-0	Widerstand Draht 68R 50W RB50 Alu-Geh. resistor wire 68R 50W RB50 alum-box				RXD Widerstand RXD resistor					4.7			
	013		R02	1589-0	Varistor S20K250 varistor S20K250				Ueberspannungsschutz overvoltage protection					2.4			
	013		R03	23204-0	Varistor SIOV B60 K680 varistor SIOV B60 K680				Ueberspannungsschutz overvoltage protection					3.3			
	013		R04	23204-0	Varistor SIOV B60 K680 varistor SIOV B60 K680				Ueberspannungsschutz overvoltage protection					3.4			

Freigegeben	Mengen-Stückliste/ quantity bill of material														
	Artikel Nr.- Index/ article no.-index		Menge/ quantity		Benennung deutsch/ title german Benennung englisch/ title english			SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english				Hersteller/manufacturer Hersteller Teile-Nr. / manufacturer part no.			
	1589-0		1		Varistor varistor			Varistor S20K250 varistor S20K250				EPCOS-TDK B72220S 251K101			
	16847-0		1		Reihenklemme terminal			Klemme Reihen- 16mm ² 101A gn/ge USLKG16N terminal block 16mm ² 101A gr/yeUSLKG16N				Phoenix Contact GmbH & Co. KG 0443023			
	17882-0		51		Dreistockklemme three level terminal			Klemme Dreistock 4mm ² grau DIKD1,5 terminal three level 4mm ² grey DIKD1,5				Phoenix Contact GmbH & Co. KG 2715979			
	17883-0		7		Dreistockklemme three level terminal			Klemme Dreistock 4mm ² blau DIKD1,5BU terminal three level 4mm ² blue DIKD1,5BU				Phoenix Contact GmbH & Co. KG 2716101			
	23204-0		2		Varistor varistor			Varistor SIOV B60 K680 varistor SIOV B60 K680				EPCOS-TDK B72260B0681K001			
	53569-0		2		Leistungsschütz power contactor			Schütz kl. 07A 1Ö 230V 3RT1015-1AP02 contactor sm. 07A 1NC 230V 3RT1015-1AP02				Siemens AG A&D 3RT1015-1AP02			
	53787-0		1		Platine PCB			Platine Optovert.Rotor Typ CAN V1.2 E112 PCB opt.distrib.rotor type CAN V1.2 E112				ENERCON GmbH 53787			
	5539-0		4		Reihenklemme terminal			Klemme Reihen- 35mm ² 1000V 150A gr UK35 terminal block 35mm ² 1000V 150A gr UK35				Phoenix Contact GmbH & Co. KG 3008012			
	7364-0		1		Lastwiderstand load resistor			Widerstand Draht 68R 50W RB50 Alu-Geh. resistor wire 68R 50W RB50 alum-box				Meggit Elektronik Components -			
	7947-0		2		Feste Klemmenbrücke terminal bridge with screw			Feste Brücke 3pol m.Schr.m.Isol.FBI3-15 bridge 3pol with screw,no insul.FBI3-15				Phoenix Contact GmbH & Co. KG 0201278			
	82787-0		1		Sensor sensor			Sensor Geräusch- KA1-Q25KB-DPÖ-X0409 sensor noise KA1-Q25KB-DPÖ-X0409				Pulsontronic GmbH & Co. KG 08340001100			
	885-0		15		Reihenklemme terminal			Klemme Reihen- 4mm ² 800V 41A gr UK5N terminal block 4mm ² 800V 41A gr UK5N				Phoenix Contact GmbH & Co. KG 3004362			
	886-0		10		Reihenklemme terminal			Klemme Reihen- 4mm ² 800V 41A bl UK5NBU terminal block 4mm ² 800V 41A bu UK5NBU				Phoenix Contact GmbH & Co. KG 3004388			
	959-0		7		Reihenklemme terminal			Klemme Reihen- 4mm ² gn/ge USLKG5 terminal block 4mm ² gr/ye USLKG5				Phoenix Contact GmbH & Co. KG 0441504			
	961-0		2		Reihenklemme terminal			Klemme Reihen- 10mm ² gn/ge USLKG10N terminal block 10mm ² gr/ye USLKG10N				Phoenix Contact GmbH & Co. KG 3003923			
				date	15.01.14	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Mengen-Stückliste quantity parts list Rotorunterverteilung rotor subdistribution	order no.: 013-53734-V1-13		Y66		= 013Y66	
				prep.by	K.Langer					ass. ID:					
				arr.by	A.Rülander					© Copyright ENERCON GmbH. All rights reserved.		document ID: D0116345		index: 13	
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by							

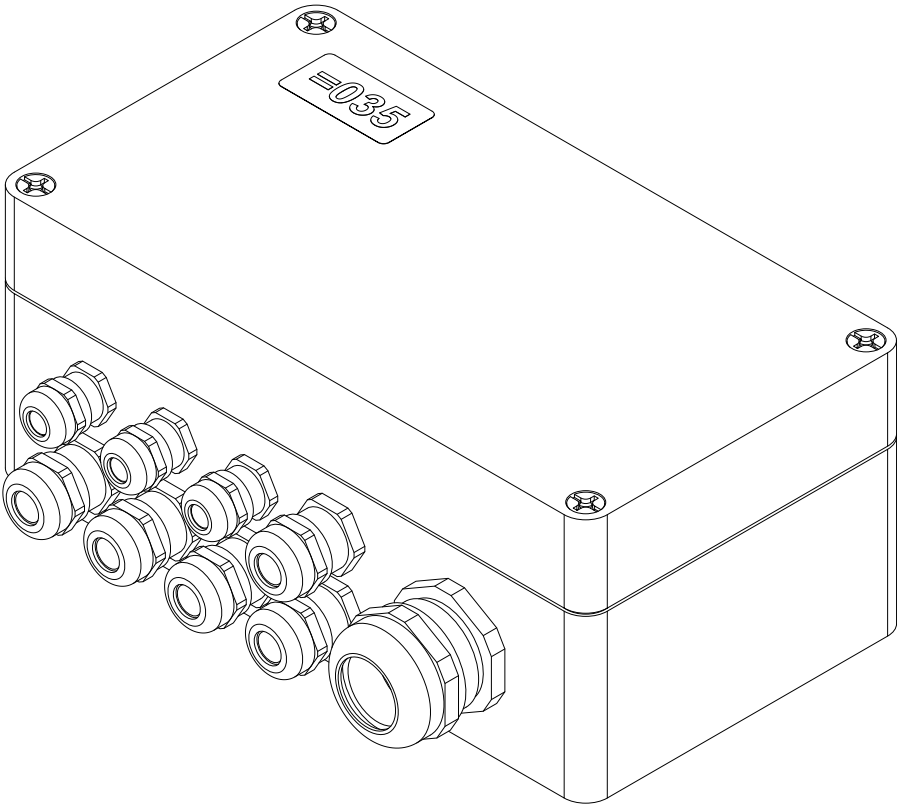
Nabenunterverteilung
hub subdistribution

SAP article no.: index:
57032 0

Reifegrad / level of maturity:
Serie
series

Zusatzinfo / additional info:

Anlagentyp / converter type:
E-44; E-48; E-53
Steuerung / control system:
CS48a
Baugruppennr. / assembly no.:
035
Version Baugr. / version of assembly:
V1
Baugr. Index / component index:
00



Baugruppenschaltplan
assembly circuit diagram

document ID: index:
D0116410 0c

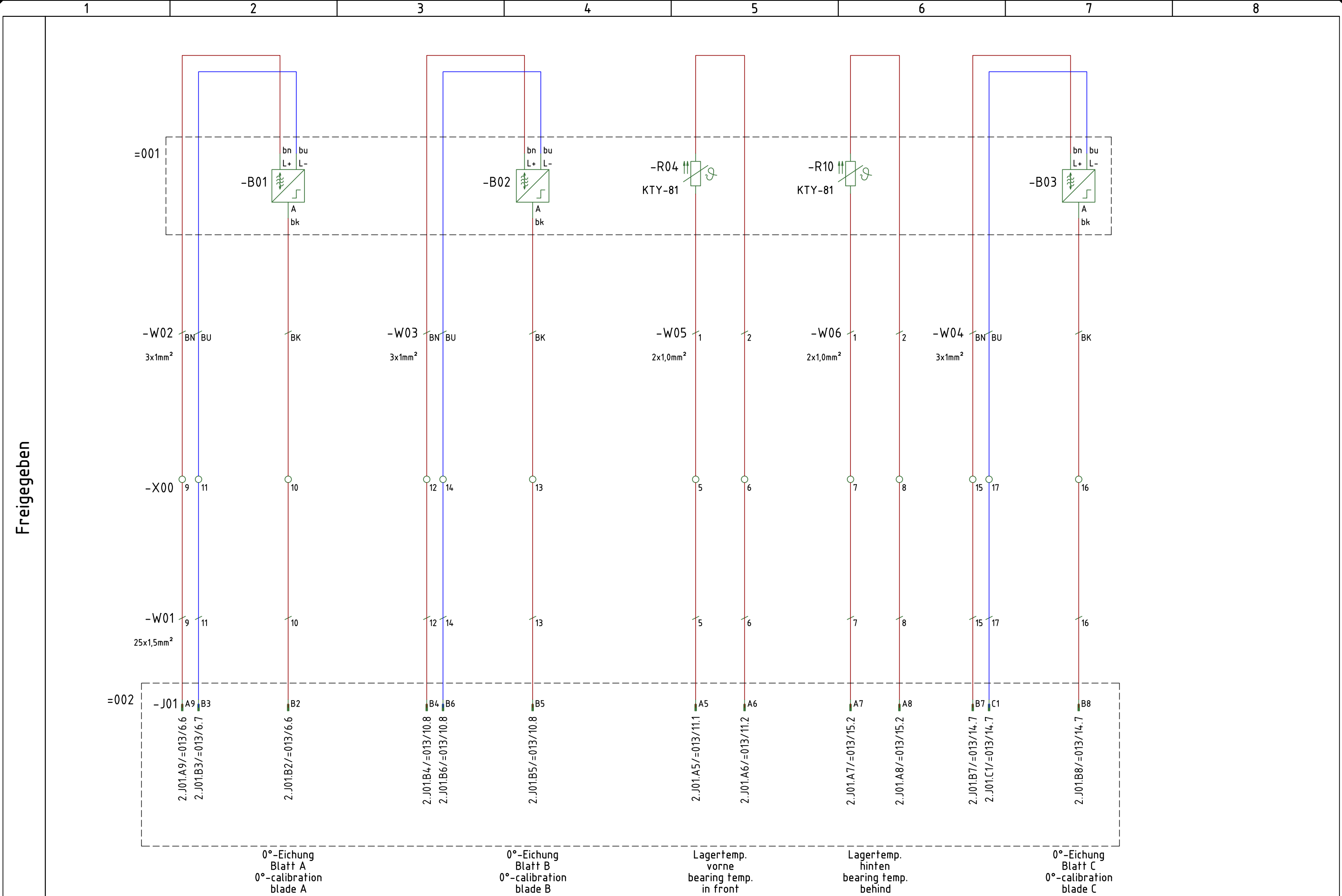
title or assembly ID:
035-57032-V100c

Alle nicht gekennzeichneten Leitungen 1mm² schwarz.
All not marked lines 1mm² black.
Alle nicht gekennzeichneten Neutralleiter 1mm² hellblau.
All not marked neutral conductor 1mm² light blue.

Technische Änderungen vorbehalten!
Technical subject to alteration!

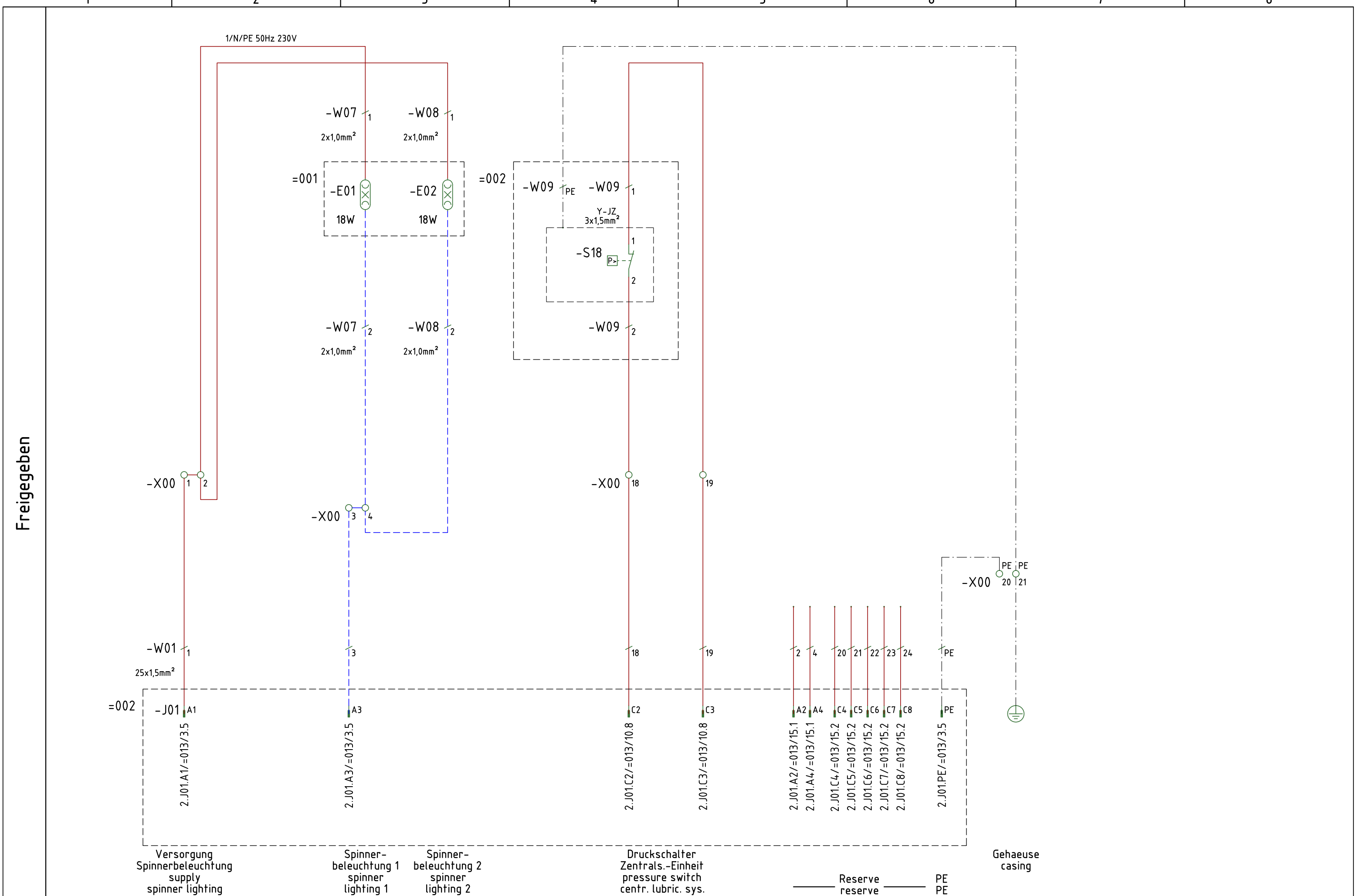
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Ob	div.	29.09.2009	ehmen	date	16.11.2004 09.10.2007	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Deckblatt cover	order no.: ass. ID: 035-57032-V100c	Y04	= 035COVER		
				prep.by	J. Rohlfen									
				arr.by	Newiger, D.									
no.	revision	date	name	appr.	14056 KG	origin: germany	repl. for	repl. by	Nabenunterverteilung hub subdistribution	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0116410	index: 0c	page 1 of 1	

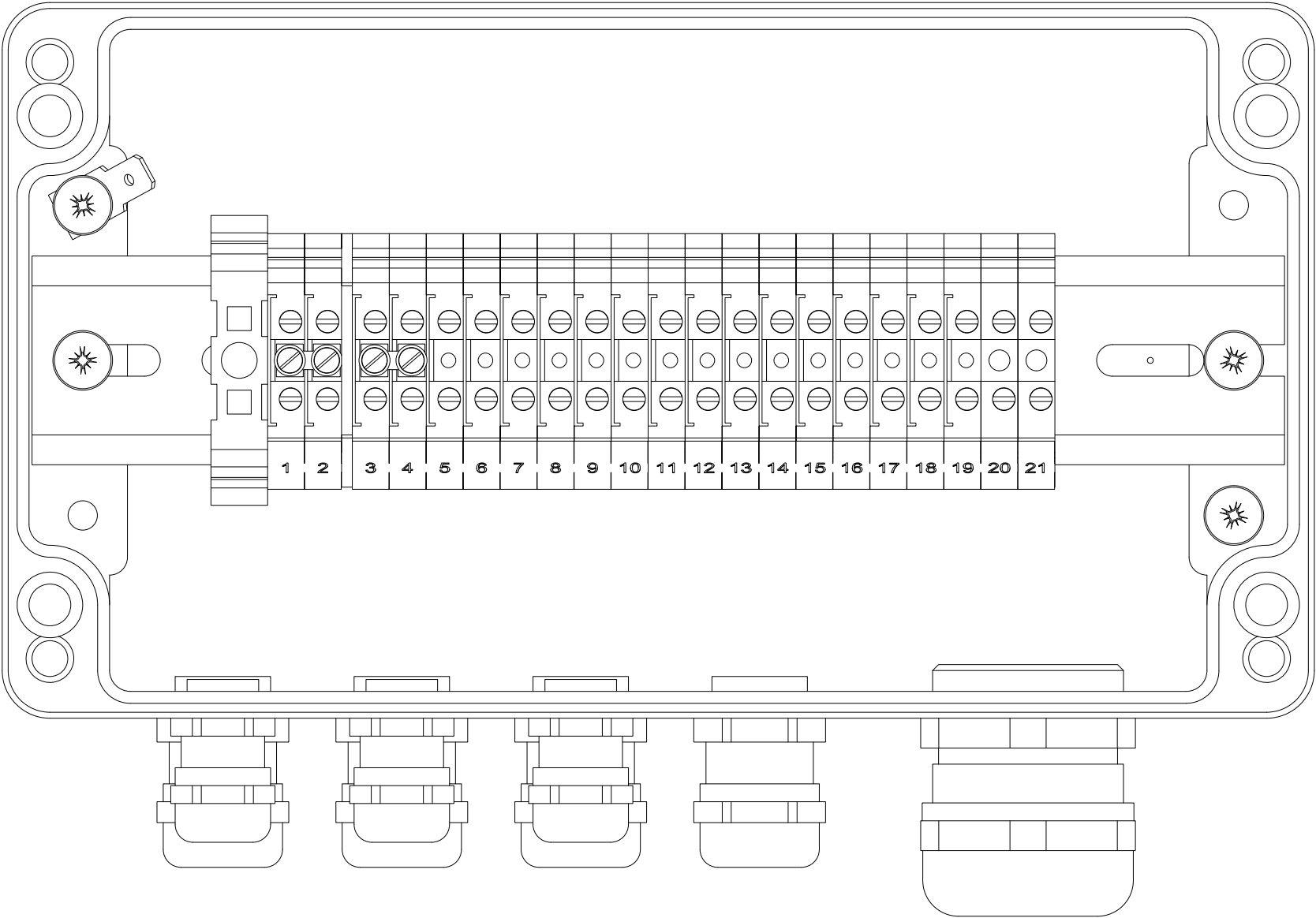
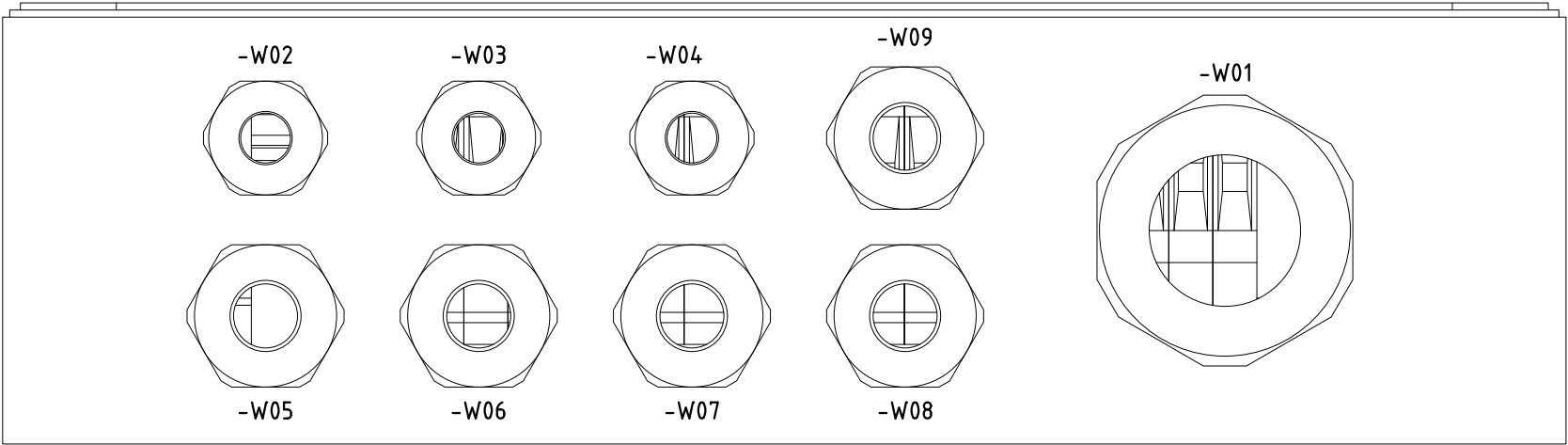


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				arr.by	FN		wind energy converter		Nabenunterverteilung	© Copyright ENERCON GmbH.			
no.	revision	date	name	appr.	J de Boer	origin: germany	repl. for	repl. by	hub subdistribution	document ID:	D0116410	index: 0c	page 1 of 2

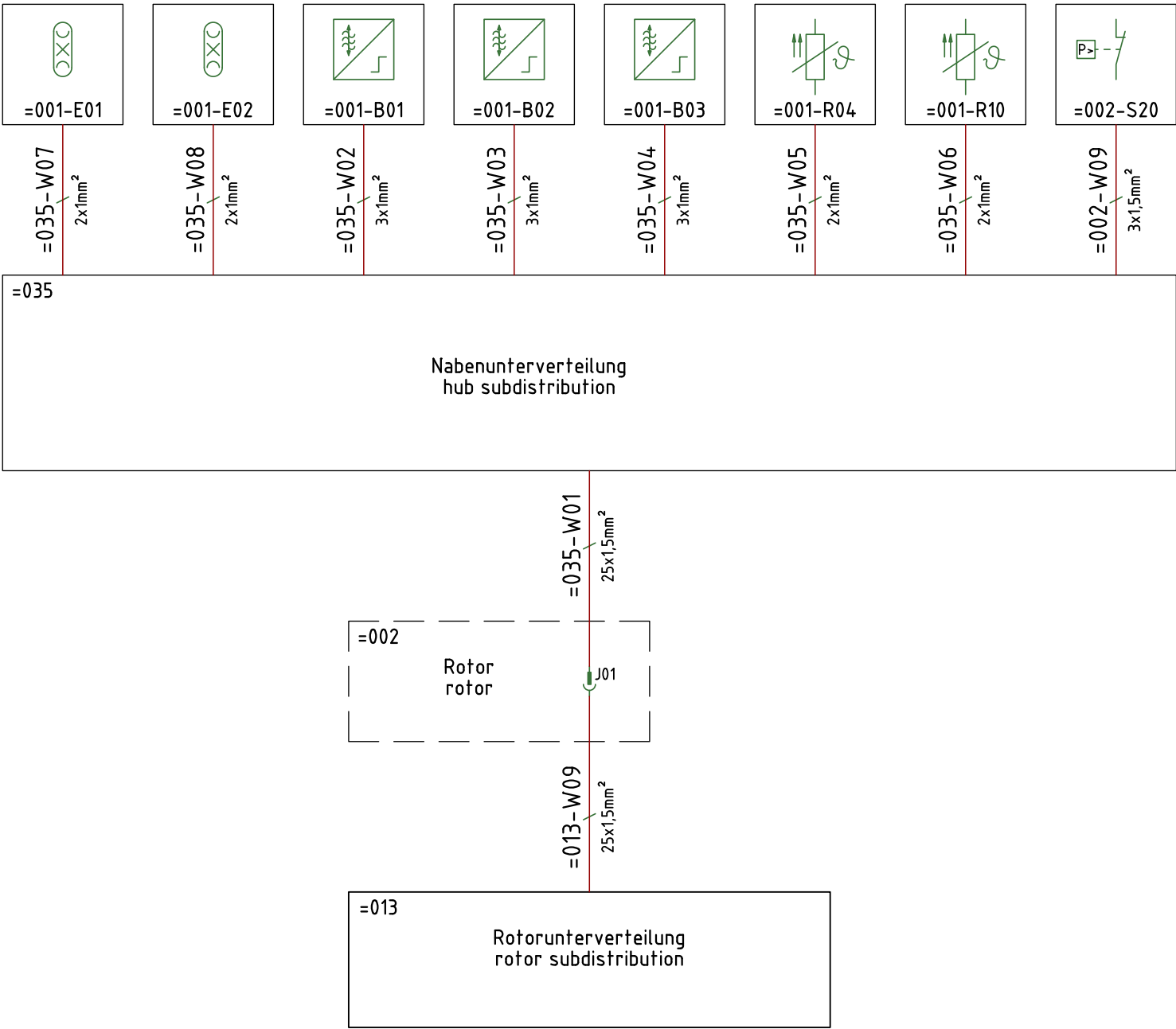
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				prep.by	Rohlfen	circuit diagram	Windenergieanlage		spinner lighting	ass. ID:			
				arr.by	FN		wind energy converter		Nabenunterverteilung	© Copyright ENERCON GmbH.			
no.	revision	date	name	appr.	J de Boer KG	origin: germany	repl. for	repl. by	hub subdistribution	document ID:	D0116410	index: 0c	page 2 of 2



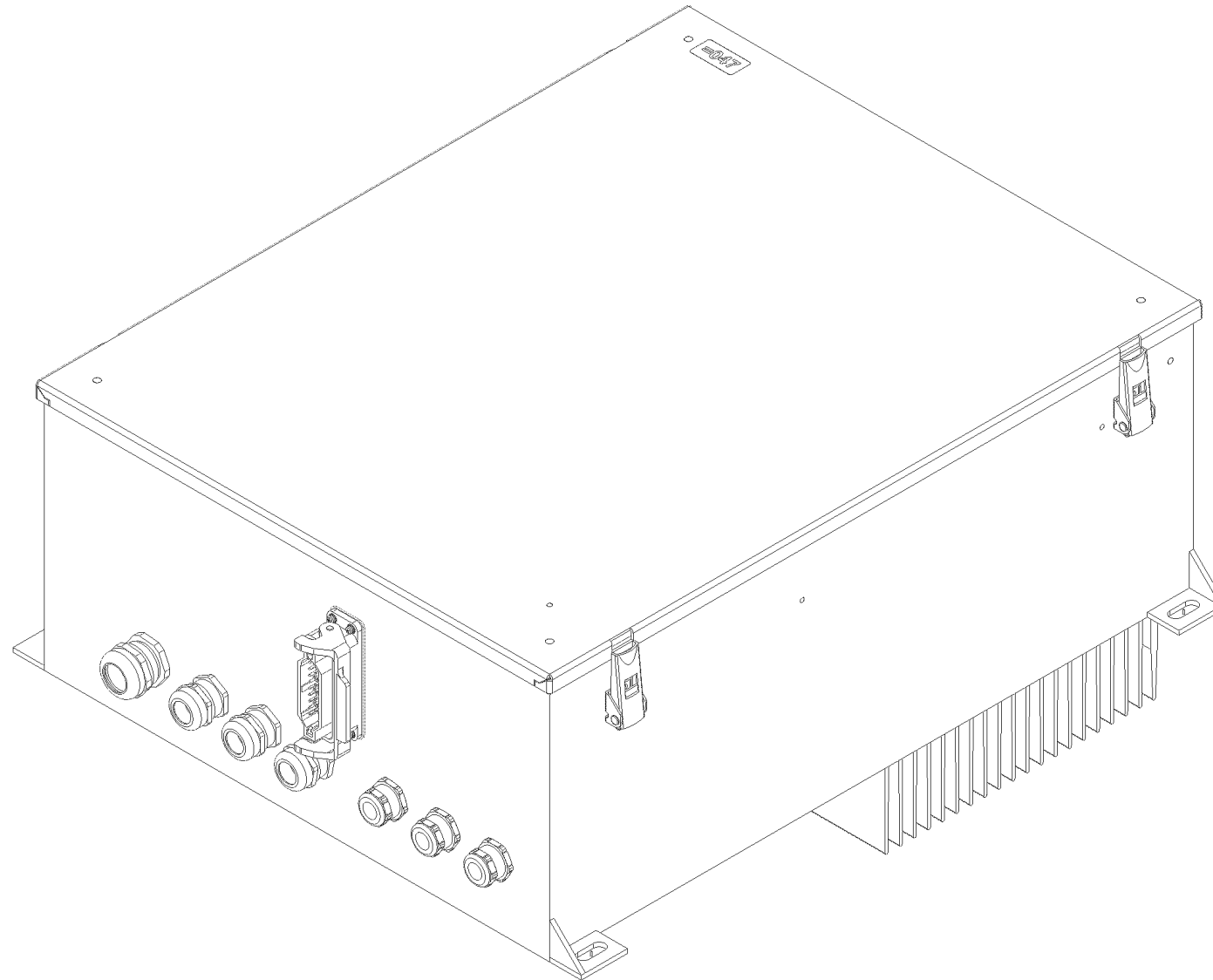
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Baugruppenschaltplan assembly circuit diagram

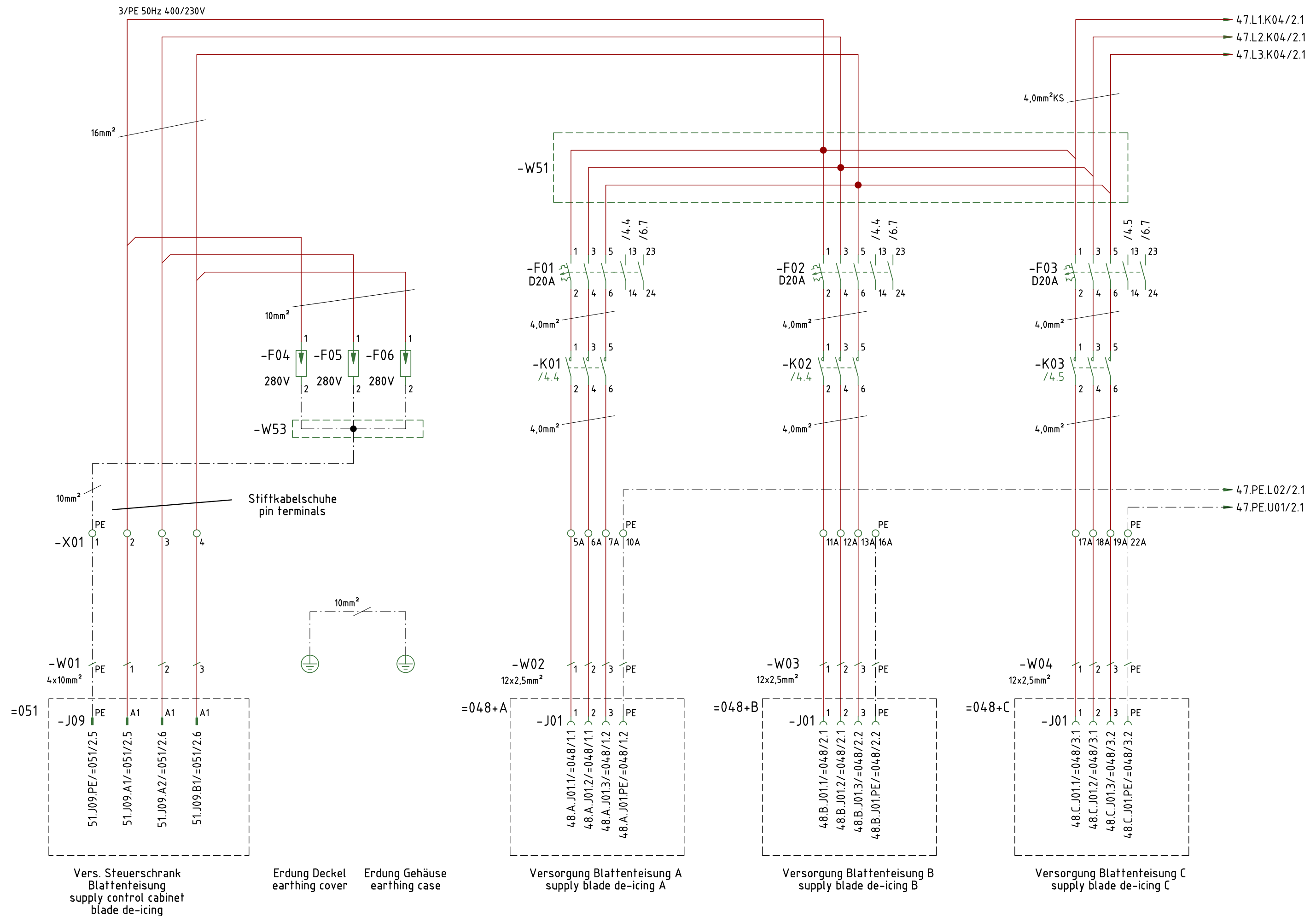
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047-88200-V1-05



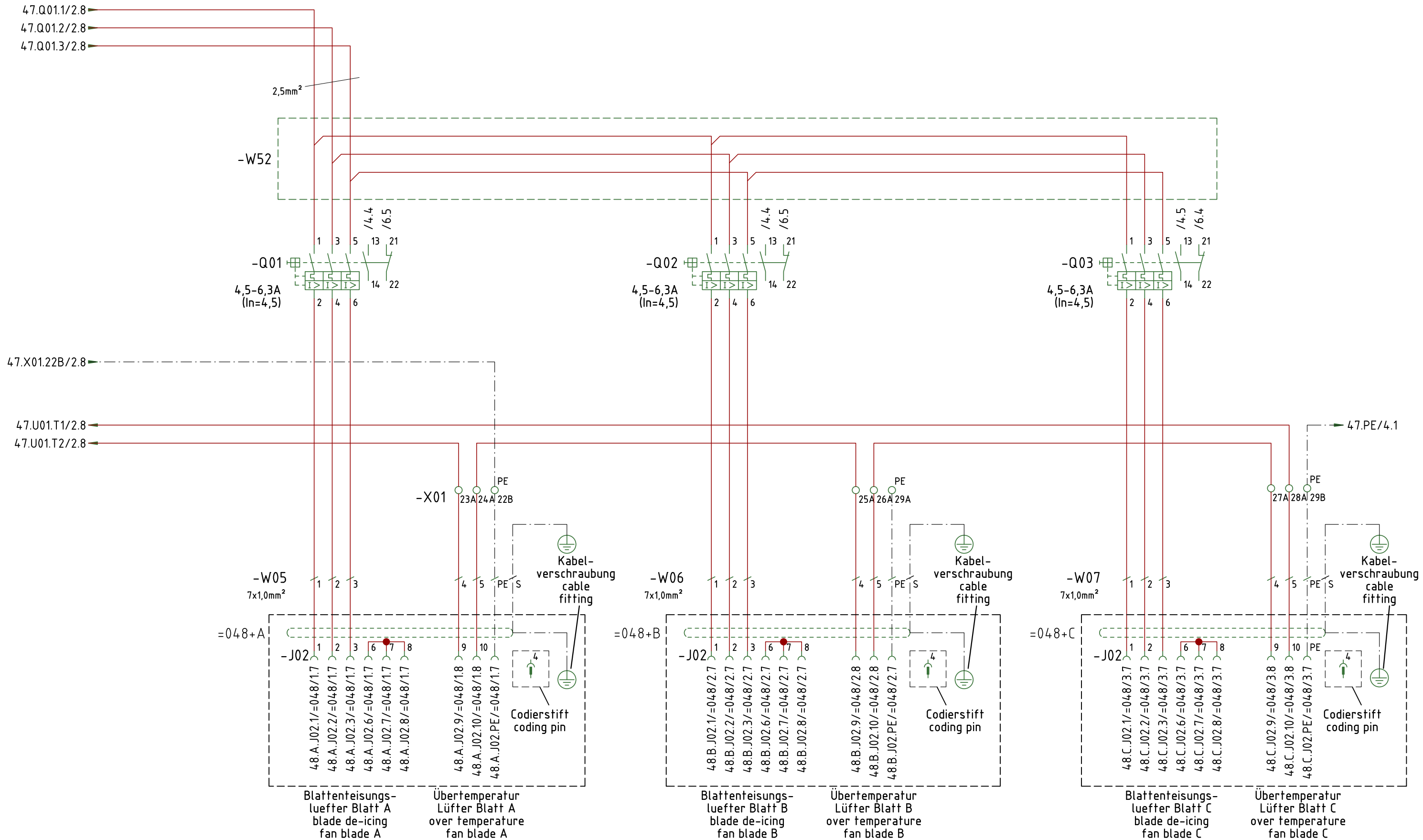
Technische Änderungen vorbehalten!
Technical subject to alteration!

5	EC00001634	01.03.2013	40681	date	18.03.13	24.06.2009	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Deckblatt cover	order no.: 047-88200-V1-05 Y04 = 047COVER			
4	EC00000339	30.09.2010	12257	prep.by	K.Langer						ass. ID: +			
3	C000598	28.09.2009	gberents	arr.by	A.Rülander									
no.	revision	date	name	appr.			origin: germany	repl. for	repl. by	Steuerschrank Rotorblattenteisung contr. cab. rotor blade de-icing system	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0136581	index: 5	page 1 of 1



5	EC00001634	01.03.2013	40681	date	18.03.13	03.09.2008	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Versorgung Blattenteisung supply blade de-icing	order no.: ass. ID: 047-88200-V1-05	Y32	= 047 +	
3	4 + 16mm ²	28.09.2009	gberents	prep.by	K.Langer									
1	F01-3, 4+16	07.07.2009	gberents	arr.by	A.Rütlander									
no.	revision	date	name	appr.			origin: germany	repl. for	repl. by	Steuerschrank Rotorblattenteisung contr. cab. rotor blade de-icing system	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0136581	index: 5	page 1 of 6

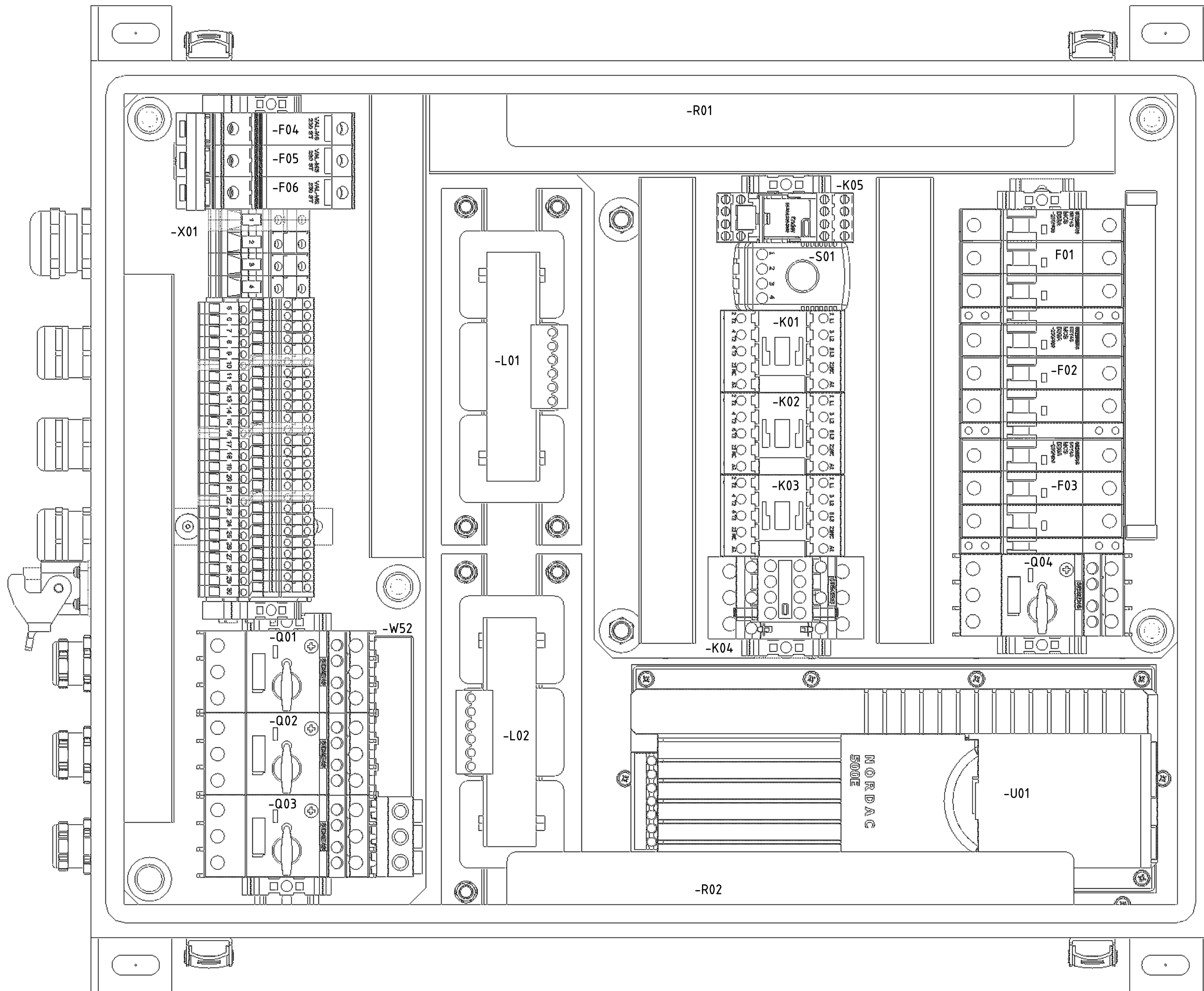




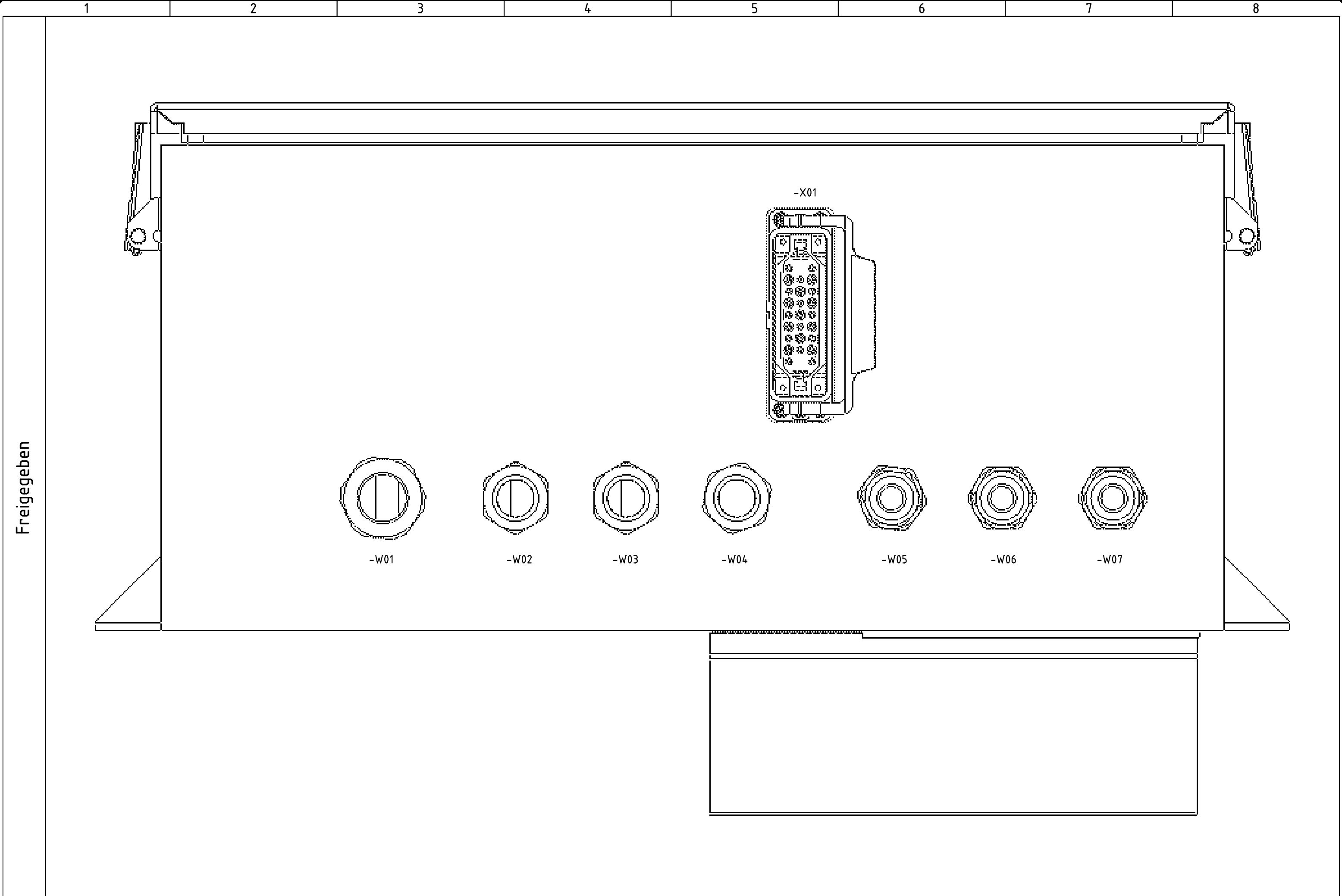
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				prep.by	K.Langer	circuit diagram	Windenergieanlage		supply blade de-icing fan	ass. ID:			
				arr.by	A.Rülander		wind energy converter		Steuerschrank Rotorblattenteisung	© Copyright ENERCON GmbH.			
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	contr. cab. rotor blade de-icing system	document ID:	D0136581	index: 5	page 3
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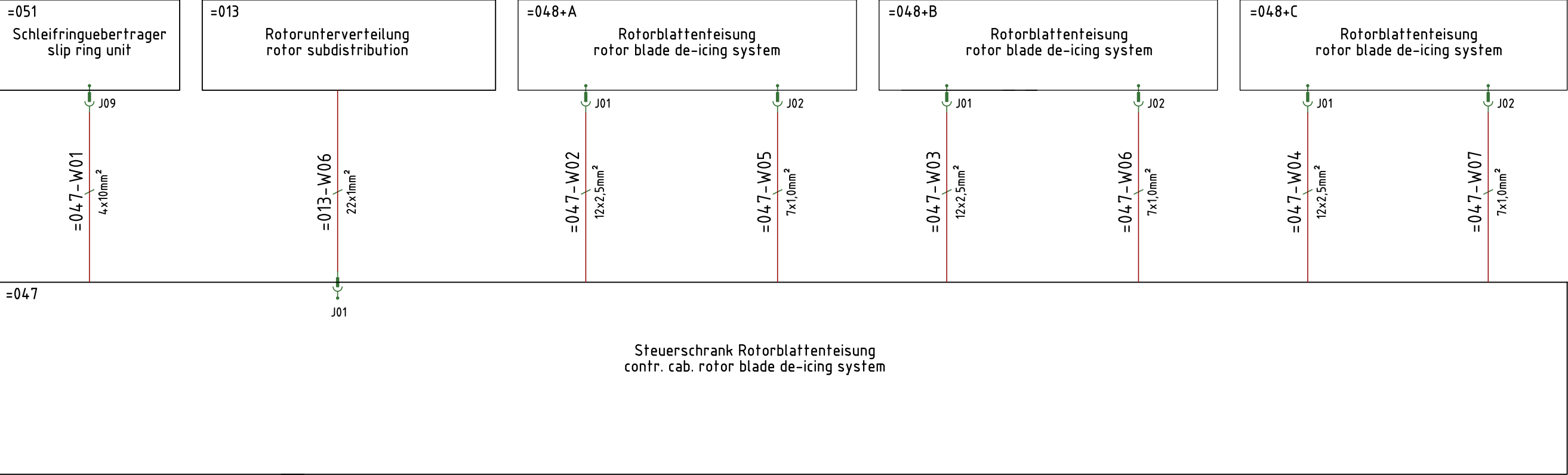


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2	R1+2,U1,K5	11.08.2009	gberents	prep.by	K.Langer	circuit diagram	Windenergieanlage		construction plan	ass. ID:			
1	F01-3, K01-3	30.07.2009	gberents	arr.by	A.Rülander		wind energy converter		Steuerschrank Rotorblattenteisung	© Copyright ENERCON GmbH.	document ID: D0136581	index: 5	page 1
no.	revision	date	name	appr.		origin: germany	repl. for		contr. cab. rotor blade de-icing system	All rights reserved.			
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5	EC00001634	01.03.2013	40681	date	18.03.13	Schaltplan	CS 48a		Aufbauplan	order no.:	047-88200-V1-05		Y52	= 047Y52
				prep.by	K.Langer	circuit diagram	Windenergieanlage		construction plan	ass. ID:				+
				arr.by	A.Rülander		wind energy converter		Steuerschrank Rotorblattenteisung	© Copyright ENERCON GmbH.	document ID:	D0136581	index: 5	page 2
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	contr. cab. rotor blade de-icing system	All rights reserved.				of 2



Freigegeben

Stückliste/ bill of material																
Anlagenkennung/ installation ID Ortskennung/ location ID BMK/ device ID				Artikel Nr. - Index/ article no. - index	SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english				Funktionstext deutsch/ function remarks german Funktionstext englisch/ function remarks english				Seite. Pfad page. column			
047		F01	75531-0	Schalter Hilfs- Anb. 2S 5ST3011 SIE☉ switch auxiliary 2NO 5ST3011 SIE☉				Versorgung Blattenteisung supply blade de-icing				1.4				
047		F01	58803-0	Sicherungsautomat 3p.D20A 400VAC SIE☉ autom.circuit break.3p. D20A 400VAC SIE☉				Versorgung Blattenteisung supply blade de-icing				1.4				
047		F02	58803-0	Sicherungsautomat 3p.D20A 400VAC SIE☉ autom.circuit break.3p. D20A 400VAC SIE☉				Versorgung Blattenteisung supply blade de-icing				1.6				
047		F02	75531-0	Schalter Hilfs- Anb. 2S 5ST3011 SIE☉ switch auxiliary 2NO 5ST3011 SIE☉				Versorgung Blattenteisung supply blade de-icing				1.6				
047		F03	58803-0	Sicherungsautomat 3p.D20A 400VAC SIE☉ autom.circuit break.3p. D20A 400VAC SIE☉				Versorgung Blattenteisung supply blade de-icing				1.7				
047		F03	75531-0	Schalter Hilfs- Anb. 2S 5ST3011 SIE☉ switch auxiliary 2NO 5ST3011 SIE☉				Versorgung Blattenteisung supply blade de-icing				1.7				
047		F04	33333-0	Überspannungsableiter kompl. VAL-MS230 overvoltage arrester compl. VAL-MS230				Ueberspannungsschutz overvoltage protection				1.3				
047		F05	33333-0	Überspannungsableiter kompl. VAL-MS230 overvoltage arrester compl. VAL-MS230				Ueberspannungsschutz overvoltage protection				1.3				
047		F06	33333-0	Überspannungsableiter kompl. VAL-MS230 overvoltage arrester compl. VAL-MS230				Ueberspannungsschutz overvoltage protection				1.3				
047		K01	510487-0	Schütz Leistungs-15,5A 1Ö DILM15-01 230V contactor power-15,5A 1NC DILM15-01 230V				Blattenteisung A blade de-icing A				4.4				
047		K01	58637-0	Schutzbeschalt.240-500VAC DILM12-XSPR500 protective dev.240-500VAC DILM12-XSPR500				Blattenteisung A blade de-icing A				4.4				
047		K02	510487-0	Schütz Leistungs-15,5A 1Ö DILM15-01 230V contactor power-15,5A 1NC DILM15-01 230V				Blattenteisung B blade de-icing B				4.4				
047		K02	58637-0	Schutzbeschalt.240-500VAC DILM12-XSPR500 protective dev.240-500VAC DILM12-XSPR500				Blattenteisung B blade de-icing B				4.4				
047		K03	510487-0	Schütz Leistungs-15,5A 1Ö DILM15-01 230V contactor power-15,5A 1NC DILM15-01 230V				Blattenteisung C blade de-icing C				4.5				
047		K03	58637-0	Schutzbeschalt.240-500VAC DILM12-XSPR500 protective dev.240-500VAC DILM12-XSPR500				Blattenteisung C blade de-icing C				4.5				

				date	18.03.13	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Stueckliste parts list	order no.: ass. ID: 047-88200-V1-05 Y64	= 047Y64 +		
				prep.by	K.Langer								
				arr.by	A.Rülander								
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Steuerschrank Rotorblattenteisung contr. cab. rotor blade de-icing system	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0136581	index: 5	page 1 of 3

Freigegeben

Stückliste/ bill of material														
Anlagenkennung/ installation ID Ortskennung/ location ID BMK/ device ID			Artikel Nr. - Index/ article no. - index	SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english			Funktionstext deutsch/ function remarks german Funktionstext englisch/ function remarks english				Seite. Pfad page. column			
047		K04	540548-0	Schütz Leistungs- 230VAC 3RT2025-1AP00 contactor power- 230VAC 3RT2025-1AP00			Blattenteisungsluefter blade de-icing fan				4.2			
047		K04	540535-0	Schutzbeschalt.240-400VAC 3RT2926-1CE00 protective dev.240-400VAC 3RT2926-1CE00			Blattenteisungsluefter blade de-icing fan				4.2			
047		K04	547563-0	Schalter Hilfs- SÖSS 3RH2911-1XA31-0MA0 switch aux. 0COO 3RH2911-1XA31-0MA0			Blattenteisungsluefter blade de-icing fan				4.2			
047		K05	97180-0	Relais FINDER 55.34.8.230.5050 /4W relay FINDER 55.34.8.230.5050 /4Ch			Blattenteisungsluefter blade de-icing fan				4.3			
047		K05	16486-0	Sockel Relais- FINDER 94.84.30 socket relay- FINDER 94.84.30			Blattenteisungsluefter blade de-icing fan				4.3			
047		K05	44623-0	Bügel Halte- FINDER 094.91.30 (Variclip) bracket holding- FINDER 094.91.30 varic.			Blattenteisungsluefter blade de-icing fan				4.3			
047		K05	78107-0	Schutzbeschalt.110-240V 99.80.0.230.09 protective dev.110-240V 99.80.0.230.09			Blattenteisungsluefter blade de-icing fan				4.3			
047		L01	91609-0	Drossel Drehstrom-Netz- DEX-LN3-025 choke three-ph.wall grid- DEX-LN3-025			Netzdrossel net choke				2.3			
047		L02	91516-0	Drossel Motor- AC DEX-LM3-016 choke motor-ac DEX-LM3-016			Motordrossel motor choke				2.4			
047		Q01	547041-0	Schalter Hilfs- 1S/1Ö 3RV2901-1E switch auxiliary 1S/1Ö 3RV2901-1E			Vers. Blattenteisungsluefter Blatt A supply blade de-icing fan blade A				3.3			
047		Q01	547187-0	Schalter Motorschutz- 4,5-6,3A 3RV2011 switch motor protect. 4,5-6,3A 3RV2011			Vers. Blattenteisungsluefter Blatt A supply blade de-icing fan blade A				3.3			
047		Q02	547041-0	Schalter Hilfs- 1S/1Ö 3RV2901-1E switch auxiliary 1S/1Ö 3RV2901-1E			Vers. Blattenteisungsluefter Blatt B supply blade de-icing fan blade B				3.5			
047		Q02	547187-0	Schalter Motorschutz- 4,5-6,3A 3RV2011 switch motor protect. 4,5-6,3A 3RV2011			Vers. Blattenteisungsluefter Blatt B supply blade de-icing fan blade B				3.5			
047		Q03	547041-0	Schalter Hilfs- 1S/1Ö 3RV2901-1E switch auxiliary 1S/1Ö 3RV2901-1E			Vers. Blattenteisungsluefter Blatt C supply blade de-icing fan blade C				3.7			
047		Q03	547187-0	Schalter Motorschutz- 4,5-6,3A 3RV2011 switch motor protect. 4,5-6,3A 3RV2011			Vers. Blattenteisungsluefter Blatt C supply blade de-icing fan blade C				3.7			

				date	18.03.13	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Stueckliste parts list	order no.: ass. ID: 047-88200-V1-05 Y64	= 047Y64 +		
				prep.by	K.Langer								
				arr.by	A.Rülander								
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Steuerschrank Rotorblattenteisung contr. cab. rotor blade de-icing system	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0136581	index: 5	page 2 of 3

Freigegeben

Stückliste/ bill of material													
Anlagenkennung/ installation ID Ortskennung/ location ID BMK/ device ID			Artikel Nr. - Index/ article no. - index	SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english		Funktionstext deutsch/ function remarks german Funktionstext englisch/ function remarks english			Seite. Pfad page. column				
047		Q04	547165-0	Schalter Motorschutz- 20-25A 3RV2021 switch motor protect. 20-25A 3RV2021		Versorgung Blattenteisungsluefter supply blade de-icing fan			2.2				
047		Q04	547041-0	Schalter Hilfs- 1S/1Ö 3RV2901-1E switch auxiliary 1S/1Ö 3RV2901-1E		Versorgung Blattenteisungsluefter supply blade de-icing fan			2.2				
047		R01	976-0	Widerstand Leistung 820R 140W 10% Rohr resistor power 820R 140W 10% pipe		Schaltschrankheizung cabinet heater			4.5				
047		R02	976-0	Widerstand Leistung 820R 140W 10% Rohr resistor power 820R 140W 10% pipe		Schaltschrankheizung cabinet heater			4.5				
047		S01	81310-0	Thermostat ALRE® RTBSS-110.250/04 thermostat ALRE® RTBSS-110.250/04		Schaltschrankheizung cabinet heater			4.5				
047		U01	95018-0	Frequenzumrichter SK500E-751-340-A-CP frequency converter SK500E-751-340-A-CP		Frequenzumrichter frequency converter			2.4				

Freigegeben

Mengen-Stückliste/ quantity bill of material				
Artikel Nr.- Index/ article no.-index	Menge/ quantity	Benennung deutsch/ title german Benennung englisch/ title english	SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english	Hersteller/manufacturer Hersteller Teile-Nr. / manufacturer part no.
16486-0	1	Relaissockel relay base	Socket Relais- FINDER 94.84.30 socket relay- FINDER 94.84.30	Finder 94.84.30
16847-0	1	Reihenklemme terminal	Klemme Reihen- 16mm ² 101A gn/ge USLKG16N terminal block 16mm ² 101A gr/yeUSLKG16N	Phoenix Contact GmbH & Co. KG 0443023
33333-0	3	Überspannungsableiter overvoltage arrester	Überspannungsableiter kompl. VAL-MS230 overvoltage arrester compl. VAL-MS230	Phoenix Contact GmbH & Co. KG 2839127
44623-0	1	Haltebügel bracket	Bügel Halte- FINDER 094.91.30 (Variclip) bracket holding- FINDER 094.91.30 varic.	Finder 094.91.30
510487-0	3	Leistungsschutz power contactor	Schutz Leistungs-15,5A 1Ö DILM15-01 230V contactor power-15,5A 1NC DILM15-01 230V	Moeller Electric GmbH 290093
540535-0	1	Schutzbeschaltung contact suppression	Schutzbeschalt.240-400VAC 3RT2926-1CE00 protective dev.240-400VAC 3RT2926-1CE00	Siemens AG A&D 3RT2926-1CE00
540548-0	1	Leistungsschutz power contactor	Schutz Leistungs- 230VAC 3RT2025-1AP00 contactor power- 230VAC 3RT2025-1AP00	Siemens AG A&D 3RT2025-1AP00
547041-0	4	Hilfsschalter auxiliary switch	Schalter Hilfs- 1S/1Ö 3RV2901-1E switch auxiliary 1S/1Ö 3RV2901-1E	Siemens AG A&D 3RV2901-1E
547165-0	1	Motorschutzschalter motor protection switch	Schalter Motorschutz- 20-25A 3RV2021 switch motor protect. 20-25A 3RV2021	Siemens AG A&D 3RV2021-4DA10
547187-0	3	Motorschutzschalter motor protection switch	Schalter Motorschutz- 4,5-6,3A 3RV2011 switch motor protect. 4,5-6,3A 3RV2011	Siemens AG A&D 3RV2011-1GA10
547563-0	1	Hilfsschalter auxiliary switch	Schalter Hilfs- SÖSS 3RH2911-1XA31-0MA0 switch aux. 0C00 3RH2911-1XA31-0MA0	Siemens AG A&D 3RH2911-1XA31-0MA0
58637-0	3	Schutzbeschaltung contact suppression	Schutzbeschalt.240-500VAC DILM12-XSPR500 protective dev.240-500VAC DILM12-XSPR500	Moeller Electric GmbH 281201
58803-0	3	Leitungsschutzschalter line safety switch	Sicherungsautomat 3p.D20A 400VAC SIE® autom.circuit break.3p. D20A 400VAC SIE®	Siemens AG A&D 5SY4320-8
75531-0	3	Hilfsschalter auxiliary switch	Schalter Hilfs- Anb. 2S 5ST3011 SIE® switch auxiliary 2NO 5ST3011 SIE®	Siemens AG A&D 5ST3011
759-0	22	Kontaktstift contact pin	Stift Steuerungsk. 10A 1,00mm ² vergol. signal contact pin 10A 1,00mm ² go.pl.	Harting GmbH 09150006122

				date	18.03.13	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Mengen-Stückliste quantity parts list	order no.: 047-88200-V1-05 ass. ID: Y66	= 047Y66 +		
				prep.by	K.Langer								
				arr.by	A.Rülander								
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Steuerschrank Rotorblattenteisung contr. cab. rotor blade de-icing system	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0136581	index: 5	page 1 of 2

Freigegeben	Mengen-Stückliste/ quantity bill of material														
	Artikel Nr.- Index/ article no.-index		Menge/ quantity		Benennung deutsch/ title german Benennung englisch/ title english			SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english				Hersteller/manufacturer Hersteller Teile-Nr. / manufacturer part no.			
	777-0		1		Anbaugehäuse mounting case			Gehäuse Anbau- 16p 96x 29 Met. 1 Bügel housing,ext.mount.16p 96x29 met.1 lever				Harting GmbH 09200160301			
	78107-0		1		Schutzbeschaltung contact suppression			Schutzbeschalt.110-240V 99.80.0.230.09 protective dev.110-240V 99.80.0.230.09				Finder 99.80.0.230.09			
	792-0		1		Kontakteinsatz Stift pin insert			Einsatz Stift 25p 250VAC 10A Crimpan. pin insert 25p 250VAC 10A crimp term.				Harting GmbH 09210253001			
	7942-0		3		Doppelstockklemme double level terminal			Klemme Doppelstock 4mm ² gn/ge UKK5-PE terminal double level 4mm ² gn/ye UKK5-PE				Phoenix Contact GmbH & Co. KG 2774211			
	81310-0		1		Thermostat thermostat			Thermostat ALRE® RTBSS-110.250/04 thermostat ALRE® RTBSS-110.250/04				ALRE-IT Regeltechnik GmbH ZN111524			
	883-0		3		Reihenklemme terminal			Klemme Reihen- 16mm ² 800V 101A gr UK16N terminal block 16mm ² 800V 101A gr UK16N				Phoenix Contact GmbH & Co. KG 3006043			
	91516-0		1		Drossel choke			Drossel Motor- AC DEX-LM3-016 choke motor-ac DEX-LM3-016				Moeller Electric GmbH 269542			
	91609-0		1		Drossel choke			Drossel Drehstrom-Netz- DEX-LN3-025 choke three-ph.wall grid- DEX-LN3-025				Moeller Electric GmbH 269504			
	95018-0		1		Umrichter converter			Frequenzumrichter SK500E-751-340-A-CP frequency converter SK500E-751-340-A-CP				Nord GmbH 275450751			
	960-0		1		Reihenklemme terminal			Klemme Reihen- 6mm ² gn/ge USLKG6N terminal block 6mm ² gr/ye USLKG6N				Phoenix Contact GmbH & Co. KG 0442079			
	965-0		22		Doppelstockklemme double level terminal			Klemme Doppelstock 4mm ² grau UKK5 terminal double level 4mm ² grey UKK5				Phoenix Contact GmbH & Co. KG 2774017			
	97180-0		1		Relais relay			Relais FINDER 55.34.8.230.5050 /4W relay FINDER 55.34.8.230.5050 /4Ch				Finder 55.34.8.230.5050			
	976-0		2		Lastwiderstand load resistor			Widerstand Leistung 820R 140W 10% Rohr resistor power 820R 140W 10% pipe				Frizlen GmbH u. Co.KG FZW265X24-820			
				date	18.03.13	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Mengen-Stückliste quantity parts list	order no.: 047-88200-V1-05		Y66		= 047Y66	
				prep.by	K.Langer					ass. ID:					
				arr.by	A.Rülander					© Copyright ENERCON GmbH. All rights reserved.		document ID: D0136581		index: 5	
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Steuerschrank Rotorblattenteisung contr. cab. rotor blade de-icing system				of 2		

Freigegeben	Kabelliste / cable list														
	Anlagenkennung/ installation ID Ortskennung/ location ID BMK/ device ID			Artikel Nr. - Index/ article no. - index	Adern / wires belegte Adern / wires used	SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english		Funktionstext deutsch - englisch/ function remarks german - english							
	047		W01	28998-0	4 4	Leitung Steuer- Y-JZ 4x10,0mm ² grau cable control- Y-JZ 4x10,0mm ² grey		Vers. Steuerschrank Blattenteisung supply control cabinet blade de-icing							
	047		W02	21335-0	12 11	Leitung Steuer- Y-JZ 12x 2,5mm ² grau cable control- Y-JZ 12x2,5mm ² grey		Blattenteisung A blade de-icing A							
	047		W03	21335-0	12 11	Leitung Steuer- Y-JZ 12x 2,5mm ² grau cable control- Y-JZ 12x2,5mm ² grey		Blattenteisung B blade de-icing B							
	047		W04	21335-0	12 11	Leitung Steuer- Y-JZ 12x 2,5mm ² grau cable control- Y-JZ 12x2,5mm ² grey		Blattenteisung C blade de-icing C							
	047		W05	88660-0	7 6	Leitung Steuer- CY-JZ 7x 1,0mm ² grau cable control- CY-JZ 7x1,0mm ² grey		Blattenteisung B blade de-icing B							
	047		W06	88660-0	7 6	Leitung Steuer- CY-JZ 7x 1,0mm ² grau cable control- CY-JZ 7x1,0mm ² grey		Blattenteisung B blade de-icing B							
	047		W07	88660-0	7 6	Leitung Steuer- CY-JZ 7x 1,0mm ² grau cable control- CY-JZ 7x1,0mm ² grey		Blattenteisung C blade de-icing C							

Blattheizung
blade heater

SAP article no.: index:
93682 0

Reifegrad / level of maturity:
Serie
series

Zusatzinfo / additional info:
Umluft
air circulation

Anlagentyp / converter type:
E-44, E-48, E-53
Steuerung / control system:
CS48
Baugruppennr. / assembly no.:
048
Version Baugr. / version of assembly:
V1
Baugr. Index / component index:
02

Baugruppens Schaltplan
assembly circuit diagram

document ID: index:
D0138553 0a

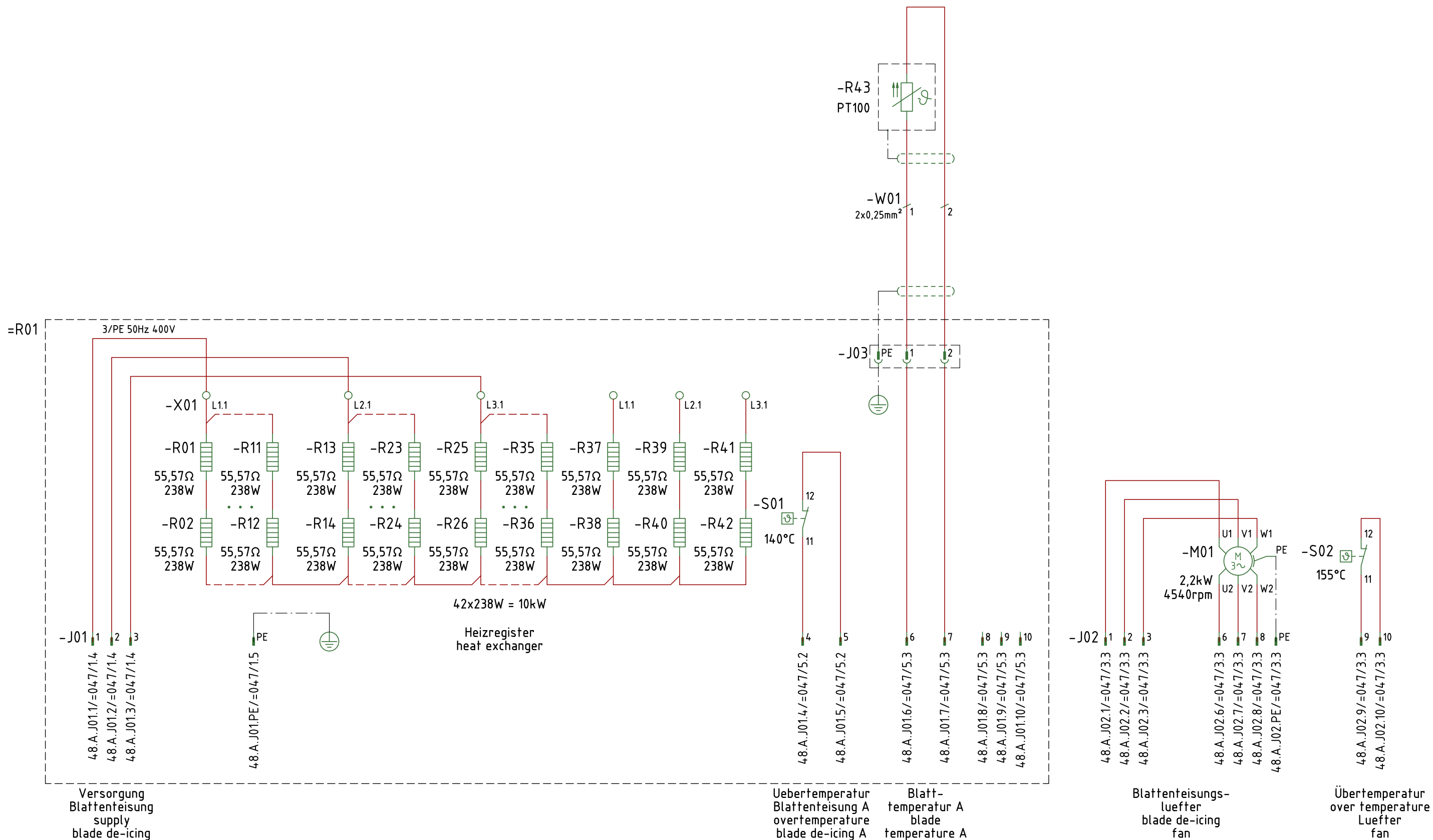
title or assembly ID:
048-93682-V100a

Alle nicht gekennzeichneten Leitungen 1mm² schwarz.
All not marked lines 1mm² black.
Alle nicht gekennzeichneten Neutralleiter 1mm² hellblau.
All not marked neutral conductor 1mm² light blue.

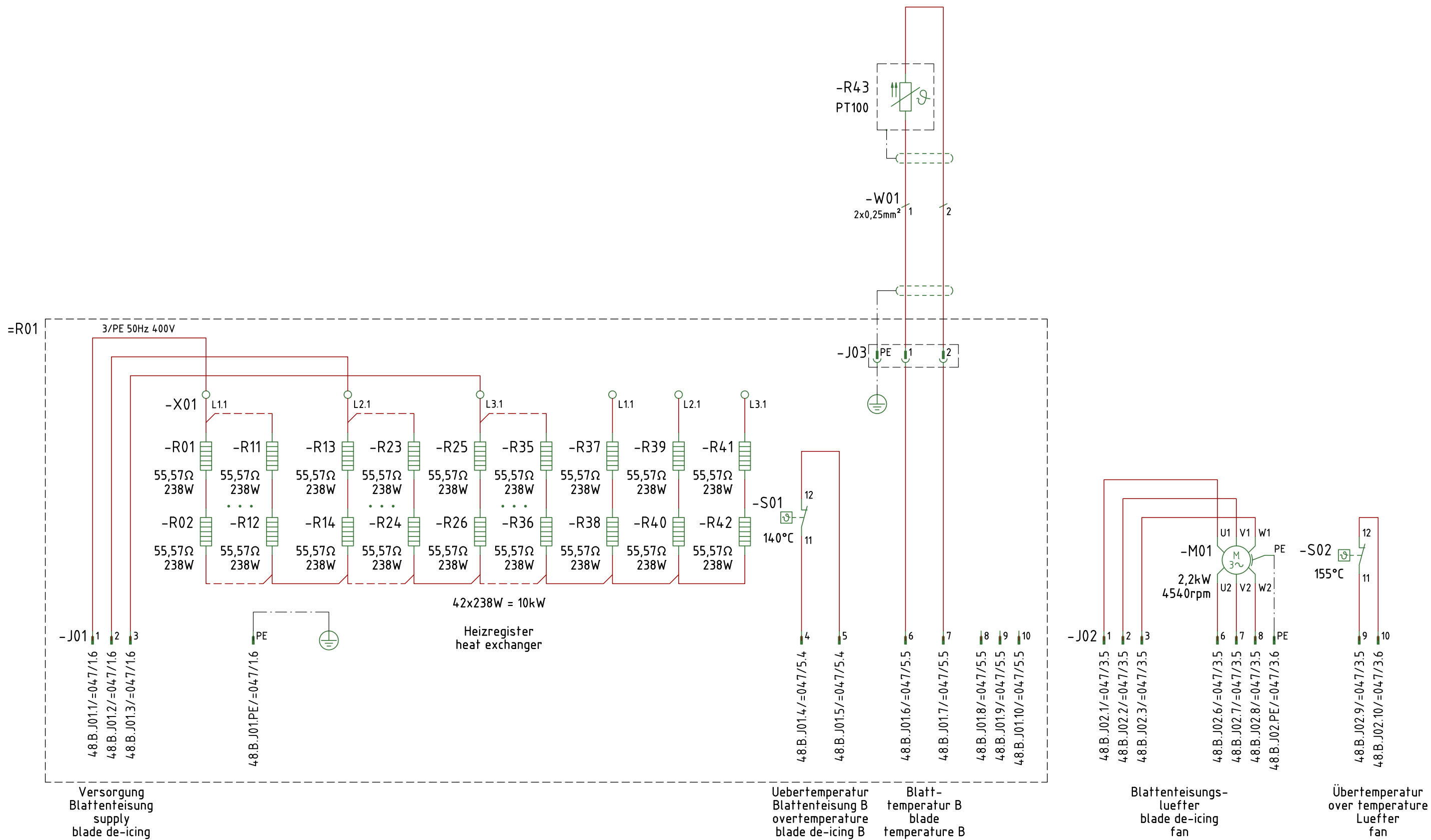
Technische Änderungen vorbehalten!
Technical subject to alteration!

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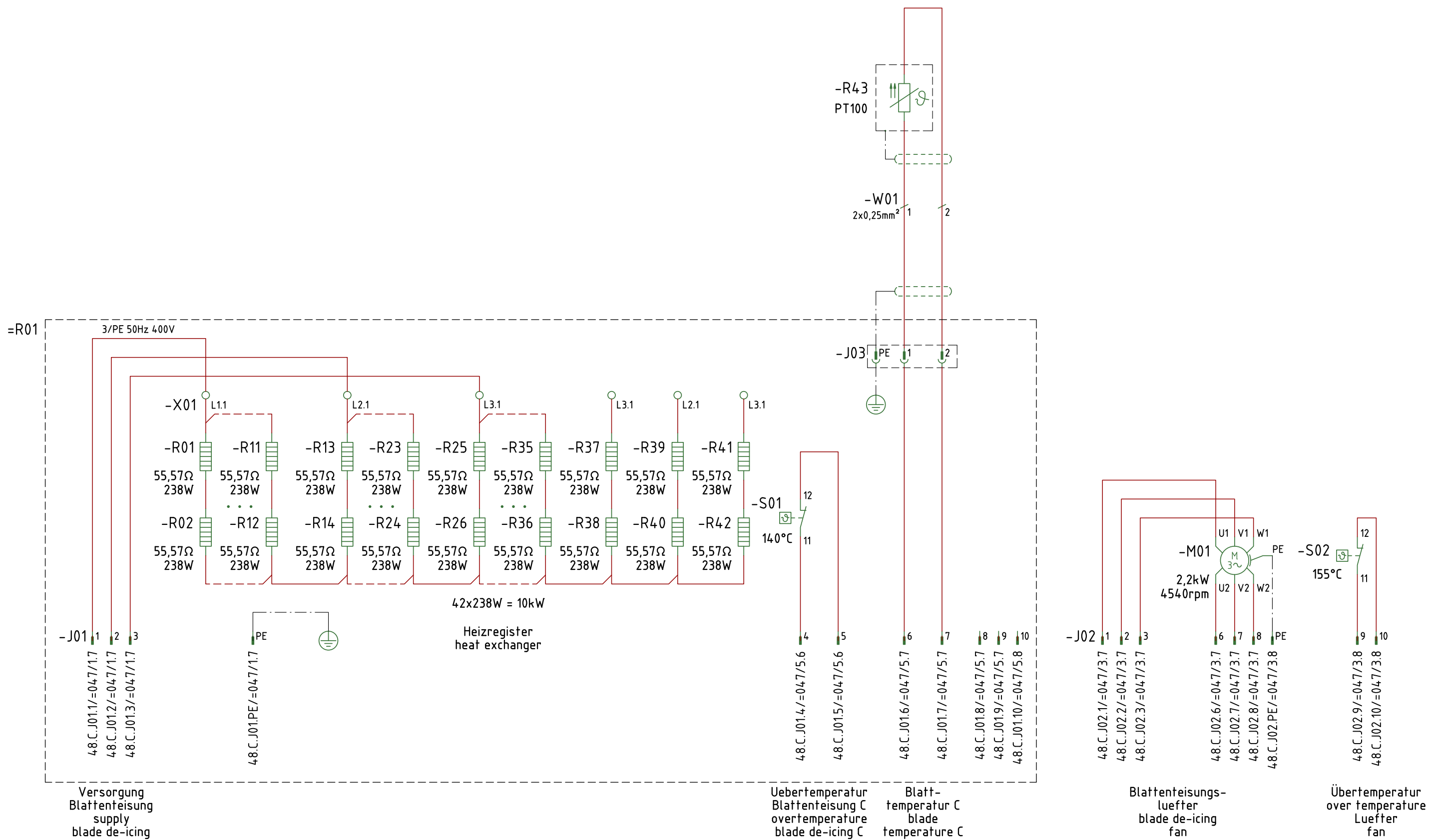
0a		23.12.2010	14056	date	30.03.2009 02.09.2009	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Deckblatt cover Rotorblattenteisung rotor blade de-icing system	order no.: 048-93682-V1-00a		= 048COVER	
				prep.by	J.Cassoli					ass. ID:		+	
				arr.by	G.Ehmen					© Copyright ENERCON GmbH. All rights reserved.		document ID: D0138553	index: 0a
no.	revision	date	name	appr.	14056	origin: germany	repl. for	repl. by	of 1				



				date	30.03.2009	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Blatt A blade A	order no.: ass. ID: 048-93682-V1-00a Y32	= 048		
				prep.by	J.Cassoli						+ A		
				arr.by	G.Ehmen								
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Rotorblattenteisung rotor blade de-icing system	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0138553	index: 0a	page 1 of 3



				date	30.03.2009	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Blatt B blade B	order no.: ass. ID: 048-93682-V1-00a Y32	= 048		
			prep.by	J.Cassoli	+ B								
			arr.by	G.Ehmen									
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Rotorblattenteisung rotor blade de-icing system	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0138553	index: 0a	page 2 of 3



				date	30.03.2009	Schaltplan	CS 48a		Blatt C	order no.:	048-93682-V1-00a		Y32	= 048
				prep.by	J.Cassoli	circuit diagram	Windenergieanlage		blade C	ass. ID:				+ C
				arr.by	G.Ehmen		wind energy converter		Rotorblattenteisung	© Copyright ENERCON GmbH.	document ID:	D0138553	index:	0a
no.	revision	date	name	appr.		origin: germany	repl. for		rotor blade de-icing system	All rights reserved.			page	3
							repl. by						of	3

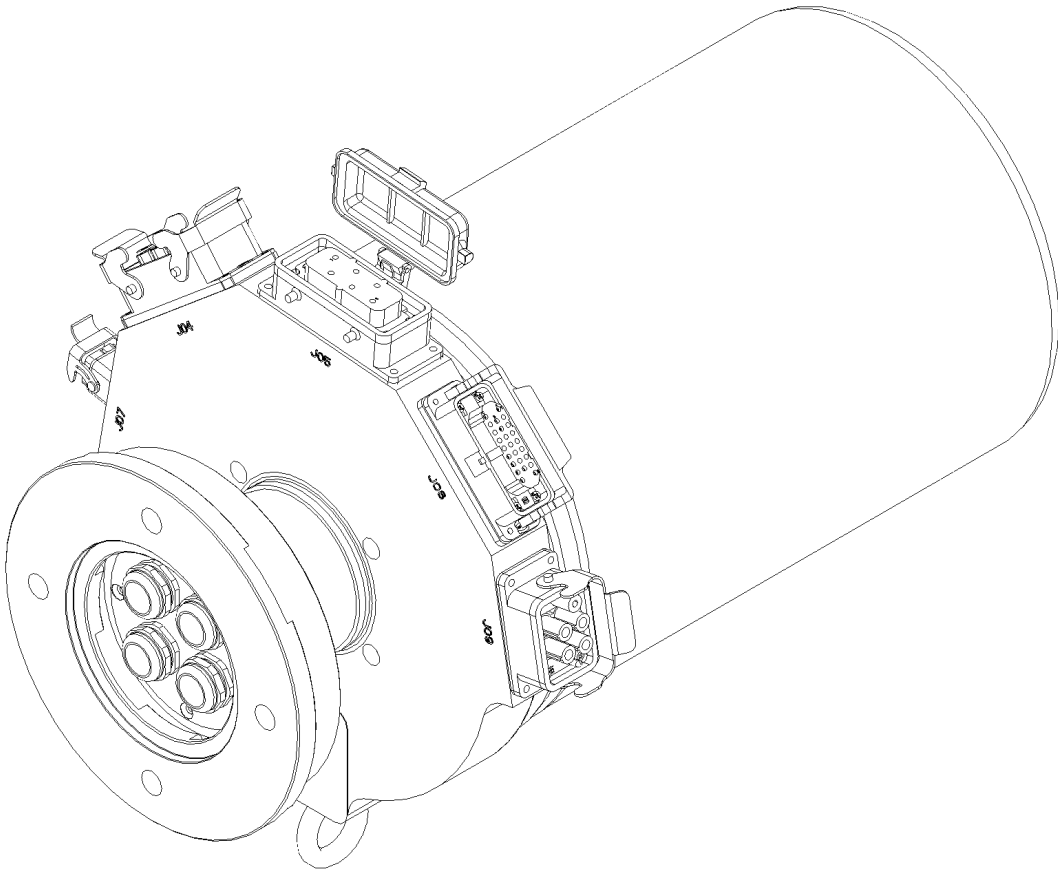
Schleifringübertrager
slip ring unit

SAP article no.: index:
112322 1

Reifegrad / level of maturity:
Serie
series

Zusatzinfo / additional info:

Anlagentyp / converter type:
E-48
Steuerung / control system:
CS48a
Baugruppennr. / assembly no.:
051
Version Baugr. / version of assembly:
5
Baugr. Index / component index:
0



Baugruppenschaltplan
assembly circuit diagram

document ID: index:
D0246851 0

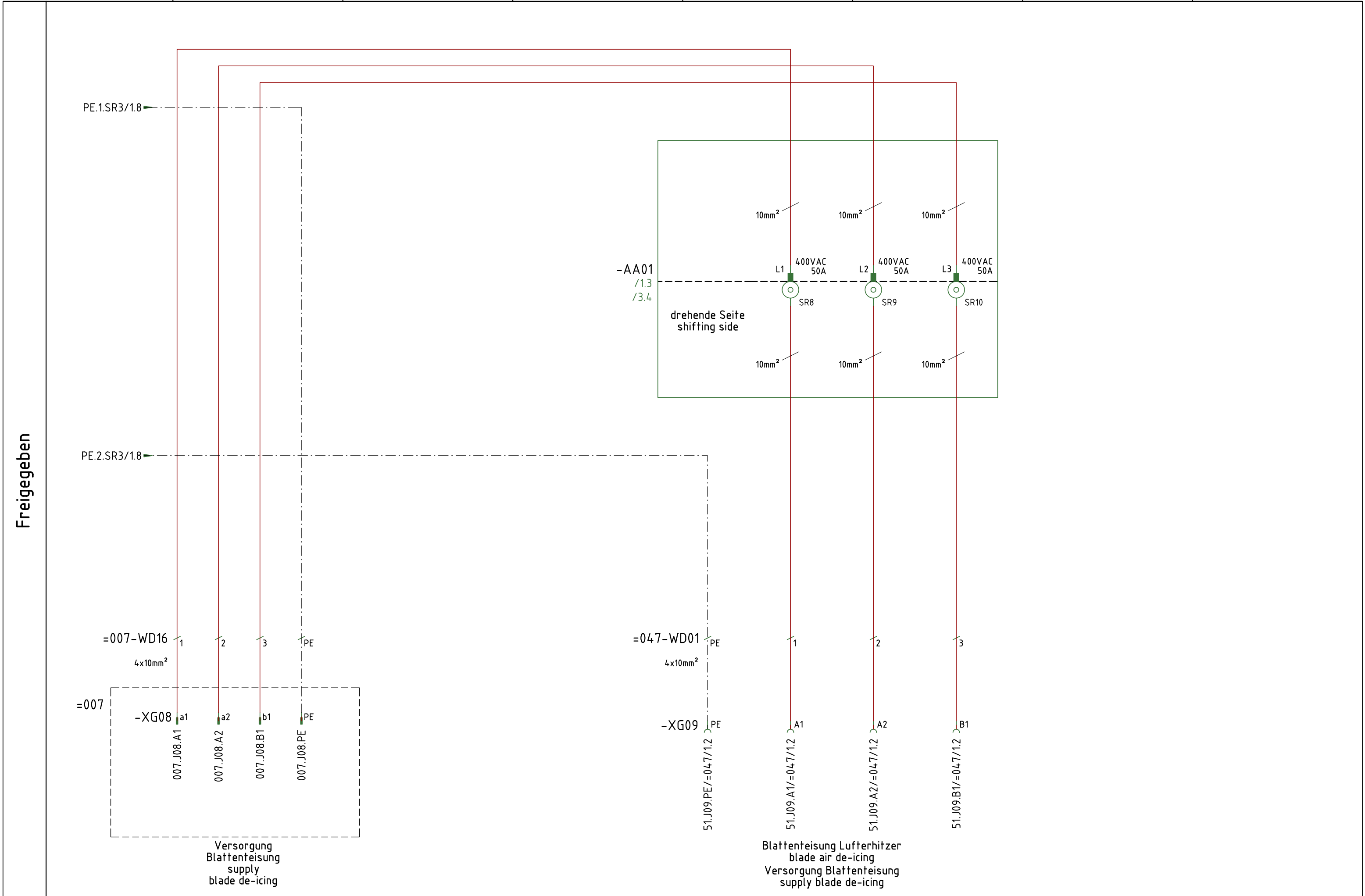
title or assembly ID:
051-112322-V5-00

Alle nicht gekennzeichneten Leitungen 1mm² schwarz.
All not marked lines 1mm² black.
Alle nicht gekennzeichneten Neutralleiter 1mm² hellblau.
All not marked neutral conductor 1mm² light blue.

Technische Änderungen vorbehalten!
Technical subject to alteration!

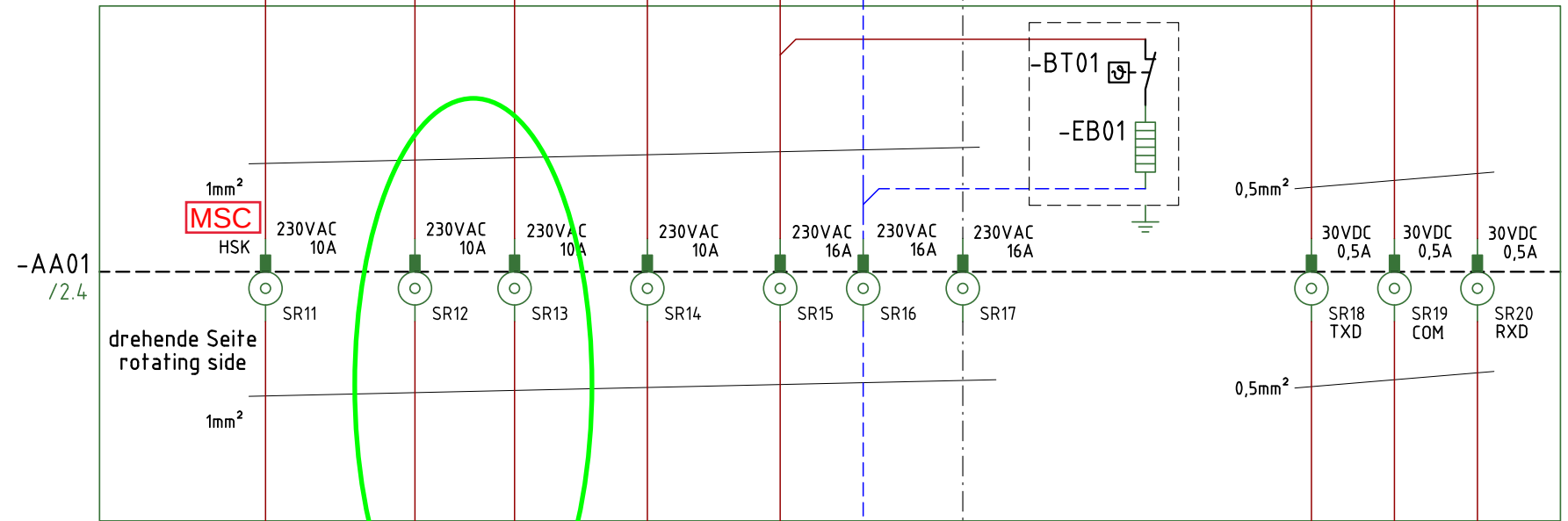
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				date	18.03.13	19.02.2013	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	Deckblatt cover	Schleifringuebertrager slip ring unit	order no.: 051-112322-V5-00		= 051COVER	
				prep.by	J.Rohlfen							ass. ID:			
				arr.by	A.Rülander									+	
no.	revision	date	name	appr.			origin: germany	repl. for	repl. by			© Copyright ENERCON GmbH. All rights reserved.	document ID: D0246851	index: 0	page 1 of 1



Freigegeben

Rotor supply
için 2 yedek
uç var



43:1 hattı

=015 te K07+4 numarasından
ve K09'un 4 numarasından
enerji kontrolü yapılabilir

=013 K02

=015 F02
sigortasından
geliyor

=015 te
optical distribution pcb
den geliyor

=013 numaralı box
içerisindeki
optical distribution
karta gidiyor
oradan A,B,C
dağılımı oluşturuyor

=013 K02

Hauptsicher-
heitskreis
main
security circuit

Reserve
reserve

Blatt aus
ES-95° fahren
blade get
out of LS-95°

Versorgung
Rotor
supply
rotor

Datenbus
data bus

Hauptsicher-
heitskreis
main
security circuit

Reserve
reserve

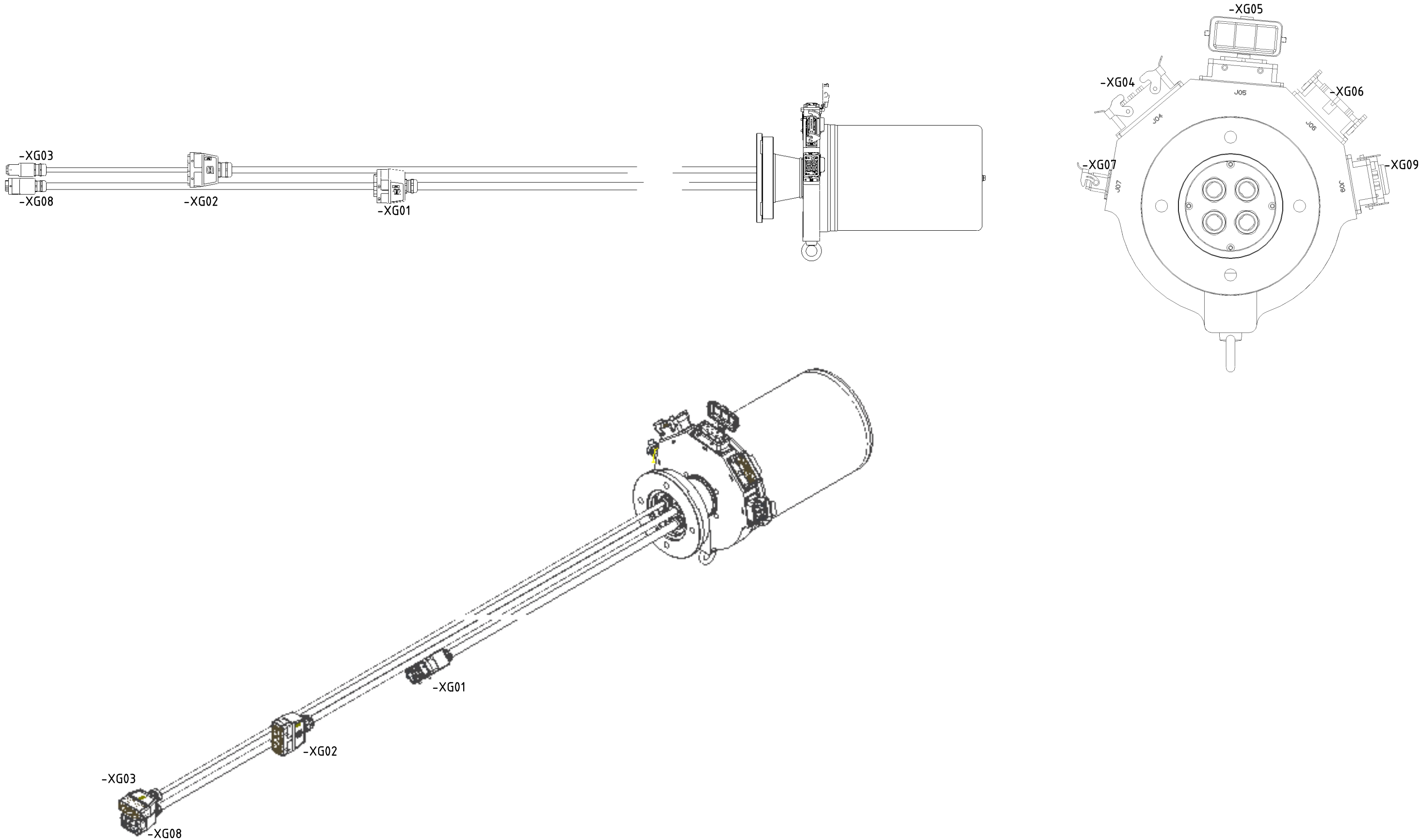
Blatt aus
ES-95° fahren
blade get
out of LS-95°

Versorgung
Rotor
supply
rotor

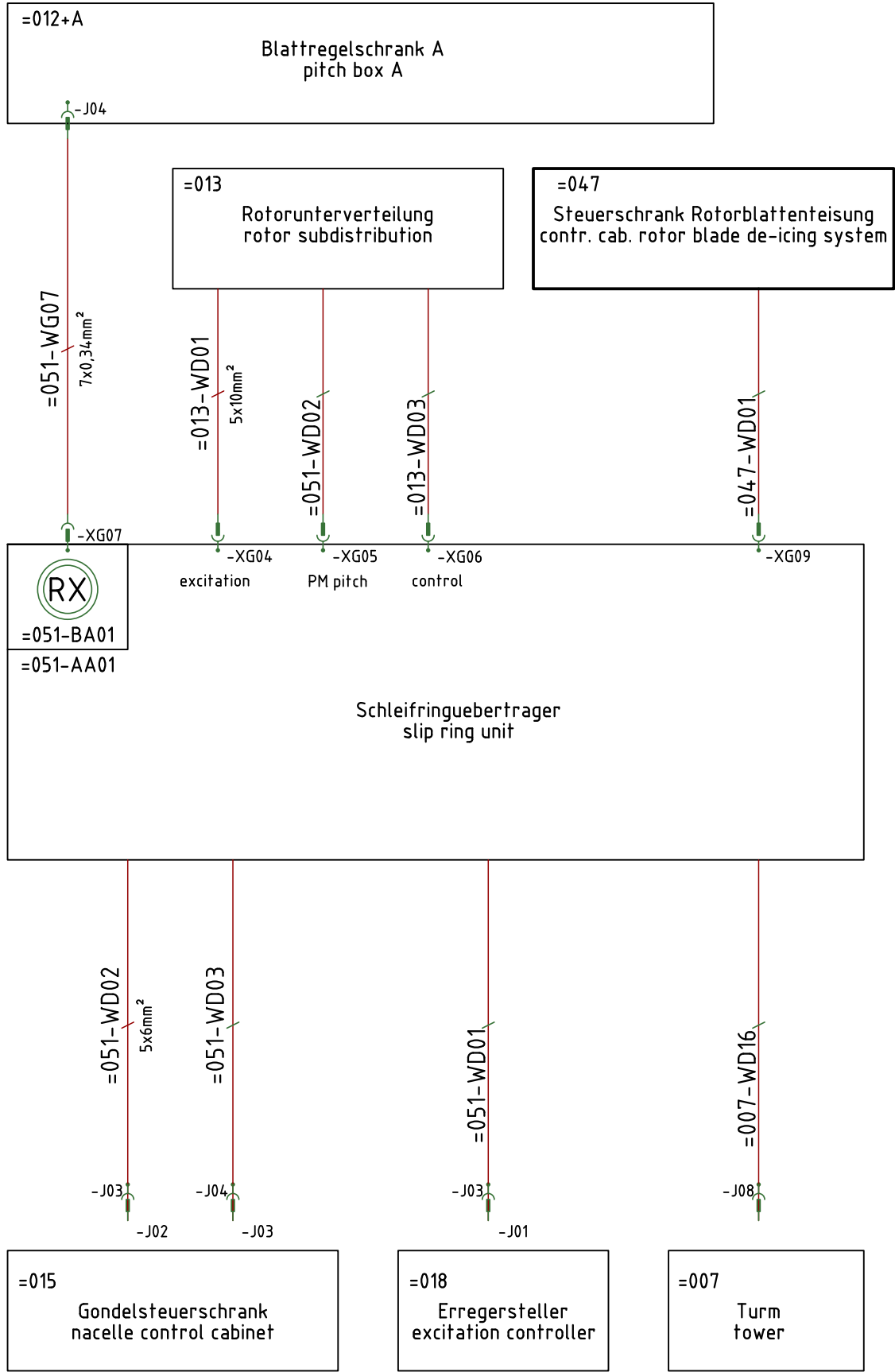
Datenbus
data bus

				date	18.03.13	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Steuerung control	order no.: ass. ID: 051-112322-V5-00	Y32	= 051		
			prep.by	J.Rohlfen	+									
			arr.by	A.Rülander										
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Schleifringuebertrager slip ring unit	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0246851	index: 0	page 3 of 3	

Freigegeben



				date	18.03.13	Schaltplan	CS 48a		Aufbauplan	order no.:	051-112322-V5-00		Y52	= 051Y52
				prep.by	J.Rohlfen	circuit diagram	Windenergieanlage		construction plan	ass. ID:				+
				arr.by	A.Rülander		wind energy converter		Schleifringuebertrager	© Copyright ENERCON GmbH.	document ID:	D0246851	index: 0	page 1
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	slip ring unit	All rights reserved.				of 1



Lastregelschrank load control box

SAP article no.: index:

27530 02

Reifegrad / level of maturity:

Serie
series

Zusatzinfo / additional info:

VAR CS44 : SAP66873
VAR CS48 : SAP68745
VAR CS82 : SAP66875

Anlagentyp / converter type:

E-40/6.44, E82, E-53

Steuerung / control system:

CS44A, CS82, CS48

Baugruppennr. / assembly no.:

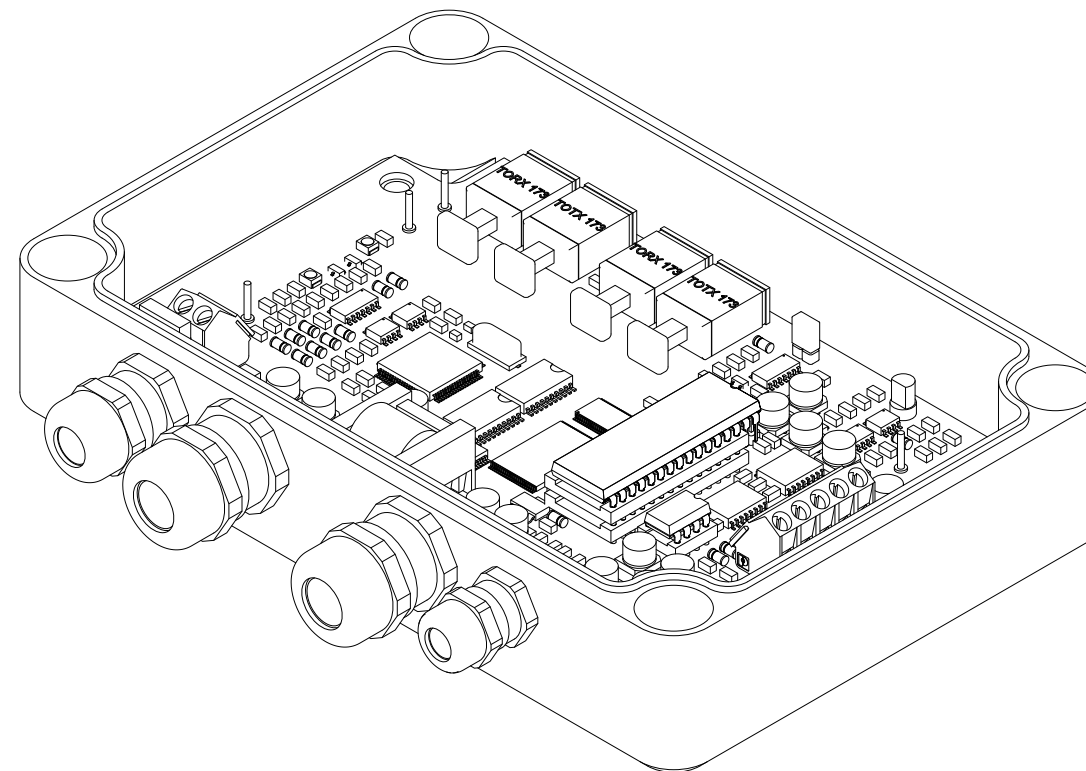
075

Version Baugr. / version of assembly:

V1

Baugr. Index / component index:

04



Baugruppens Schaltplan assembly circuit diagram

document ID: index:

D0116513 3a

title or assembly ID:

075-27530-V1-03a

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All not marked lines 1mm² black.

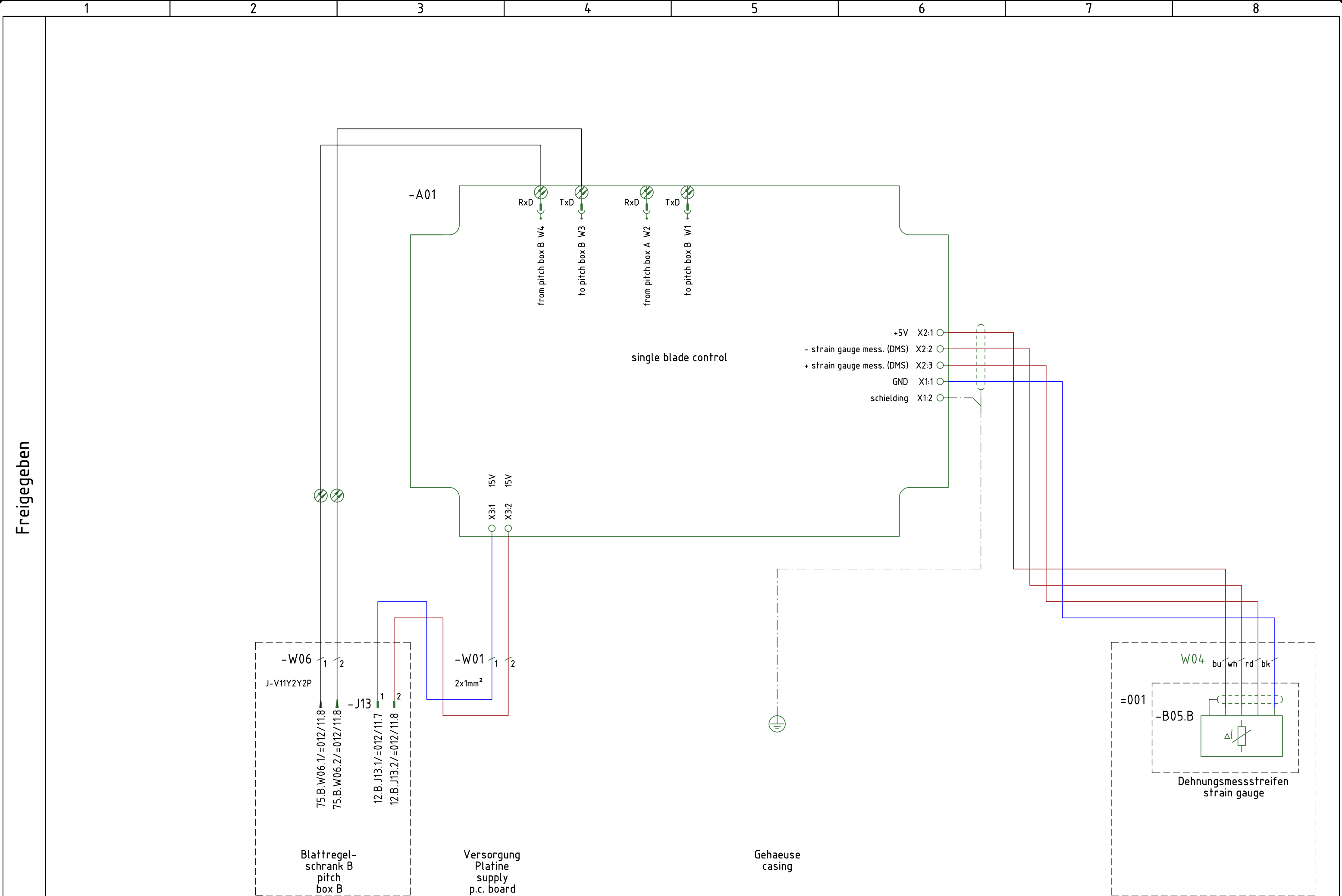
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All not marked neutral conductor 1mm² light blue.

Technische Änderungen vorbehalten!
Technical subject to alteration!

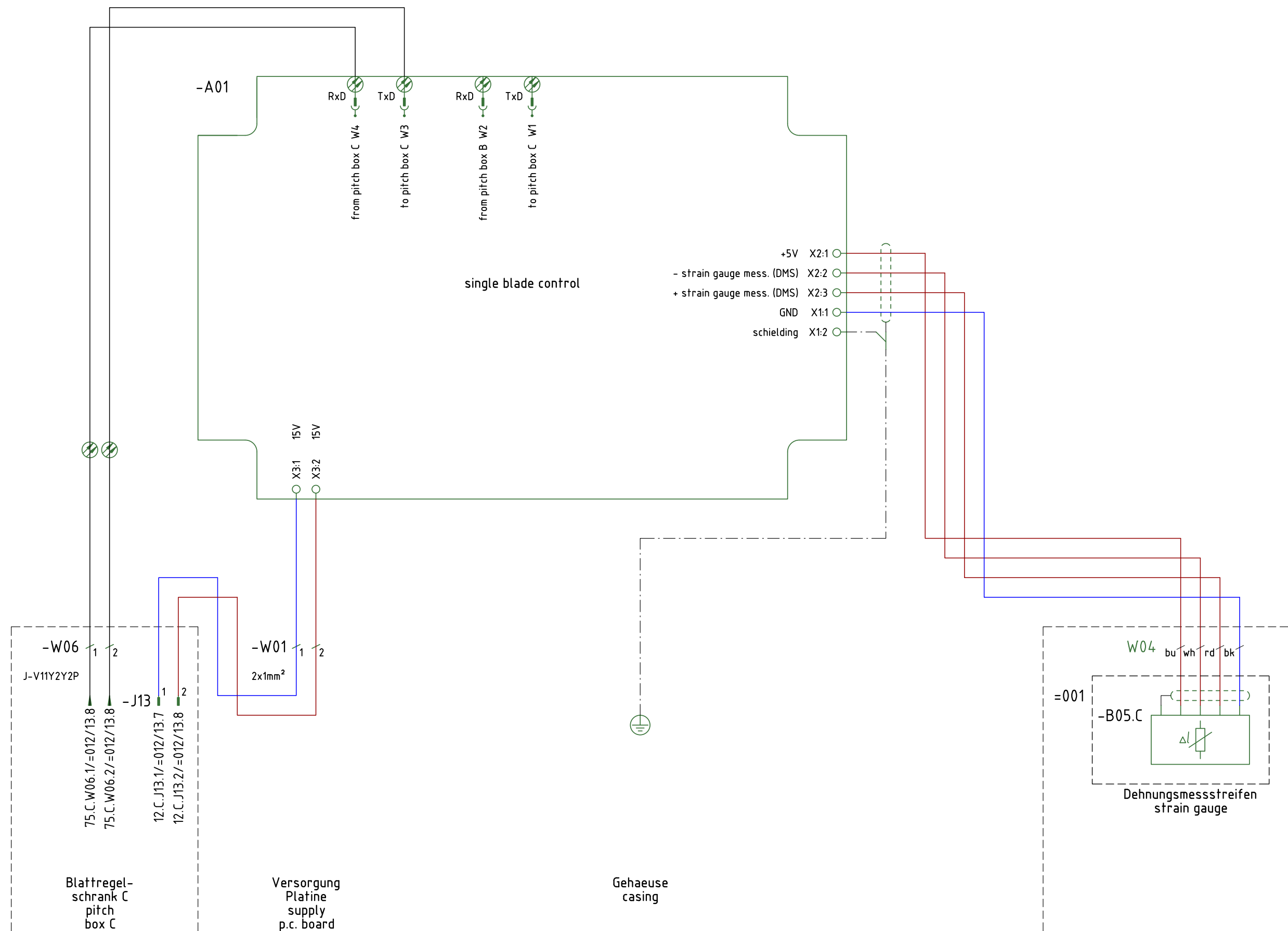
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				date	18.08.2005	10.10.2007	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	Deckblatt cover	order no.: 075-27530-V1-03a		Y08		= 075COVER					
				prep.by	G.Ehmen						ass. ID:				+					
				arr.by	D.Newiger															
no.	revision	date	name	appr.	14056	KG	origin: germany	repl. for	repl. by	Lastregelschrank load control box	© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0116513		index:	3a	page of	1 1		

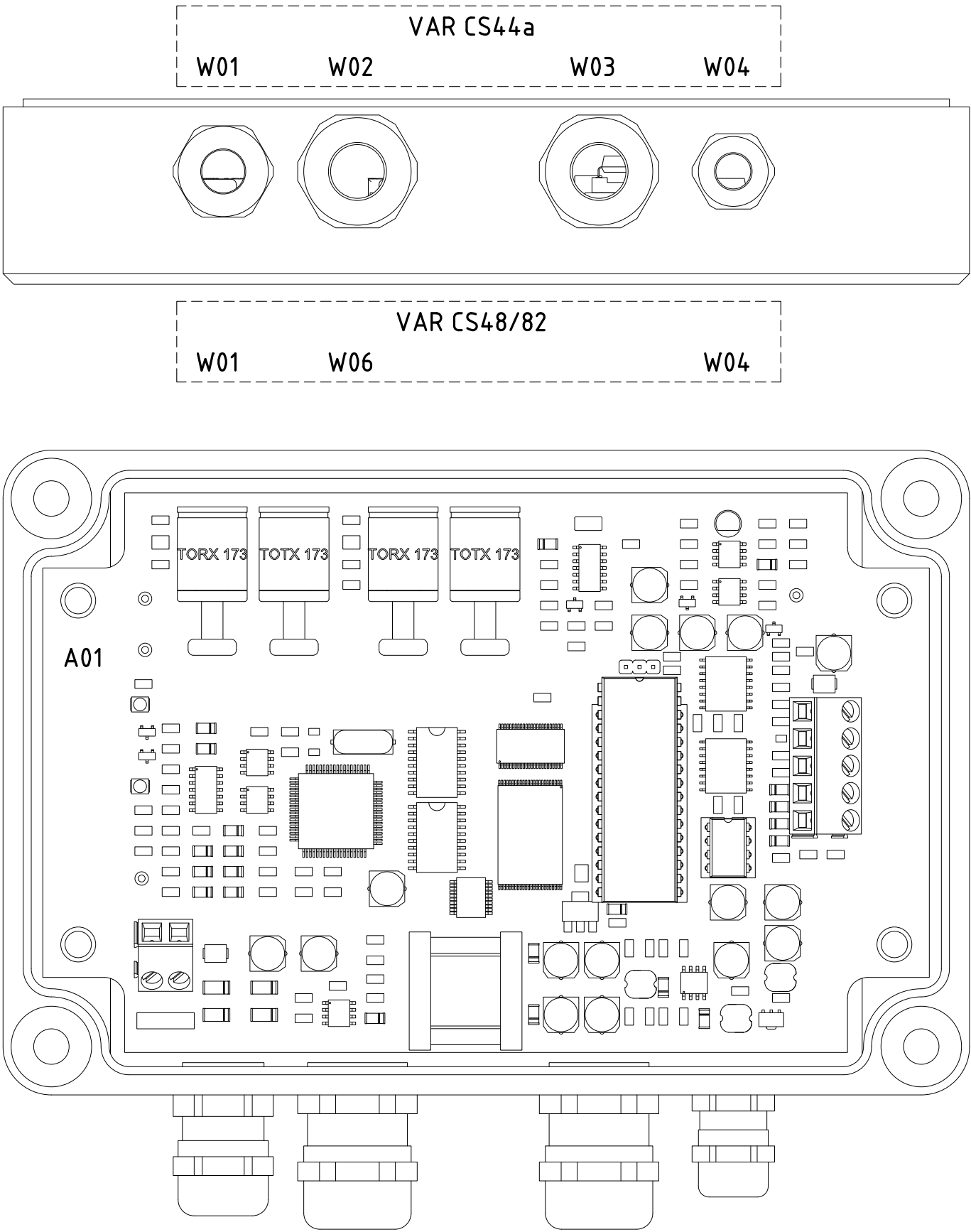




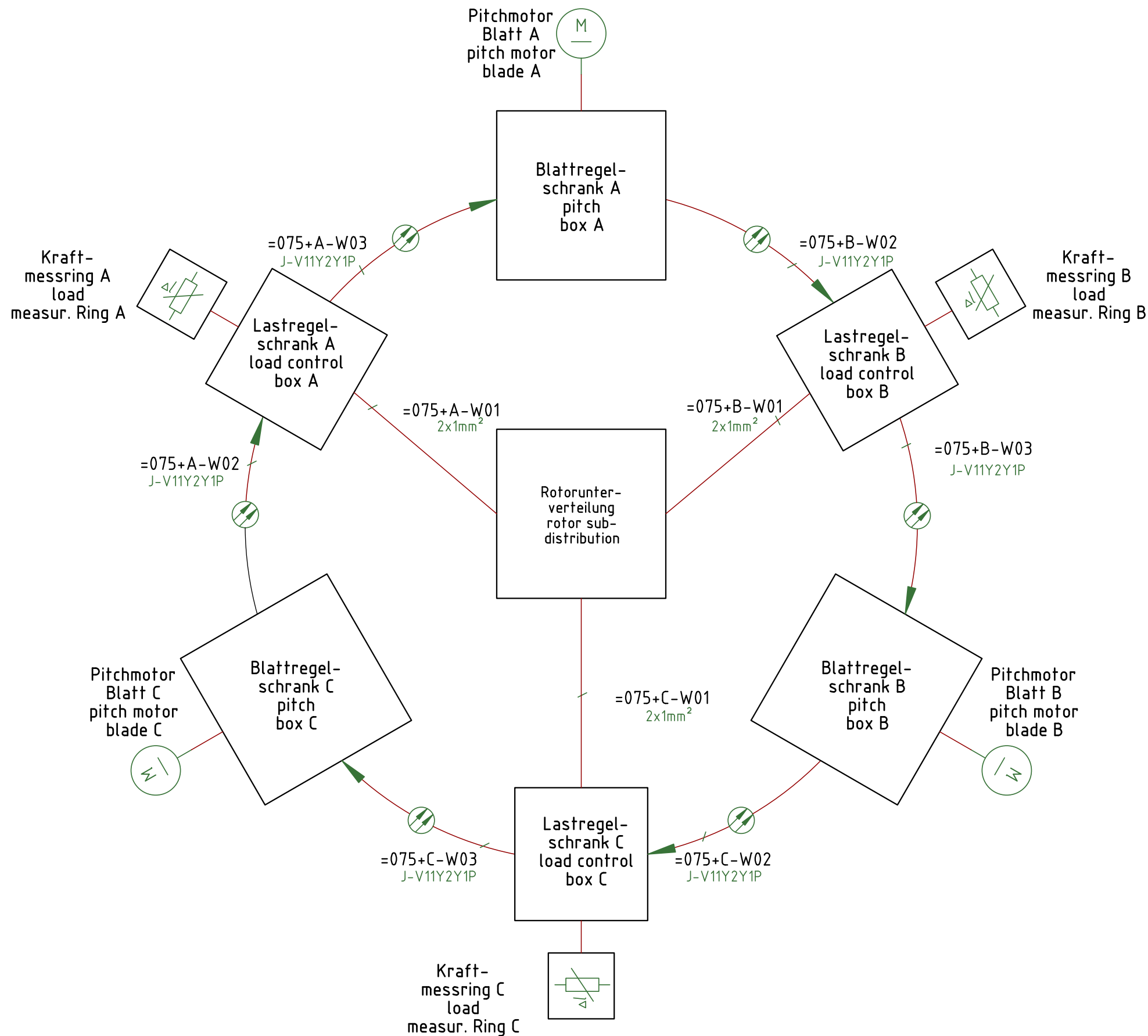
3	C000311	28.03.08	GEH	date	18.08.2005	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Blattlastbegrenzung blade load limitation	order no.: 075-27530-V1-03a		= 075	
2	text	06.02.07	GEH	prep.by	G.Ehmen					ass. ID: 075-27530-V1-03a		+ B	
1	E-82	18.08.05	KL	arr.by	D.Newiger					© Copyright ENERCON GmbH. All rights reserved.		document ID: D0116513	index: 3a
no.	revision	date	name	appr.	G.Eden	origin: germany	repl. for	repl. by	Lastregelschrank load control box				page 2 of 3

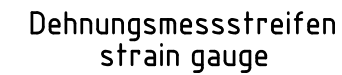


3	C000311	28.03.08	GEH	date	18.08.2005	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Blattlastbegrenzung blade load limitation	order no.: 075-27530-V1-03a		= 075	
2	text	06.02.07	GEH	prep.by	G.Ehmen					ass. ID: 075-27530-V1-03a		+ C	
1	E-82	18.08.05	KL	arr.by	D.Newiger					© Copyright ENERCON GmbH. All rights reserved.		document ID: D0116513	index: 3a
no.	revision	date	name	appr.	G.Eden	KG	origin: germany	repl. for	repl. by	of 3			
									Lastregelschrank load control box				



				date	18.08.2005	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	Aufbauplan construction plan Lastregelschrank load control box	order no.:	= 075Y52			
				prep.by	G.Ehmen					ass. ID:	075-27530-V1-03a			+
				arr.by	D.Newiger					© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0116513		index: 3a
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by				of	1	





				date	18.08.2005	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Anschlussplan connecting plan E-82/E-53	order no.: ass. ID: 075-27530-V1-03a	= 075Y58		
				prep.by	G.Ehmen						+		
				arr.by	D.Newiger								
no.	revision	date	name	appr.	G.Eden KG	origin: germany	repl. for	repl. by	Lastregelschrank load control box	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0116513	index: 3a	page 2 of 2

Freigegeben	Stückliste/ parts list														
	Anlagenkennung/ installation ID Ortskennung/ location ID BMK/ device ID				SAP-Artikel Nr. / SAP-article no. EDM-Artikel ID / EDM-article ID		SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english				Funktionstext deutsch/ function remarks german Funktionstext englisch/ function remarks english				Seite. Pfad page. column
	001	A	B04.A		29831 A009317-0		Kraftmeßring KMR 400kN force sensing ring KMR 400kN								1.6
	075	A	A01		71404 A018806-0		Platine Load Control V2.0 Typ CS44/E40/6 PCB load control V2.0 type CS44/E40/6				Platine Load Control PCB Load Control				1.3
	075	A	R01		13899 A005466-0		Widerstand-MS 66k5 0,25W 0,1% TK±25 resistor,m.-f. 66k5 0,25W 0,1% TK±25								1.7
	075	A	R02		13899 A005466-0		Widerstand-MS 66k5 0,25W 0,1% TK±25 resistor,m.-f. 66k5 0,25W 0,1% TK±25								1.7

Freigegeben	Stückliste/ parts list																			
	Anlagenkennung/ installation ID Ortskennung/ location ID BMK/ device ID				SAP-Artikel Nr. / SAP-article no. EDM-Artikel ID / EDM-article ID		SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english				Funktionstext deutsch/ function remarks german Funktionstext englisch/ function remarks english				Seite. Pfad page. column					
	001		A		B05.A		63666		Knochen ges. Dehnungssensor 6608227-3 compl. strain sensor 6608227-3								1.6			
	075		A		A01		71405 A018806-0		Platine Load Control V2.0 Typ E53/E82 PCB load control V2.0 type E53/E82				Platine Load Control PCB Load Control				1.3			
				date	18.08.2005		Schaltplan circuit diagram		CS 48a Windenergieanlage wind energy converter				Stueckliste parts list CS48/CS82 Lastregelschrank load control box				order no.: 075-27530-V1-03a Y64 = 075Y64			
				prep.by	G.Ehmen												ass. ID: +			
				arr.by	D.Newiger															
no.	revision	date	name	appr.			origin: germany	repl. for	repl. by							© Copyright ENERCON GmbH. All rights reserved.	document ID: D0116513	index: 3a	page 2 of 2	

Kondensatorschrank Rotor

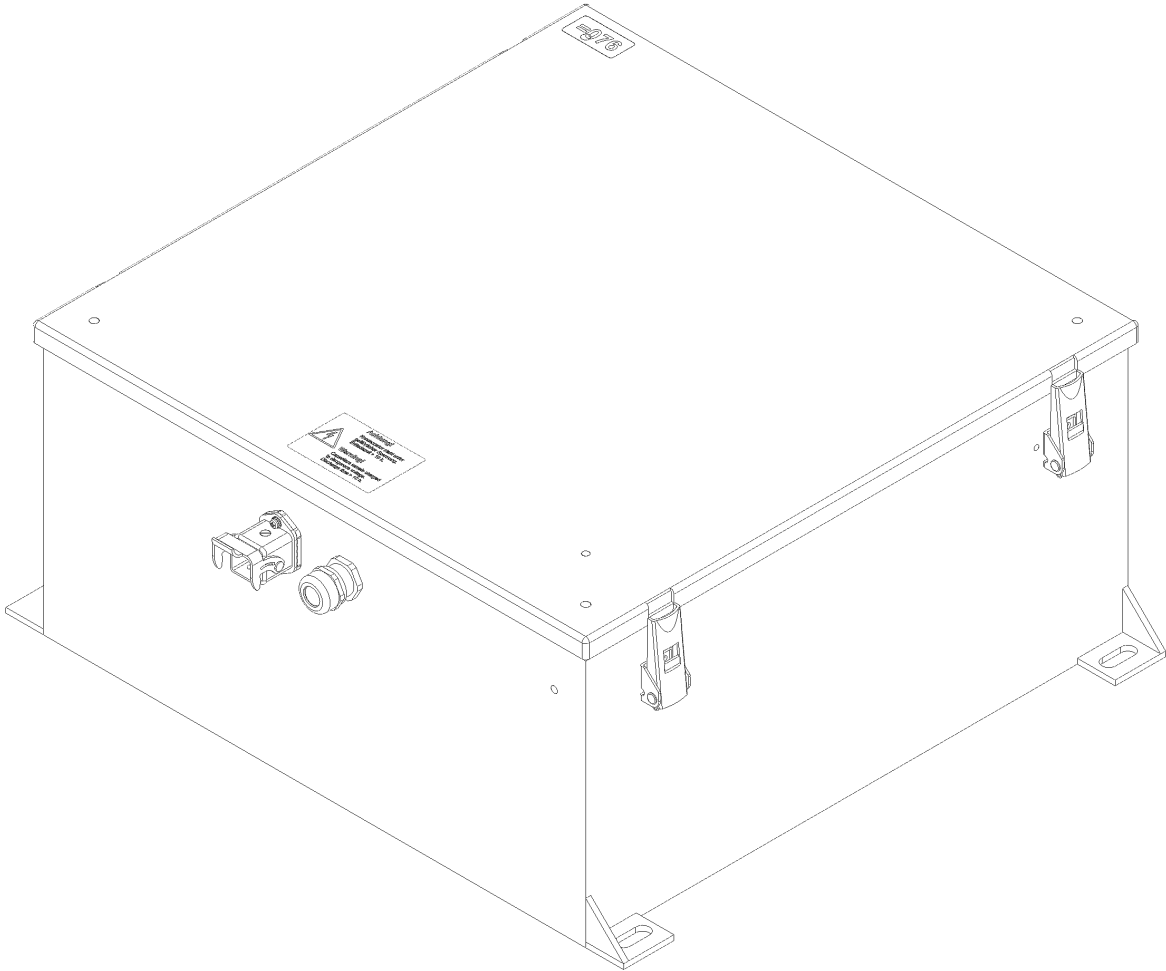
capacitor box rotor

SAP article no.: 517400 index: 3

Reifegrad / level of maturity:
Serie
series

Zusatzinfo / additional info:
Var. Nesscap: 512993 (Index 2)
Var. Maxwell : 517401 (Index 3)
Var. Mtron : 517402 (Index4)
Var. Samwha : 552656 (Index 5)

Anlagentyp / converter type:
E-44; E-48; E-53
Steuerung / control system:
CS48a-01
Baugruppennr. / assembly no.:
076
Version Baugr. / version of assembly:
2
Baugr. Index / component index:
05



Baugruppens Schaltplan

assembly circuit diagram

document ID: D0171019 index: 1

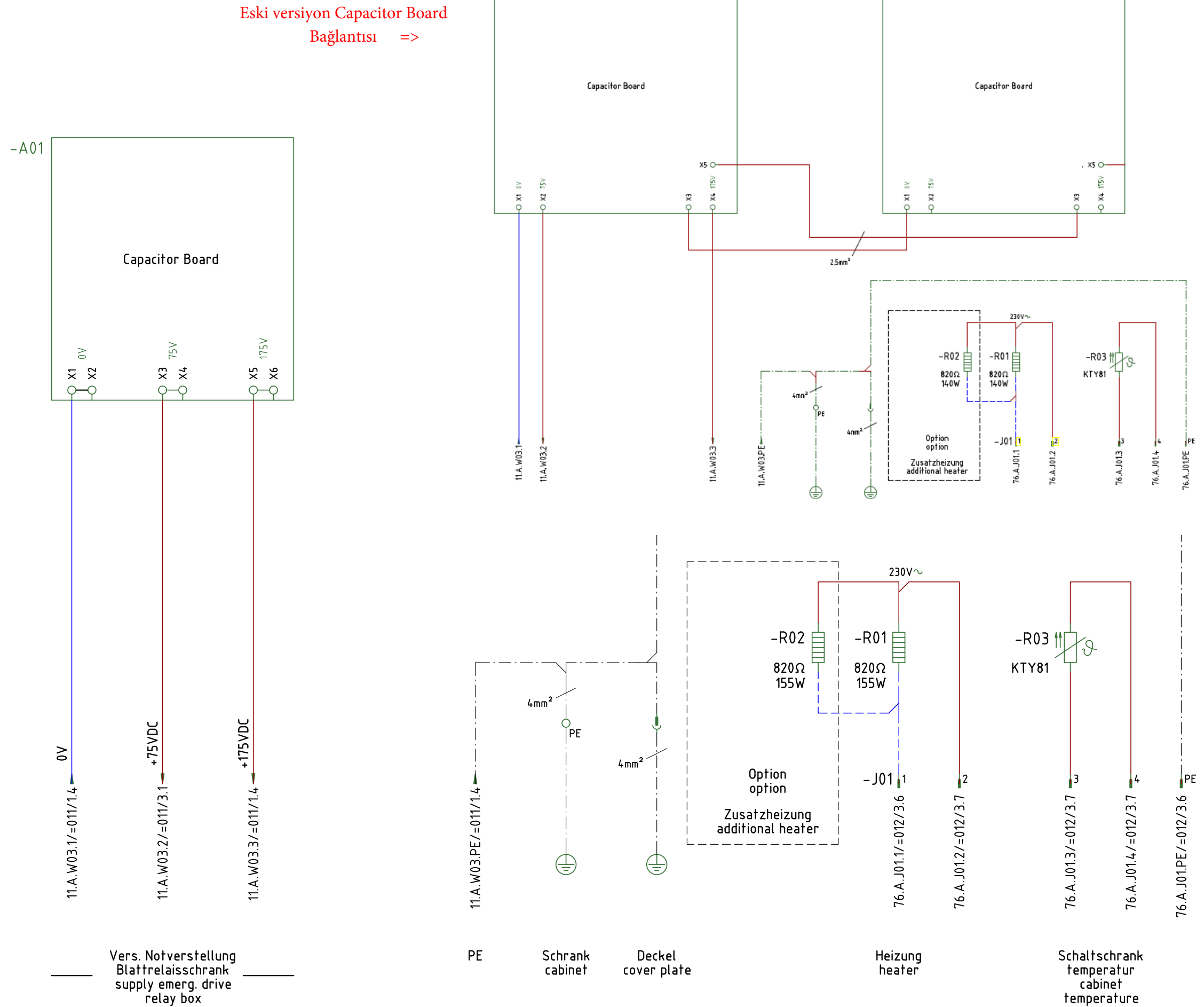
title or assembly ID:
076-517400-V2-01

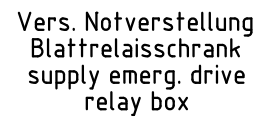
Alle nicht gekennzeichneten Leitungen 1mm² schwarz.
All not marked lines 1mm² black.
Alle nicht gekennzeichneten Neutralleiter 1mm² hellblau.
All not marked neutral conductor 1mm² light blue.

Technische Änderungen vorbehalten!
Technical subject to alteration!

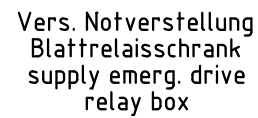
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© Copyright ENERCON GmbH. All rights reserved. This drawing is the property of ENERCON GmbH. It is protected by copyright law and, at the same time, constitutes a trade secret according to §§ 17 et seq. of the German Act Against Unfair Competition (UWG).

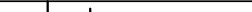
1	EC00002098	25.04.2013	40681	date	15.01.14	25.04.2013	Schaltplan	CS 48a		Deckblatt	order no.:	076-517400-V2-01		Y04	= 076COVER
0	EC00000421	14.06.2011	2467	prep.by	K.Langer		circuit diagram	Windenergieanlage		cover	ass. ID:				+
				arr.by	A.Rülander			wind energy converter		Kondensatorschrank Rotor	© Copyright ENERCON GmbH.	document ID:	D0171019	index: 1	page 1
no.	revision	date	name	appr.	P00028531		origin: germany	repl. for	repl. by	capacitor box rotor	All rights reserved.				of 1

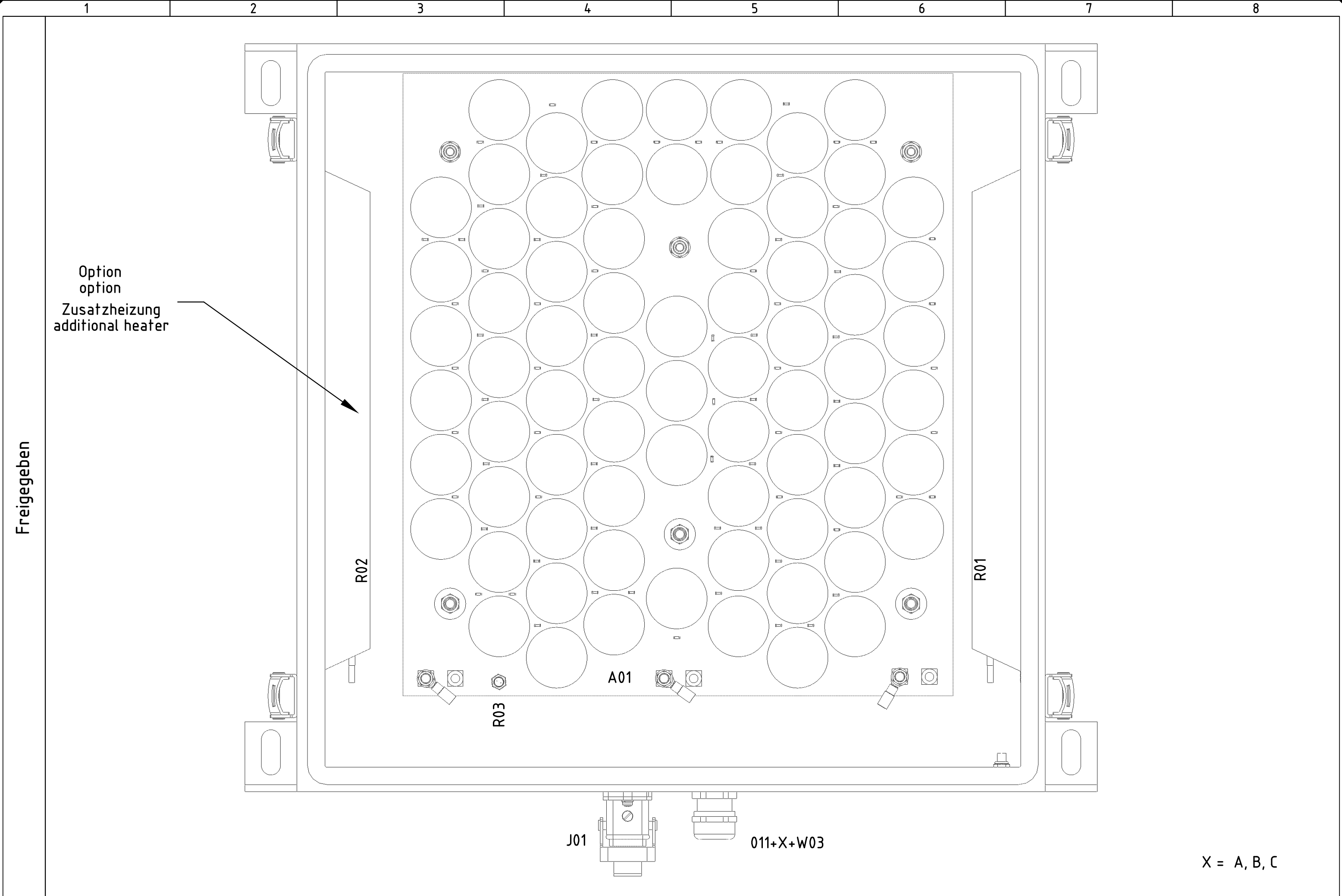




1	EC00002098	25.04.2013	40681	date	15.01.14	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Kondensatorschrank Rotor Blatt A capacitor box rotor blade A	order no.: ass. ID: 076-517400-V2-01 Y32	= 076 + B		
				prep.by	K.Langer								
				arr.by	A.Rülander								
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Kondensatorschrank Rotor capacitor box rotor	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0171019	index: 1	page 2 of 3



1	EC00002098	25.04.2013	40681	date	15.01.14	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Kondensatorschrank Rotor Blatt A capacitor box rotor blade A	order no.: ass. ID: 076-517400-V2-01 Y32	= 076 + C		
				prep.by	K.Langer								
				arr.by	A.Rülander								
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Kondensatorschrank Rotor capacitor box rotor	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0171019	index: 1	page 3 of 3



X = A, B, C

				date	15.01.14	Schaltplan	CS 48a		Aufbauplan	order no.:	= 076Y52	
				prep.by	K.Langer	circuit diagram	Windenergieanlage		construction plan	ass. ID:	+ 076-517400-V2-01	
				arr.by	A.Rülander		wind energy converter		Kondensatorschrank Rotor	document ID:	D0171019	index: 1
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	capacitor box rotor	© Copyright ENERCON GmbH. All rights reserved.		page of

$$X = A, B, C$$

				date	15.01.14	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Anschlussplan connecting plan	order no.: 076-517400-V2-01		= 076Y58	
				prep.by	K.Langer					ass. ID:		+	
				arr.by	A.Rülander					Kondensatorschrank Rotor capacitor box rotor		© Copyright ENERCON GmbH. All rights reserved.	
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by		document ID: D0171019	index: 1	page 1	
												of 1	

Freigegeben	Mengen-Stückliste/ quantity bill of material														
	Artikel Nr.- Index/ article no.-index		Menge/ quantity	Benennung deutsch/ title german Benennung englisch/ title english		SAP-Materialkurztext deutsch/ SAP-material description german SAP-Materialkurztext englisch/ SAP-material description english			Hersteller/manufacturer Hersteller Teile-Nr. / manufacturer part no.						
	3442-0		3	Anbaugehäuse mounting case		Gehäuse Anbau- 4p 28x 40 SWF Gerade housing ext.mount. 4p 28x40 SWR str.			Harting GmbH 09370030301						
	507847-0		6	Lastwiderstand load resistor		Widerstand Leist.820R 155W GMAD 300x20 resistor power 820R 155W GMAD 300x20			Frizlen GmbH u. Co.KG GMAD 300x20-820						
	514482-0		3	Platine PCB		Platine Capacitor-Board V3.2 NessCap PCB capacitor-board V3.2 NessCap			ENERCON GmbH -						
	514484-0		3	Platine PCB		Platine Capacitor-Board V4.3 Maxwell PCB capacitor-board V4.3 Maxwell									
	514678-0		3	Platine PCB		Platine Capacitor-Board V3.2 LS-Mtron PCB capacitor-board V3.2 LS-Mtron			ENERCON GmbH -						
	552507-0		3	Platine PCB					ENERCON GmbH -						
	7183-0		3	Kontakteinsatz Stift pin insert		Einsatz Stift 5p+PE 16A Crimpan. pin insert 5p+PE 16A crimp term.			Harting GmbH 09120053001						
	8914-1		3	Sensor sensor		Sensor AKKU6610 KTY81-110 E66 sensor AKKU6610 KTY81-110 E66			Birko -						
				date	15.01.14	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		Mengen-Stückliste quantity parts list Kondensatorschrank Rotor capacitor box rotor	order no.:	076-517400-V2-01	Y66	= 076Y66		
				prep.by	K.Langer							ass. ID:			+
				arr.by	A.Rülander										
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by			© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0171019	index: 1	page 1 of 1

Freigegeben	Kabelbezeichnung intern cable designation internal										Klemmenleiste: terminal strip:					=011+A-X01					Kabelbezeichnung extern cable designation external											
													Ziel intern destination internal					Ziel extern destination external	Aderbezeichnung wire designation													Funktionstext function remarks
													=Anlage+Ort -BMK:Anschluß =installation+location -identification:connection	Pin-Nummer pin-number	Brücken bridges	Seite,Pfad sheet,column	=Anlage+Ort -BMK:Anschluß =installation+location -identification:connection		no. funktion	-W03	-W01											PE Kondensatorschrank Rotor PE capacitor box rotor
														1		1.4	=076-PE:PE	PE	X													+175VDC Kondensatorschrank Rotor +175VDC capacitor box rotor
														2		1.4	=076-A01:X5	3	X													+175VDC Kondensatorschrank Rotor +175VDC capacitor box rotor
													-K02:1	3		1.4																+175VDC Kondensatorschrank Rotor +175VDC capacitor box rotor
													-K01:1	4		1.4	-K12:3															+175VDC Kondensatorschrank Rotor +175VDC capacitor box rotor
													-K12:6	5		1.4	=076-A01:X1	1	X													0V Kondensatorschrank Rotor 0V capacitor box rotor
													-K03:1	6		1.4																0V Kondensatorschrank Rotor 0V capacitor box rotor
													-F02	7		1.4	-R01															0V Kondensatorschrank Rotor 0V capacitor box rotor
													-F07	8		3.1	=076-A01:X3	2	X													+75VDC Kondensatorschrank Rotor +75VDC capacitor box rotor
													-F07	9		3.2	-J03:11															+75VDC Kondensatorschrank Rotor +75VDC capacitor box rotor
														10		3.3	-J03:12															+75VDC Bruecke +75VDC bridge
													-F03	11		3.3	-J02:A7															Messung 75V, ES-97° measuring 75V, LS-97°
													-F08	12		3.5	-J02:A1															Blatt A ES-97° blade A LS-97°
													-K06:1	13		3.7	-J02:A3															Selbsth. n. ES--2°: ES-90° - Bruecke lock to LS --2°: LS-86° - bridge
														14		5.2	=012-J05:A1	1	X													L1 Vers. Steuerung Blattrelaisschrank A L1 supply relay box A control
													-J02:B4	15		5.2	-J02:A5															L1 Vers. Kompaktends. Heizung, ES 95° L1 supply compact limit sw. Heater, LS-95°
													-K10:A2	16		5.2	=012-J05:A2	2	X													N Vers. Blattrelaisschrank A N relay box A
													-K11:A2	17		5.3	-J02:B5															N Vers.Heizung v. Blattrelaisschrank A N heater from relay box A
														18		5.4	-J03:14															N Vers. Blattrelaisschrank A N relay box A
														19		5.2	=012-J05:PE	PE	X													PE Blattrelaisschrank PE relay box
													-L01:PE	20		5.4	-J02:PE															PE Ladeschaltung PE charging circuit
													-J03:PE	21		2.6	-PE2															PE Pitchmotor, Blattrelaisschrank PE pitch motor, relay box</

[illegible]

Freigegeben

Kabelbezeichnung intern cable designation internal										Klemmenleiste: terminal strip: =013-X01A					Kabelbezeichnung extern cable designation external										Funktionstext function remarks					
										Ziel intern destination internal	Pin-Nummer pin-number	Brücken bridges	Seite.Pfad sheet.column	Ziel extern destination external	Aderbezeichnung wire designation	no. funktion	-W10	-W06	-W60											
										=Anlage+Ort -BMK:Anschluß =installation+location -identification:connection										=Anlage+Ort -BMK:Anschluß =installation+location -identification:connection										
										-X02:12	1		1.4	=012+A-J01:A1	1														L1 Vers. Leistungsteil Blattregelschrank A L1 supply power module pitch box A	
											2		1.4	=012+A-J01:A2	2														L1 Vers. Leistungsteil Blattregelschrank A L1 supply power module pitch box A	
										-X02:18	3		1.4	=012+A-J01:A3	3														N Vers. Leistungsteil Blattregelschrank A N supply power module pitch box A	
										-X01B:3	4		1.4	=012+A-J01:A4	4														N Vers. Leistungsteil Blattregelschrank A N supply power module pitch box A	
										-X02:3	5		5.2	=012+A-J01:A5	5														L1 Versorgung Blattregelschrank A L1 supply pitch box A	
										-X01B:5	6		5.2															L1 Versorgung Blattregelschrank B L1 supply pitch box B		
										-X02:6	7		5.2	=012+A-J01:A6	6														N Vers. Blattregelschrank A N sup. pitch box A	
											8		5.2	-X01B:7														N Versorg. Blattregelschrank B + Blattenteisung A N supply pitch box B + blade de-icing A		
										=051-XG06:A1	9		5.3	=012+A-J01:A7	7														Hauptsicherheitskreis Blatt A main security circuit blade A	
										-X01B:9	10		5.4	=012+A-J01:A8	8														Hauptsicherheitskreis Blatt A main security circuit blade A	
										-K01:2	11		5.4	=012+A-J01:A9	9														Blattsicherheitskreis Blatt A blade security circuit Blatt A	
										-K02:2	12		5.5	=012+A-J01:A10	10														Blatt aus ES-95° fahren blade get out of LS-95°	
										=002-S01:22	13		6.3	=012+A-J01:B2	12														Luftspaltschalter A1 air gap switch A1	
										=002-S02:22	14		6.3	=012+A-J01:B3	13														Luftspaltschalter A1 air gap switch A1	
											15		8.5														Reserve reserve			
											16		8.5														Reserve reserve			
										=002-S01:21	17		6.1	=012+A-J01:B1	11														+15VDC Sensorik +15VDC sensor technology	
										=002-S02:21	18		6.2	=047-J01:A1	1														+15VDC Sensorik +15VDC sensor technology	
										-K01:21	19		6.2															+15VDC Sensorik +15VDC sensor technology		
										=002-J01:A9	20		6.2	=002-B01	BN														+15VDC Sensorik +15VDC sensor technology	
										=002-J01:B2	21		6.4	=012+A-J01:B4	14														0°-Eichung Blatt A 0°-calibration blade A	
										=002-J01:B3	22		6.4	=012+A-J01:B5	15														GND Sensorik GND sensor technology	
										-K01:22	23		6.5	=012+A-J01:B6	16														HSK geschl. MSC closed	
											24		7.1	=012+A-J01:B7	17														Reserve reserve	
										=002-B01	25		7.2	=012+A-J01:B8	18														Akustiksensord acoustic sensor	
										=047-J01:C1	26		7.2	=012+A-J01:B9	19														Heizung in Betrieb heating activated	

[illegible]

[illegible]

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	=013-X02	order no.:	= 000Y44			
				prep.by	A.Rülander					ass. ID:	CS48a-01-R-V1-06	Y44	+	
				arr.by						© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:	6
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Klemmenplan terminal strip			of	16	

Freigegeben

Kabelbezeichnung intern cable designation internal										Klemmenleiste: =013-X02 terminal strip:					Kabelbezeichnung extern cable designation external										Funktionstext function remarks																																																																																																																																																																																																																																																																																																																																																		
										Ziel intern destination internal	Pin-Nummer pin-number	Brücken bridges	Seite.Pfad sheet.column	Ziel extern destination external																																																																																																																																																																																																																																																																																																																																																													
										=Anlage+Ort -BMK:Anschluß =installation+location -identification:connection				=Anlage+Ort -BMK:Anschluß =installation+location -identification:connection																																																																																																																																																																																																																																																																																																																																																													
										-W30 no. funktion											Aderbezeichnung wire designation	no. funktion	-W49	-W48																																																																																																																																																																																																																																																																																																																																																			

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	=047-X01	order no.:	= 000Y44			
				prep.by	A.Rülander					ass. ID:	CS48a-01-R-V1-06	Y44	+	
				arr.by						© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:	6
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Klemmenplan terminal strip			of	16	

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	=047-X01	order no.:	= 000Y44			
				prep.by	A.Rülander					ass. ID:	CS48a-01-R-V1-06	Y44	+	
				arr.by						© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:	6
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Klemmenplan terminal strip			of	16	

Freigegeben

[illegible]

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		=076+B-PE	order no.: ass. ID: CS48a-01-R-V1-06	Y44	= 000Y44 + B			
				prep.by	A.Rülander										
				arr.by											
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Klemmenplan terminal strip	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0129411	index: 6	page 10 of 16		

Freigegeben

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				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		=076+C-PE	order no.: 000Y44			
				prep.by	A.Rülander					ass. ID: CS48a-01-R-V1-06 Y44		+ C	
				arr.by						© Copyright ENERCON GmbH. All rights reserved.		document ID: D0129411	index: 6
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Klemmenplan terminal strip		of 16		

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		=002-J01	order no.:	= 000Y48			
				prep.by	A.Rülander					ass. ID:	CS48a-01-R-V1-06	Y48	+	
				arr.by						Steckerplan plugging chart	© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by					of	51

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	=007-XG08	order no.: Y48 = 000Y48			
				prep.by	A.Rülander					ass. ID: CS48a-01-R-V1-06 Y48 +			
				arr.by						Steckerplan plugging chart			
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by		© Copyright ENERCON GmbH. All rights reserved.	document ID: D0129411	index: 6	page 2 of 51

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Kabelbezeichnung intern cable designation internal										Steckerbelegung: pin assignment: =012+A-J01					Kabelbezeichnung extern cable designation external										Funktionstext function remarks																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
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				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		=012+A-J01	order no.:	= 000Y48			
				prep.by	A.Rülander					ass. ID:	CS48a-01-R-V1-06	Y48	+ A	
				arr.by						Steckerplan plugging chart	© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:
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				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		=012+A-J03	order no.:	= 000Y48			
				prep.by	A.Rülander					ass. ID:	CS48a-01-R-V1-06	Y48	+ A	
				arr.by						Steckerplan plugging chart	© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:
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				prep.by	A.Rülander					ass. ID:	CS48a-01-R-V1-06	Y48	+ B	
				arr.by						Steckerplan plugging chart	© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:
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				arr.by						Steckerplan plugging chart	© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by					of	51

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			arr.by		Steckerplan plugging chart					© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by				of	51

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	=012+C-J03	order no.:	= 000Y48			
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				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		=012+C-J05	order no.:	= 000Y48			
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				arr.by						Steckerplan plugging chart	© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by					of	51

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	=012+C-J13	order no.:	= 000Y48			
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no.	revision	date	name	appr.		origin: germany	repl. for	repl. by				of	51	

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	=015-XG02	order no.: = 000Y48		
			prep.by	A.Rülander	ass. ID: CS48a-01-R-V1-06 Y48 +							
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				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	=015-XG03	order no.:	= 000Y48			
				prep.by	A.Rülander					ass. ID:	CS48a-01-R-V1-06	Y48	+	
				arr.by						© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:	6
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Steckerplan plugging chart			of	51	

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	=018-XG01	order no.: = 000Y48		
				prep.by	A.Rülander					ass. ID: CS48a-01-R-V1-06 Y48 +		
				arr.by						© Copyright ENERCON GmbH. All rights reserved.		
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Steckerplan plugging chart	document ID: D0129411	index: 6	page 22 of 51

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	=048+A-J02	order no.: Y48 = 000Y48			
			prep.by	A.Rülander	ass. ID: CS48a-01-R-V1-06 + A								
			arr.by										
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Steckerplan plugging chart	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0129411	index: 6	page 24 of 51

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		=048+B-J02	order no.:	CS48a-01-R-V1-06	Y48	= 000Y48	
				prep.by	A.Rülander					ass. ID:			+ B	
				arr.by										
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Steckerplan plugging chart	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0129411	index: 6	page 25 of 51	

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	=051-XG05	order no.:	= 000Y48		
			prep.by	A.Rülander	ass. ID:					CS48a-01-R-V1-06	Y48	+	
			arr.by		Steckerplan plugging chart					© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by				of	51

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				arr.by						Steckerplan plugging chart	© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by				of	51

				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter	 ENERCON ENERGY FOR THE WORLD	=051-XG09	order no.:	= 000Y48			
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				arr.by						Steckerplan plugging chart		© Copyright ENERCON GmbH. All rights reserved.	
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				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		=076+C-J01	order no.: ass. ID: CS48a-01-R-V1-06 Y48	= 000Y48		
				prep.by	A.Rülander						+ C		
				arr.by									
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by	Steckerplan plugging chart	© Copyright ENERCON GmbH. All rights reserved.	document ID: D0129411	index: 6	page 34 of 51

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				date	16.01.2014	Schaltplan circuit diagram	CS 48a Windenergieanlage wind energy converter		=096-J01	order no.:	= 000Y48			
				prep.by	A.Rülander					ass. ID:	CS48a-01-R-V1-06	Y48	+	
				arr.by						Steckerplan plugging chart	© Copyright ENERCON GmbH. All rights reserved.	document ID:	D0129411	index:
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				arr.by						Steckerplan plugging chart			
no.	revision	date	name	appr.		origin: germany	repl. for	repl. by		© Copyright ENERCON GmbH. All rights reserved.	document ID: D0129411	index: 6	page 37 of 51

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